

LEBANESE UNIVERSITY
REAL ANALYSIS

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INTRODUCTION

Analysis is a profound subject; its neither easy to understand nor summarize. However real analysis can be discovered by solving problems. These notes aim to give you opportunities to discover the basic ideas of differential calculus on the real line. Any suggestions for improvements and additions are very much appreciated. Email: ziad.khalil.1@ul.edu.lb

1.1 Sets of Numbers.

Let us briefly examine the main sets of numbers that we are interested in.

$\mathbb{N} := \{0, 1, 2, 3 \dots\}$	The set of natural numbers.
$\mathbb{N}^* := \{1, 2, 3 \dots\}$	The set of positive integers.
$\mathbb{Z} := \{0, \pm 1, \pm 2, \pm 3 \dots\}$	The set of integer numbers.
$\mathbb{Q} := \{\frac{m}{n}; m \in \mathbb{Z} \text{ and } n \in \mathbb{N}^*\}$	The set of rational numbers.
$\mathbb{R} := \{ \text{The unique complete ordered field } \supset \mathbb{Q} \}$	The set of real numbers.
$\mathbb{R} \setminus \mathbb{Q} := \{x : x \in \mathbb{R} \text{ and } x \notin \mathbb{Q}\}$	The set of irrational numbers.

Without loss of generality, in the definition of $\mathbb{Q}$, one can assume that the fraction is reduced, that is, m and n have no common divisors (other than ± 1). Also one may assume that $n \in \mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$. The set of rationals $(\mathbb{Q}, +, \cdot)$ with the basic operations addition and multiplication is a highly satisfactory algebraic system, that is, it satisfies the field axioms:

- ① Closure law: $a + b \in \mathbb{Q}$ and $ab \in \mathbb{Q}$ for all $a, b \in \mathbb{Q}$.
- ② Commutative law: $a + b = b + a$ and $ab = ba$ for all $a, b \in \mathbb{Q}$.
- ③ Associative law: $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ for all $a, b, c \in \mathbb{Q}$.
- ④ Distributive law: $a(b + c) = ab + ac$ for all $a, b, c \in \mathbb{Q}$.
- ⑤ Existence of Identity Elements: $\exists 0, 1 \in \mathbb{Q}$ such that $0 + a = a$ and $1a = a$ for all $a \in \mathbb{Q}$.
- ⑥ Existence of Inverse Elements:
 - Additive inverse: $\forall a \in \mathbb{Q}, \exists -a \in \mathbb{Q}$ such that $a + (-a) = 0$.
 - Multiplicative inverse: $\forall a \in \mathbb{Q}^* = \mathbb{Q} \setminus \{0\}, \exists a^{-1} \in \mathbb{Q}$ such that $aa^{-1} = 1$.

we can use the above structure to define the concepts of subtraction and division, as $a - b = a + (-b)$ and $\frac{a}{b} = ab^{-1}$ (assuming $b \neq 0$).

It would appear that the set $\mathbb{Q}$ is a nice algebraic system until one tries to solve the equation $x^2 = 2$. It turns out that no rational number satisfies this equation, but there is a good reason to believe that there is some kind of numbers satisfying this equation. Consider a square with sides having length one. If x is the length of the diagonal, then from geometry we know that $x^2 = 1^2 + 1^2$, i.e., $x^2 = 2$. Apparently there is a positive length whose square is 2, which we write as $\sqrt{2}$. But $\sqrt{2}$ cannot be a rational number.

Let us introduce some methods of proofs we use in these notes.

Proof By Direct method. To show $p \implies q$, we assume that p is true and proceed through a sequence of logical steps to arrive at the conclusion that q is also true.

Proof By Contrapositive Method. To show $p \implies q$ it is equivalent to show $\neg q \implies \neg p$.

Proof By Contradiction. In order to prove a statement s is true one can proceed as follows: We suppose s is false and try to obtain a contradiction i.e., a statement which is always false, for example $r \wedge \neg r$ where r is an additional statement. In other words $s \iff [\neg s \implies (r \wedge \neg r)]$.

For example $[p \implies q] \iff [p \wedge \neg q \implies r \wedge \neg r]$. That is, to show $[p \implies q]$ we assume q is false and try to obtain $r \wedge \neg r$. Note that the contrapositive method is a particular case of contradiction.

An integer is said to be even (odd) if $n = 2k$ ($n = 2k + 1$) for some $k \in \mathbb{Z}$.

1.1 Example. Show that: $[n \text{ is an even integer}] \iff [n^2 \text{ is an even integer}]$.

Solution.

$\implies$ Direct method: Assume that n is an even integer, then $n = 2k$, $k \in \mathbb{Z}$.

Now $n^2 = 4k^2 = 2(2k^2)$. Since $2k^2 \in \mathbb{Z}$, then n^2 is an even integer.

$\impliedby$ Contrapositive Method: To show $[n^2 \text{ is an even integer}] \implies [n \text{ is an even integer}]$ is equivalent to show $[n \text{ is an odd integer}] \implies [n^2 \text{ is an odd integer}]$.

Assume that n is an odd integer, then $n = 2k + 1$, $k \in \mathbb{Z}$.

Now $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $(2k^2 + 2k) \in \mathbb{Z}$, then n^2 is an odd integer.

1.2 Property. $\sqrt{2}$ is not a rational number i.e., $\sqrt{2} \notin \mathbb{Q}$.

Proof. By Contradiction: Suppose that $\sqrt{2} \in \mathbb{Q}$. Then $\sqrt{2} = \frac{m}{n}$, $m \in \mathbb{Z}$, $n \in \mathbb{N}^*$ with m and n have no common divisors (other than ± 1). Hence $m^2 = 2n^2$ (by taking squares). Thus m^2 is an even integer which gives m is even and can be written $m = 2k$, $k \in \mathbb{Z}$.

Now $n^2 = 2k^2$ is an even integer which gives n is even. Therefore m and n have 2 as common divisor which contradicts the fact that m and n have no common divisors. Thus $\sqrt{2} \notin \mathbb{Q}$. $\square$

This shows that there are 'holes' in the rational numbers, so we want enough numbers to fill the holes. Informally, what we want to do is 'complete' the rational numbers to include numbers like $\sqrt{2}$. We call numbers in $\mathbb{R} \setminus \mathbb{Q} := \{x; x \in \mathbb{R} \text{ and } x \notin \mathbb{Q}\}$ irrational numbers.

1.3 Property. There exists a nonempty subset $\mathbb{Q}_+^* := \{\frac{m}{n}; m, n \in \mathbb{N}^*\}$ of $\mathbb{Q}$, called the set of positive elements of $\mathbb{Q}$, satisfying the following properties:

- ⑦ $\mathbb{Q}_+^*$ is closed under addition i.e., $x + y \in \mathbb{Q}_+^* \forall x, y \in \mathbb{Q}_+^*$.
- ⑧ $\mathbb{Q}_+^*$ is closed under multiplication $xy \in \mathbb{Q}_+^* \forall x, y \in \mathbb{Q}_+^*$.
- ⑨ For each $x \in \mathbb{Q}$ exactly one of the following holds: $x \in \mathbb{Q}_+^*$, $-x \in \mathbb{Q}_+^*$, $x = 0$.

1.4 Definition. On $\mathbb{Q}$ we define the relation " $\geq$ " by

$$x \geq y \iff x - y \geq 0 \text{ (i.e., } x - y > 0 \text{ or } x - y = 0 \text{)}.$$

1.5 Theorem. $(\mathbb{Q}, +, \cdot, \geq)$ is an ordered field i.e., it satisfies the nine properties ①... ⑨.

Also $\mathbb{Q}(\sqrt{2}) := \{x + y\sqrt{2}; x, y \in \mathbb{Q}\}$ is an ordered field. This suggests that there are infinitely many ordered sub fields of $\mathbb{R}$. The property that ensures the existence of a unique ordered field $\mathbb{R}$ containing $\mathbb{Q}$ (up to renaming the elements) and distinguishes $\mathbb{R}$ from all other ordered fields is the Completeness Property described in section 1.4.

1.2 Order Structure of $\mathbb{R}$.

$(\mathbb{R}, +, \cdot, \geq)$ is an ordered field i.e., it satisfies the algebraic properties ①... ⑥ and the Order Axiom: There exists a nonempty subset $\mathbb{R}_+^*$ of $\mathbb{R}$, called the set of positive elements of $\mathbb{R}$, satisfying the following properties:

- ⑦ $\mathbb{R}_+^*$ is closed under addition i.e., $x + y \in \mathbb{R}_+^* \forall x, y \in \mathbb{R}_+^*$.
 - ⑧ $\mathbb{R}_+^*$ is closed under multiplication $xy \in \mathbb{R}_+^* \forall x, y \in \mathbb{R}_+^*$.
 - ⑨ $\forall x \in \mathbb{R}$, exactly one of the following holds: $x \in \mathbb{R}_+^*$, $-x \in \mathbb{R}_+^*$, $x = 0$. (Trichotomy Property)
- A real number x is called positive if $x \in \mathbb{R}_+^*$ and negative if $-x \in \mathbb{R}_+^*$.

The binary relation $\geq$ is derived from the above Order Axiom:

$$x \geq y \iff x - y \geq 0 \text{ (i.e., } x - y > 0 \text{ or } x - y = 0 \text{)}.$$

Hence, if either $x > y$ or $x = y$, we write $x \geq y$ or $y \leq x$ and say that x is greater than or equal to y or y is less than or equal to x . If $S \subset \mathbb{R}$, we define $S_+ := \{x \in S : x \geq 0\}$ and $S_+^* := \{x \in S : x > 0\}$.

1.6 Property. The binary relation $\geq$ on $\mathbb{R}$ has the following properties:

1. Reflexive: $x \geq x$ for all $x \in \mathbb{R}$.
2. Anti-symmetric: $x \geq y$ and $y \geq x$ implies that $x = y$.
3. Transitive: $x \geq y \geq z$ implies $x \geq z$.
4. Totally ordered: $x \geq y$ or $y \geq x$, $\forall x, y \in \mathbb{R}$ (any two elements of $\mathbb{R}$ are comparable).
5. Compatible with addition: $x \geq y \implies x + a \geq y + a$ for all $a \in \mathbb{R}$.
6. Compatible with multiplication: $x \geq y \implies ax \geq ay$ for all $a \geq 0$.
7. Reflection: $x \geq y \implies ax \leq ay$ for all $a \leq 0$.

Proof.

1. Let $x \in \mathbb{R}$. Since $x - x = 0$, then $x \geq x$.
2. Contrapositive Method: This follows by applying the trichotomy property for $x - y$.
3. If $x = y$ or $y = z$, then $x \geq z$ follows easily.
Assume $x \neq y$ and $y \neq z$, then $x - y > 0$ and $y - z > 0$. Hence, by closure under addition $x - z = (x - y) + (y - z) > 0$.
4. This follows by applying the trichotomy property for $x - y$.
5. Let $a \in \mathbb{R}$. Then $(x + a) - (y + a) = x - y \geq 0$.
6. If $x = y$, then $ax = ay$.
Assume $x \neq y$, then $x - y > 0$. Hence, by closure under multiplication $ax - ay = a(x - y) > 0$.
7. Let $a \leq 0$. If $x = y$, then $ax = ay$.
Assume $x \neq y$, then $x - y > 0$. Hence, by closure under multiplication $ay - ax = -a(x - y) > 0$.

□

The binary relation $\geq$ with properties 1., 2., and 3. is called **partially ordered relation** and with 4. is called **totally ordered relation**. Note that 4. implies 1., hence a totally ordered relation is a relation satisfies 2., 3. and 4.

1.7 Property. The totally ordered relation $\geq$ on $\mathbb{R}$ has the following properties:

1. $(a \leq b \text{ and } c \leq d) \implies a + c \leq b + d$.
2. $(0 \leq a \leq b \text{ and } 0 \leq c \leq d) \implies ac \leq bd$.
3. $(a \leq b \leq 0 \text{ and } c \leq d \leq 0) \implies ac \geq bd$.
4. $\alpha.) 0 < a < b \implies a^2 < b^2$.
 $\beta.) a < b < 0 \implies a^2 > b^2$.
5. $(0 < a < b \text{ or } a < b < 0) \implies \frac{1}{a} > \frac{1}{b}$.

Proof.

1. $(a \leq b \text{ and } c \leq d) \implies (a + c \leq b + c \text{ and } b + c \leq b + d)$ (compatible with addition)
 $\implies a + c \leq b + d.$ (Transitive)
2. $(0 \leq a \leq b \text{ and } 0 \leq c \leq d) \implies (ac \leq bc \text{ and } bc \leq bd)$ (compatible with multiplication)
 $\implies ac \leq bd.$ (Transitive)
3. $(a \leq b \leq 0 \text{ and } c \leq d \leq 0) \implies (-a \geq -b \geq 0 \text{ and } -c \geq -d \geq 0)$ (Reflection)
 $\implies ac \geq bd.$ (Using 2.)
4. $\alpha.)$ and $\beta.)$ are particular cases of 2. and 3.
5. $(0 < a < b \text{ or } a < b < 0) \implies (ab > 0 \text{ and } b - a > 0) \implies (b - a)\frac{1}{ab} > 0 \implies \frac{1}{a} > \frac{1}{b}.$

□

1.8 Remark. Let $a, b \in \mathbb{R}$ such that $a < b$.

$$\begin{aligned} (x - a)(x - b) &\leq 0 \iff a \leq x \leq b \text{ (Equivalently)} \\ (x - a)(x - b) &> 0 \iff x < a \text{ or } x > b. \end{aligned}$$

1.9 Remark. Let $a, b \in \mathbb{R}$.

$$\begin{aligned} ab &= 0 \iff a = 0 \text{ or } b = 0 \text{ (Equivalently)} \\ ab &\neq 0 \iff a \neq 0 \text{ and } b \neq 0. \end{aligned}$$

1.3 Absolute Value And Intervals.

1.10 Definition. The absolute value of a real number x , denoted by $|x|$ is defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

Note that

- $|x| \geq 0 \quad \forall x \in \mathbb{R}$ (by definition). For example $|0| = 0$ and $|-2| = |2| = 2$.
- $|x| = \max\{x, -x\} \quad \forall x \in \mathbb{R}.$
- $|x| = \sqrt{x^2} \quad \forall x \in \mathbb{R} (\because |x|^2 = x^2).$ While $(\sqrt{x})^2 = x$ for all $x \geq 0$.

1.11 Examples.

1. $|x - 2| = \begin{cases} x - 2 & \text{if } x - 2 \geq 0 \\ -(x - 2) & \text{if } x - 2 < 0 \end{cases} = \begin{cases} x - 2 & \text{if } x \geq 2 \\ -x + 2 & \text{if } x < 2. \end{cases}$
2. $|x^2 - x - 2| = \begin{cases} x^2 - x - 2 & \text{if } (x + 1)(x - 2) \geq 0 \\ -(x^2 - x - 2) & \text{if } (x + 1)(x - 2) < 0 \end{cases} = \begin{cases} x^2 - x - 2 & \text{if } x \leq -1 \text{ or } x \geq 2 \\ -x^2 + x + 2 & \text{if } -1 < x < 2. \end{cases}$

1.12 Property. Let $x, y \in \mathbb{R}$. The following properties of absolute value are true.:

- | | |
|--|---|
| <p>1. $x = a \iff x = \pm a$ (provided $a \geq 0$).
 1'. $x = 0 \iff x = 0$.</p> <p>2. $xy = x y$.
 2'. $\frac{x}{y} = \frac{ x }{ y }$. (provided $y \neq 0$)</p> <p>3. $x = -x$.
 3'. $x - y = y - x$.</p> <p>4. $x^2 = y^2 \iff x = y$.
 4'. $x^2 = y^2 \iff x = y$. (provided $xy \geq 0$)</p> <p>5. $x^2 < y^2 \iff x < y$.
 5'. $x^2 < y^2 \iff x < y$. (provided $x, y \geq 0$)</p> | <p>6. $x \leq a \iff -a \leq x \leq a$. (provided $a \geq 0$)
 6'. $x < a \iff -a < x < a$. ($a \geq 0$)</p> <p>7. $x \geq a \iff x \leq -a$ or $x \geq a$. ($a \in \mathbb{R}$)
 7'. $x > a \iff x < -a$ or $x > a$. ($a \in \mathbb{R}$)</p> <p>8. $- x \leq x \leq x$ for all $x \in \mathbb{R}$.
 8'. $- x \leq -x \leq x$ for all $x \in \mathbb{R}$.</p> <p>9. $x + y \leq x + y$ for all $x, y \in \mathbb{R}$.
 9'. $x - y \leq x + y$ for all $x, y \in \mathbb{R}$.</p> <p>10. $x - y \geq x - y$ for all $x, y \in \mathbb{R}$.
 10'. $x + y \geq x - y$ for all $x, y \in \mathbb{R}$.</p> |
|--|---|

Proof.

1. is obvious from the definition of absolute value and 1'. is a particular case of 1. ($a = 0$).
2. We consider four cases $(x \geq 0 \wedge y \geq 0) \vee (x \geq 0 \wedge y \leq 0) \vee (x \leq 0 \wedge y \geq 0) \vee (x \leq 0 \wedge y \leq 0)$. For example if $(x \geq 0 \wedge y \leq 0)$, then $|xy| = -xy = |x||y|$. Similarly, we establish the other cases. Now 2'. is an application of o 2.; Since $y \neq 0$, then $|x| = |\frac{x}{y}y| = |\frac{x}{y}||y|$ and $|\frac{x}{y}| = \frac{|x|}{|y|}$.
3. We consider two cases $(x \geq 0 \vee x < 0)$. If $x \geq 0$, $|x| = x = |-x|$ and if $x < 0$, $|x| = -x = |-x|$. Now 3'. is an application of o 3.; $|x - y| = |-(x - y)| = |y - x|$.
4. $x^2 = y^2 \iff |x|^2 = |y|^2 \iff |x|^2 - |y|^2 = 0 \iff (|x| - |y|)(|x| + |y|) = 0 \iff |x| = |y|$.
Now for 4'. , the product $xy > 0$ is equivalent to x and y have same sign i.e., both positive or both negative. In both cases $x = y \iff |x| = |y| \iff x^2 = y^2$.
5. $x^2 < y^2 \iff x^2 - y^2 < 0 \iff (|x| - |y|)(|x| + |y|) < 0 \iff |x| - |y| < 0 \iff |x| < |y|$. If we assume $x, y \geq 0$, then we establish 5'..

Important notes:

- For $x, y \geq 0$, to show $x < y$ it is equivalent to show $x^2 < y^2$.
- For $x, y \geq 0$, to show $x \leq y$ it is equivalent to show $x^2 \leq y^2$. (combining 4'. and 5'. together)

6. $|x| \leq a \iff x^2 \leq a^2 \iff (x - a)(x + a) \leq 0 \iff -a \leq x \leq a$. Similarly for 6'..
7. and 7'. may be deduced from 6. and 6'. using $(p \iff q) \iff (\neg p \iff \neg q)$.
8. is a particular case of 6. ($a = |x| \geq 0$) or using $x = \pm|x|$. Since property 8. is true $\forall x \in \mathbb{R}$, then we may replace x by $-x \in \mathbb{R}$ to obtain 8'.
9. First way: Let $x, y \in \mathbb{R}$. We have
 $(-|x| \leq x \leq |x|) \wedge (-|y| \leq y \leq |y|) \implies -|x| - |y| \leq x + y \leq |x| + |y| \implies |x + y| \leq |x| + |y|$.
Second way: Let $x, y \in \mathbb{R}$, $a = |x + y| \geq 0$ and $b = |x| + |y| \geq 0$. To show $a \leq b$ is equivalent to show $a^2 \leq b^2$. Now $a^2 - b^2 = (x^2 + 2xy + y^2) - (x^2 + 2|x||y| + y^2) = 2(xy - |xy|) \leq 0$.
10. First way: Using 6., to show 10. is equivalent to show $-|x - y| \leq |x| - |y| \leq |x - y|$. Now,
 $|x| = |(x - y) + y| \leq |x - y| + |y| \implies |x| - |y| \leq |x - y|$. Similarly,
 $|y| = |(y - x) + x| \leq |y - x| + |x| \leq |x - y| + |x| \implies |x| - |y| \geq -|x - y|$.
Second way: $a = ||x| - |y||$, $b = |x - y| \geq 0$. $a^2 - b^2 = 2(xy - |xy|) \leq 0 \implies a^2 \leq b^2 \implies a \leq b$.

□

1.13 Definition. A set I of real numbers is called an interval if any real number that lies between two numbers in I is also in I i.e.:

$$\forall a \in I \forall b \in I, a < x < b \implies x \in I.$$

It follows clearly from the definition that the set $\mathbb{R} = (-\infty, \infty)$ is an interval. In addition to $(-\infty, \infty)$, there are 8 other types of intervals:

Open Interval: $(a, b) = \{x \in \mathbb{R} : a < x < b\}$

Closed Interval: $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$

Half-open Intervals: $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$

Infinite Open Intervals: $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$ $(a, \infty) = \{x \in \mathbb{R} : a < x\}$

Infinite Closed Intervals: $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$ $[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$.

It's easy to check that each of these nine types of sets are intervals. Moreover, one may show that every interval has one of these nine forms. Note that, some references use equivalent notations

$$]a, b[= (a, b), [a, b[= [a, b) \text{ and }]a, b] = (a, b].$$

Important notes:

- It's very useful to associate real numbers as points on a number line to understand many concepts and to arrive at ideas for a proof. In fact they will help us translating geometric ideas into analytic proofs.



- For $a, b \in \mathbb{R}$, $|a - b| = \text{dist}(a, b)$ represents the distance between a and b . In particular $|a| = |a - 0| = \text{dist}(a, 0)$. For example $\text{dist}(-1, -3) = |-1 - (-3)| = 2$ and $\text{dist}(-1, 3) = |-1 - 3| = 4$.



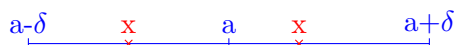
- $(a + b)/2$ represents the midpoint of the segment $[a, b]$ or $[b, a]$.



- $\min\{a, b\} = (a + b)/2 - |a - b|/2$ i.e., if we move to the left half the distance from the midpoint we get the minimum.
- $\max\{a, b\} = (a + b)/2 + |a - b|/2$ i.e., if we move to the right half the distance from the midpoint we get the maximum.
- Let $a \in \mathbb{R}$ and $\delta > 0$.

$$|x - a| < \delta \iff -\delta < x - a < \delta \iff a - \delta < x < a + \delta \iff x \in]a - \delta, a + \delta[.$$

So the set $I_\delta(a) := \{x \in \mathbb{R} : |x - a| < \delta\} =]a - \delta, a + \delta[$ consists of the points on the real line whose distance from a is less than δ .



$I_\delta(a) :=]a - \delta, a + \delta[$ is called a δ -neighborhood of a .

1.14 Property. Let $x, x_0 \in \mathbb{R}$. The following properties are true:

1. $(\forall \varepsilon > 0, x < \varepsilon) \implies x \leq 0$.
2. $(\forall \varepsilon > 0, |x| < \varepsilon) \implies x = 0$.
3. $(\forall \varepsilon > 0, |x - x_0| < \varepsilon) \implies x = x_0$.

Proof.

1. Given: $\forall \varepsilon > 0, x < \varepsilon$. R.T.S(Require To Show): $x \leq 0$.
Suppose to the contrary that $x > 0$. Take $\varepsilon = x/2 > 0$, then $x < \varepsilon = x/2$ (equivalently $1 < 1/2$) which is a contradiction. Thus $x \leq 0$.
2. $\forall \varepsilon > 0, |x| < \varepsilon \implies |x| \leq 0$ (using 1.) $\implies |x| = 0$ ($\because 0 \leq |x|$) $\implies x = 0$.
3. $\forall \varepsilon > 0, |x - x_0| < \varepsilon \implies x - x_0 = 0$ (using 2.) $\implies x = x_0$.

□

1.4 Completeness Property of $\mathbb{R}$.

In this section we discuss the notion of boundedness of a set and we give the completeness axiom that assure us $\mathbb{R}$ has no 'holes'.

1.15 Definition. Let $A \subseteq \mathbb{R}$.

1. We say that A is **bounded from above** in $\mathbb{R}$ if $(\exists M \in \mathbb{R})(\forall x \in A)(x \leq M)$.
Any M with this property is called an **upper bound** of A .
2. We say that A is **bounded from below** in $\mathbb{R}$ if $(\exists m \in \mathbb{R})(\forall x \in A)(m \leq x)$.
Any m with this property is called a **lower bound** of A .
3. We say that A is **bounded** in $\mathbb{R}$ if its bounded from above and below i.e.,

$$(\exists m, M \in \mathbb{R})(\forall x \in A)(m \leq x \leq M).$$

In terms of intervals, A is bounded above if $A \subseteq (-\infty, M]$, bounded below if $A \subseteq [m, \infty)$ and bounded if $A \subseteq [m, M]$.

1.16 Examples.

1. Let $A = (0, 1]$.

$\text{set of lower bounds } (-\infty, 0] \quad \textcolor{red}{0=m} \quad \textcolor{red}{A} \quad \textcolor{red}{M=1} \quad \textcolor{blue}{\text{set of upper bounds } [1, \infty)}$

$M = 1$ is an upper bound of A ($\because (\forall x \in (0, 1])(x \leq 1)$) and every number larger than M is also an upper bound of A . $m = 0$ is a lower bound of A ($\because (\forall x \in (0, 1])(0 \leq x)$) and every number smaller than m is also a lower bound of A .

2. Let $B = [0, 1)$.

$\text{set of lower bounds } (-\infty, 0] \quad \textcolor{red}{0=m} \quad \textcolor{red}{B} \quad \textcolor{red}{M=1} \quad \textcolor{blue}{\text{set of upper bounds } [1, \infty)}$

Note that upper bound (lower bound) of a set may and may not belongs to the set.

3. The set $\{1/n^2 : n \in \mathbb{N}^*\}$ is a bounded set because $0 < 1/n^2 \leq 1$ for all $n \in \mathbb{N}^*$.

4. The set $A = (0, \infty)$ is **not bounded above** because $(\forall M \in \mathbb{R})(\exists x \in A)(x > M)$.



5. The set $A = (-\infty, 0]$ is **not bounded below** because $(\forall m \in \mathbb{R})(\exists x \in A)(x < m)$.



Any set that is not bounded (either above or below) is said to be **unbounded**.

1.17 Definition. (max-min) Let $A \subseteq \mathbb{R}$ and $M, m \in \mathbb{R}$. We say that:

1. M is the **maximum** of A "and we write $M = \max A$ " if
 - ① M is an upper bound of A and ② $M \in A$. (i.e., M is the largest element in A)
2. m is the **minimum** of A "and we write $M = \min A$ " if
 - ① m is a lower bound of A and ② $m \in A$. (i.e., m is the smallest element in A)

A set may not have a largest or smallest member. For example, the set $A = (0, 1]$ admits a maximum ($\max A = 1$ because 1 is an u.b. of A and $1 \in A$) but it does not admit a minimum. While, the set $B = [0, 1)$ admits a minimum ($\min B = 0$ because 0 is a l.b. of B and $0 \in B$) but it does not admit a maximum. **Note that a finite set always admits a maximum and minimum.**

1.18 Definition. (sup-inf) Let A be a nonempty subset of $\mathbb{R}$.

1. An upper bound M of A with the property that $M \leq u$ for all upper bounds u of A is called a **least upper bound** or **supremum** of A , and is denoted by $\sup A$.
2. A lower bound m of A with the property that $l \leq m$ for all lower bounds l of A is called a **greatest lower bound** or **infimum** of A , and is denoted by $\inf A$.
3. If A is not bounded above, one defines $\sup A = \infty$.
4. If A is not bounded below, one defines $\inf A = -\infty$.

Note that $\sup A, \inf A \in \bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$

1.19 Remark.

- ① If $\max A$ exists in $\mathbb{R}$, then $\sup A = \max A$.
- ② If $\min A$ exists in $\mathbb{R}$, then $\inf A = \min A$.

Proof.

① Let $M = \max A$. Then $\underbrace{M}_{\text{element of } A} \leq \underbrace{\sup A}_{\text{l.u.b } A} \leq \underbrace{M}_{\text{u.b } A}$.

② Let $m = \min A$. Then $\underbrace{m}_{\text{l.b } A} \leq \underbrace{\inf A}_{\text{g.l.b } A} \leq \underbrace{m}_{\text{element of } A}$. □

In the previous examples (look at the figures above):

- Let $A = (0, 1]$, $\sup A = 1$ and $\inf A = 0$. **Note that $\min A$ doesn't exist in $\mathbb{R}$. Because if it exists, then $\min A = \inf A = 0 \in (0, 1]$ which is a contradiction.**
- Let $B = [0, 1)$, $\sup B = 1$ and $\inf B = 0$. **Note that $\max B$ doesn't exist in $\mathbb{R}$. Because if it exists, then $\max B = \sup B = 1 \in [0, 1)$ which is a contradiction.**
- $\sup(0, \infty) = \infty$ and $\inf(-\infty, 0] = -\infty$.

We may now state The property that ensures the existence of a unique ordered field $\mathbb{R}$ containing $\mathbb{Q}$ (up to renaming the elements) and distinguishes $\mathbb{R}$ from all other ordered fields.

1.20 Axiom. (*Completeness Property*)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. Equivalently, Every nonempty subset of $\mathbb{R}$ that is bounded below has a greatest lower bound.

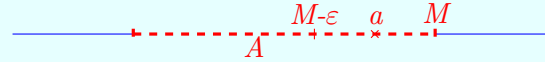
1.21 Theorem. *There is exactly one complete ordered field say $\mathbb{R}$ (up to isomorphism i.e., renaming the elements).*

The following important consequences of the completeness property is useful in determining the infimum or supremum of certain sets.

1.22 Property. (ε - characterization of supremum) Let A be a non empty bounded above subset of $\mathbb{R}$. Then $M = \sup A$ if and only if

1. M is an upper bound of A .

2. $(\forall \varepsilon > 0)(\exists a \in A)(M - \varepsilon < a)$.



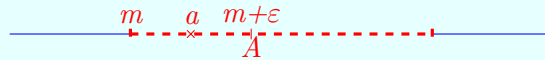
Proof. Assume that $M = \sup A$. Then, by definition, M is an upper bound for A . Hence 1. holds. For 2. assume to the contrary that $(\exists \varepsilon > 0)(\forall a \in A)(M - \varepsilon \geq a)$. Then, $M - \varepsilon$ is an upper bound for A which is smaller than $M = \sup A$, a contradiction.

For the converse, assume that 1. and 2. hold. Since A is bounded above, it has a supremum, say it S (by the completeness property). Since M is an upper bound for A , we must have that $S \leq M$. If $S < M$, then with $\varepsilon = M - S > 0$ and by 2. there exists $a \in A$ such that $M - (M - S) < a \leq S$ i.e., $S < S$ which is contradiction. Thus $\sup A = M$. $\square$

1.23 Property. (ε - characterization of infimum) Let A be a non empty bounded below subset of $\mathbb{R}$. Then $m = \inf A$ if and only if

1. m is a lower bound of A .

2. $(\forall \varepsilon > 0)(\exists a \in A)(a < m + \varepsilon)$.



1.24 Property. (*Archimedean Principle*) For all $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$ i.e., the set $\mathbb{N}$ of natural numbers is not bounded above.

Proof. Assume that $\mathbb{N}$ is bounded above. By the Completeness Property, $\sup \mathbb{N}$ exists. Let $M = \sup \mathbb{N}$. By the ε - characterization with $\varepsilon = 1$, there exists $k \in \mathbb{N}$ such that $M - 1 < k$. This implies that $M < k + 1 \leq M$ which is contradiction. Thus $\mathbb{N}$ is not bounded above. $\square$

1.25 Corollary. For all $x > 0$ for all $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $y < nx$.

Proof. Given that $\frac{y}{x} \in \mathbb{R}$. By the Archimedean Principle, there exists $n \in \mathbb{N}$ such that $n > \frac{y}{x}$. This implies that there exists $n \in \mathbb{N}$ such that $y < nx$. $\square$

1.26 Corollary. For all $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.

Proof. Given that $\frac{1}{\varepsilon} \in \mathbb{R}$. By the Archimedean Principle, there exists $n \in \mathbb{N}$ such that $n > \frac{1}{\varepsilon}$. This implies that there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. $\square$

1.27 Corollary. $(\forall n \in \mathbb{N}^*, a - \frac{1}{n} \leq b \leq a + \frac{1}{n}) \implies a = b$.

Proof. The Corollary will follow if we show that $(\forall \varepsilon > 0, |a - b| < \varepsilon)$. Let $\varepsilon > 0$. By the Archimedean Principle, there exists $n_0 \in \mathbb{N}$ such that $\frac{1}{n_0} < \varepsilon$. We have $a - \frac{1}{n_0} \leq b \leq a + \frac{1}{n_0}$, or equivalently $|a - b| \leq \frac{1}{n_0}$. This implies that $|a - b| < \varepsilon$. $\square$

1.28 Property. Every nonempty subset A of $\mathbb{Z}$ which is bounded above admits a maximum .

Proof. Since A is bounded above, $\sup A$ exists (By the Completeness Property). Let $M = \sup A$. The property will follow if we show that $M \in A$. Suppose to the contrary that $M \notin A$. By the ε -characterization with $\varepsilon = 1$, there exist $a \in A$ such that $M - 1 < a < M$. By the ε -characterization again with $\varepsilon = \frac{M-a}{2}$, there exist $a' \in A$ such that $M - \frac{M-a}{2} < a' < M$. Hence $M - 1 < a < M - \frac{M-a}{2} < a' < M$. This implies that $a' - a$ is an integer strictly between 0 and 1 which is a contradiction. $\square$

1.29 Corollary. Every nonempty subset A of $\mathbb{Z}$ which is bounded below admits a minimum .

Proof. $\min A = -\max(-A)$ $\square$

In particular, we have the following principle.

The Well Ordering Principle: every nonempty subset A of $\mathbb{N}$ admits a minimum.

1.30 Theorem. (Principle of Mathematical Induction)

For each $n \in \mathbb{N}$, let $P(n)$ be a property depends on n . If

1. $P(0)$ is true i.e., the property is true for $n = 0$.
2. $P(n) \implies P(n+1)$ i.e., the truth of $P(n)$ implies the truth of $P(n+1)$.

Then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. Let $A := \{n \in \mathbb{N} : P(n) \text{ is true}\}$, the theorem will follow if we show $A = \mathbb{N}$. Assume to the contrary that $A \neq \mathbb{N}$, then $\mathbb{N} \setminus A = \{n \in \mathbb{N} : n \notin A\}$ admits a minimum (By the well ordering principle). Let $m = \min(\mathbb{N} \setminus A) \in \mathbb{N} \setminus A$. Since $0 \in A$, then $m \neq 0$. Hence $m - 1 \in A$ i.e., $P(m - 1)$ is true. This implies, using 2., $P(m)$ is true i.e., $m \in A$ which is contradiction. Thus $A = \mathbb{N}$. $\square$

Principle of Mathematical Induction(modified)

Let $n_0 \in \mathbb{N}$ and $P(n)$ be a property depends on n . If

1. $P(n_0)$ is true i.e., the property is true for $n = n_0$.
2. $P(n) \implies P(n+1)$ for $n \geq n_0$

Then $P(n)$ is true for all $n \geq n_0$.

1.31 Examples. 1. Let $r \geq -1$. Prove Bernoulli's inequality: $(1+r)^n \geq 1+nr \quad \forall n \in \mathbb{N}$.

Solution. In this case, $P(n)$ is $(1+r)^n \geq 1+nr$.

• For $n = 0$ we have $(1+r)^0 = 1 \geq 1+0r$. Hence $P(0)$ is true.

• Suppose $P(n)$ is true i.e., $(1+r)^n \geq 1+nr$.

Let us show $P(n+1)$ is true i.e., $(1+r)^{n+1} \geq 1+(n+1)r$.

Using $1+r \geq 0$ and $P(n)$ is true, we obtain

$$(1+r)^{n+1} = (1+r)(1+r)^n \geq (1+r)(1+nr) \geq 1+(n+1)r+nr^2 \geq 1+(n+1)r.$$

Thus $P(n)$ is true for all $n \in \mathbb{N}$.

2. Let $x_1, \dots, x_n \in \mathbb{R}$. Prove the following triangle inequality:

$$|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n| \quad \forall n \geq 2.$$

Solution. In this case; $|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n| \quad (P(n))$.

• For $n = 2$ we have $|x_1 + x_2| \leq |x_1| + |x_2|$ for all $x_1, x_2 \in \mathbb{R}$. Hence $P(2)$ is true.

• Assume for a given $n \geq 2$, $P(n)$ is true and let us show $P(n+1)$ is true.

$$\begin{aligned} \underbrace{|x_1 + x_2 + \dots + x_n|}_{X_1} + \underbrace{|x_{n+1}|}_{X_2} &\leq \underbrace{|x_1 + x_2 + \dots + x_n|}_{X_1} + \underbrace{|x_{n+1}|}_{X_2} && (\because P(2) \text{ is true}) \\ &\leq |x_1| + |x_2| + \dots + |x_n| + |x_{n+1}| && (\because P(n) \text{ is true}) \end{aligned}$$

Thus $P(n)$ is true for all $n \geq 2$.

3. Let $x \in \mathbb{R}$. Show that $|x^n| = |x|^n$ for any integer $n \in \mathbb{N}^*$.

Solution. $|x^n| = |x|^n \quad (P(n))$.

• For $n = 1$ we have $|x^1| = |x|^1$ for all $x \in \mathbb{R}$. Hence $P(1)$ is true.

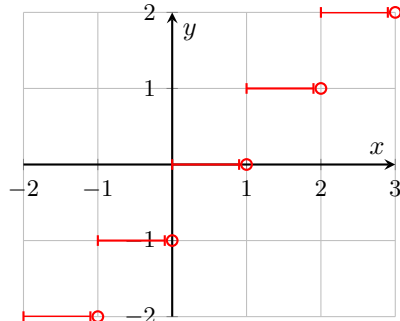
• Assume for a given $n \geq 1$, $P(n)$ is true and let us show $P(n+1)$ is true.

$$\begin{aligned} |x^{n+1}| &= \underbrace{|x^n|}_a \underbrace{|x|}_b = \underbrace{|x^n|}_a \underbrace{|x|}_b && (\because |ab| = |a||b|) \\ &= |x|^n |x| && (\because P(n) \text{ is true}) \\ &= |x|^{n+1}. \end{aligned}$$

Thus $P(n)$ is true for all $n \geq 1$.

1.32 Definition. Let $x \in \mathbb{R}$. The integer part of x , denoted by $E(x)$, is the greatest integer less or equal to x i.e., $E(x) := \max\{n \in \mathbb{Z} : n \leq x\}$.

$E(x)$ is well defined because the set $\{n \in \mathbb{Z} : n \leq x\}$ is a nonempty bounded above subset of $\mathbb{Z}$. If $x \in \mathbb{Z}$, then $E(x) = x$. In general, If we move to the left from x , the first integer is $E(x)$. For example $E(1) = 1$, $E(1.7) = 1$ and $E(-1.7) = -2$.



Greatest integer function.

1.33 Property. The integer part function $E : \mathbb{R} \mapsto \mathbb{Z}$ satisfy the following properties:

1. If $k \in \mathbb{Z}$ and $k \leq x$, then $k \leq E(x)$.
2. $E(x) \leq x < E(x) + 1$ or equivalently, $0 \leq x - E(x) < 1$.(The inequality is strict if $x \notin \mathbb{Z}$)
3. $x \leq y \implies E(x) \leq E(y)$.(i.e., E is an increasing function)
4. $E(x + k) = E(x) + k \ \forall k \in \mathbb{Z}$.

Proof.

1. Let $k \in \mathbb{Z}$ be such that $k \leq x$. Since $E(x)$ is the greatest integer less or equal to x , then $k \leq E(x)$.
2. $E(x)$ is the greatest integer less or equal to x . This implies that $E(x) \leq x$. For the second inequality, suppose to the contrary that $E(x) + 1 \leq x$, then $E(x) + 1 \leq E(x)$ which is contradiction. Hence $x < E(x) + 1$
3. $x \leq y \xrightarrow{\text{using 2.}} E(x) \leq x \leq y \implies E(x) \leq y \xrightarrow{\text{using 1.}} E(x) \leq E(y)$.
4. Let $k \in \mathbb{Z}$.
 - $E(x) \leq x \implies E(x) + k \leq x + k \xrightarrow{1.\text{or}3.} E(x) + k \leq E(x + k)$.
The equality will follow if we show $E(x + k) \leq E(x) + k$.
 - $E(x + k) \leq x + k < E(x) + k + 1 \implies E(x + k) \leq E(x) + k$.

□

1.34 Theorem. (Density of the Rationals)

Between any pair of distinct real numbers there is a rational number.

Proof. Let $a < b$. By the Archimedean Principle, $n(b - a) > 1$ for some $n \in \mathbb{N}$. Let $m := E(na) + 1$. Then $na < m \leq na + 1 < nb$, hence $a < \frac{m}{n} < b$. □

1.35 Theorem. (Density of the Irrationals)

Between any pair of distinct real numbers there is irrational number.

Proof. Let $a < b$, then $a + \sqrt{2} < b + \sqrt{2}$. By the density of the rationals, There exists $q \in \mathbb{Q}$ such that $a + \sqrt{2} < q < b + \sqrt{2}$. Hence $a < q - \sqrt{2} < b$, where $q - \sqrt{2}$ is irrational. □

1.36 Property. If n is positive integer that is not a perfect square, then $\sqrt{n}$ is irrational.

Proof. Since n is not a perfect square, then $\sqrt{n} \notin \mathbb{Z}$. Hence $0 < \sqrt{n} - E(\sqrt{n}) < 1$. Suppose to the contrary that $\sqrt{n}$ is rational and let $A := \{m \in \mathbb{N} : m\sqrt{n} \in \mathbb{N}\}$. The set A is a nonempty subset of $\mathbb{N}$, by the well ordering principle, A admits a minimum m_0 . Hence $m_0\sqrt{n} \in \mathbb{N}$. Now;

- $m := m_0(\sqrt{n} - E(\sqrt{n})) \in \mathbb{N}$ and $m < m_0$.
- $m\sqrt{n} = m_0n - m_0\sqrt{n}E(\sqrt{n}) \in \mathbb{N}$.

Hence $m \in A$. This implies $m_0 \leq m$ which is a contradiction($\because m_0 > m$). Thus $\sqrt{n}$ is irrational. □

With the help of the completeness property one can prove the existence in $\mathbb{R}$ of the n th root.

1.37 Property. (n -th root) Consider the equation $x^n = a$ ($\star$) where $n \in \mathbb{N}^*$ and $a \in \mathbb{R}$.

- If n is even and $a \geq 0$, then equation ($\star$) has a unique solution in $\mathbb{R}_+$ denoted by $\sqrt[n]{a}$ or $a^{\frac{1}{n}}$ and called the n -th root of a . (consequently, $-\sqrt[n]{a}$ is also a solution in $\mathbb{R}$).
- If n is odd and $a \in \mathbb{R}$, then equation ($\star$) has a unique solution in $\mathbb{R}$ denoted by $\sqrt[n]{a}$ or $a^{\frac{1}{n}}$ and called the n -th root of a .

We will see, in chapter three, that the existence of the n -th root is a simple application of the intermediate value theorem.

1.5 Exercises

Exercise 1. Let $a, b, c, d \in \mathbb{R}$.

1. Assume $bd > 0$. Show that $\frac{a}{b} \leq \frac{c}{d} \iff ad \leq bc$.
2. Assume $bd < 0$. Show that $\frac{a}{b} \leq \frac{c}{d} \iff ad \geq bc$.
3. Show that: $(0 \leq a \leq c) \wedge (b \geq d > 0) \implies \frac{a}{b} \leq \frac{c}{d}$.
4. Deduce that: $(a \leq c \leq 0) \wedge (0 > b \geq d) \implies \frac{a}{b} \geq \frac{c}{d}$.
5. Show that: $(a < 0) \wedge (a \leq c) \wedge (0 < b \leq d) \implies \frac{a}{b} \leq \frac{c}{d}$.

Exercise 2. Show that $\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|} \quad \forall x, y \in \mathbb{R}$.

Exercise 3. Solve the following:

- | | |
|---|--|
| <ol style="list-style-type: none"> a. $\sqrt{(x-1)(x-5)} = x-2$. b. $\frac{2x+3}{x+5} \leq \frac{x+1}{ x-1 }$. c. $x-3 \leq \sqrt{x^2-x}$ | <ol style="list-style-type: none"> d. $\sqrt{ x^2-4 } \geq x$. e. $\sqrt{x-1} = \sqrt{3-x}$. f. $\sqrt{x-1} = \sqrt{3-x} - 1$. |
|---|--|

Exercise 4. Let $x \in \mathbb{R}$ and $n \in \mathbb{N}^*$. Prove that:

- | | |
|--|---|
| <ol style="list-style-type: none"> a. $E(x) = -E(-x) \iff x \in \mathbb{Z}$. b. $E(x) + E(-x) = 0$ or -1. | <ol style="list-style-type: none"> c. $0 \leq E(nx) - nE(x) \leq n-1$. d. $E\left(\frac{E(nx)}{n}\right) = E(x)$. |
|--|---|

Exercise 5. Let $a \in \mathbb{R}$.

1. Show that: $0 < a \leq 1 \implies a^n \leq a \quad \forall n \in \mathbb{N}^*$.
2. Deduce that $a \geq 1 \implies a^n \geq a \quad \forall n \in \mathbb{N}^*$.

Exercise 6. Let $\emptyset \neq A \subset \mathbb{R}$. Show that A is bounded iff $\exists C > 0$ such that $\forall a \in A, |a| \leq C$.

Exercise 7. Let $A := \{\frac{1}{n} : n \in \mathbb{N}^*\}$.

1. Show that $\sup A$ and $\inf A$ exist in $\mathbb{R}$.
2. Find $\sup A$ and $\inf A$.
3. Does $\min A$ exist. Justify your answer.

Exercise 8. Let $A := \{x \in \mathbb{Q} : 0 \leq x \leq \sqrt{2}\} = [0, \sqrt{2}] \cap \mathbb{Q}$.

1. Show that $\sup A$ and $\inf A$ exist in $\mathbb{R}$.
2. Find $\sup A$.
3. Does $\mathbb{Q}$ complete. Justify your answer.
4. Does $\max A$ exist. Justify your answer.
5. Find $\inf A$.

Exercise 9.(Density of the dyadic rationals). Let $a < b$.

1. Show that $2^n > n$ for all $n \in \mathbb{N}^*$.
2. Show that there exists $n \in \mathbb{N}$ such that $2^n a + 1 < 2^n b$.
3. Let $m := E(2^n a) + 1$. Show that $2^n a < m \leq 2^n a + 1$.
4. Deduce that for each pair of real numbers $a < b$, there exists $m \in \mathbb{Z}$ and there exists $n \in \mathbb{N}$ such that $a < \frac{m}{2^n} < b$ (i.e., between two distinct real numbers, there exists a dyadic rational)

Exercise 10. Let $A := \{u_n = \frac{1+(-1)^n}{n} - n^2 : n \in \mathbb{N}^*\}$.

1. Show that $\sup A = -1$.
2. Now, the aim is to show that A is not bounded below:
 - (a) Let $m < 0$. Show that there exists $n_0 \in \mathbb{N}^*$ such that $n_0^2 > 1 - m$.
 - (b) Show that $u_n \leq 1 - n^2$ for all $n \in \mathbb{N}^*$.
 - (c) Deduce that A is not bounded below.

Exercise 11. Let $A := \{u_n = \frac{n-1}{n+1} : n \in \mathbb{N}\}$. Show that $\inf A = -1$ and $\sup A = 1$.

Exercise 12. Let $E := \{\frac{x^2}{x^2+1} : x \in \mathbb{R}\}$. Determine, if existent, $\sup E$, $\inf E$, $\max E$ and $\min E$.

Exercise 13. Let $E := \{\frac{x^2+2}{x^2+1} : x \in \mathbb{R}\}$. Determine, if existent, $\sup E$, $\inf E$, $\max E$ and $\min E$.

Exercise 14. Let A and B be two nonempty bounded subsets of $\mathbb{R}$ and $\lambda \in \mathbb{R}$. Define, $A + B := \{a + b : a \in A \text{ and } b \in B\}$ and $\lambda A := \{\lambda a : a \in A\}$.

1. Show that $\sup(A + B)$ exists.
2. Show that $\sup(A + B) = \sup A + \sup B$.
3. Show that $\sup(-A) = -\inf A$.
4. Show that $\sup(\lambda A) = \lambda \sup A$ for $\lambda > 0$.
5. Assume $\lambda < 0$. Deduce a formula for $\sup(\lambda A)$, $\sup(A - B)$, $\inf(A + B)$, $\inf(A - B)$.

Exercise 15. Let $A := \{xy : |x| + |y| < 1\}$. Show that $\sup A$ exists and $\sup A = \frac{1}{4}$. (Hint: $4ab \leq (a+b)^2$.)

Exercise 16. Assume A is nonempty bounded above subset of $\mathbb{R}$ and B is nonempty bounded below subset of $\mathbb{R}$ with $M_A = \sup A$ and $m_B = \inf B > 0$. Let $C := \{a - \sqrt{b} : a \in A \text{ and } b \in B\}$.

1. Show that $\sup C$ exists in $\mathbb{R}$.
2. Show that for all $\varepsilon > 0$, $\sqrt{m_B + (\frac{\varepsilon}{2})^2} < \sqrt{m_B} + \frac{\varepsilon}{2}$.
3. Show that $\sup C = M_A - \sqrt{m_B}$.

Exercise 17. Let A be a nonempty bounded subset of $\mathbb{R}$. Define $|A| := \{|a| : a \in A\}$. Show that $\sup |A| - \inf |A| \leq \sup A - \inf A$. (Hint: Use $|x| - |y| \leq |x - y|$.)

2.1 Limits Of Sequences

2.1 Definition. A sequence of real numbers is a function $u : \{n \in \mathbb{N} : n \geq N\} \xrightarrow[n]{\mapsto} \mathbb{R}$ (the natural number N is usually 0 or 1). We call u_n the n th term of the sequence and we write the sequence as $(u_n)_{n=N}^{\infty}$ or simply (u_n) .

For example, $u_n = n - 3$ is the general term of the sequence $(u_n)_{n=0}^{\infty}$ and $v_n = \frac{1}{n-3}$ is the general term of the sequence $(v_n)_{n=4}^{\infty}$. We are interested in studying the behavior of the sequence when n is very large (goes to infinity), hence the index N can be chosen very large natural number.

2.2 Definition. A property (p) of a sequence u_n is said to hold **eventually** if there exists an index $N \in \mathbb{N}$ such that u_n has property (p) for all $n \geq N$ (i.e., after a certain rank).

For example, by the Archimedean principle, $\forall \varepsilon > 0$ the sequence $(\frac{1}{n})$ is eventually less than ε .

2.3 Definition. Let $(u_n)_{n=N}^{\infty}$ be a sequence of real numbers. We say that:

1. (u_n) is bounded above if $(\exists M \in \mathbb{R}) (\forall n \geq N) (u_n \leq M)$.
2. (u_n) is bounded below if $(\exists m \in \mathbb{R}) (\forall n \geq N) (m \leq u_n)$.
3. (u_n) is bounded if $(\exists m, M \in \mathbb{R}) (\forall n \geq N) (m \leq u_n \leq M)$.

i.e., (u_n) is bounded (above, below) if the set of its terms $\{u_n : n \geq N\}$ is bounded (above, below).

All properties of bounded sets can be restated for bounded sequences. In particular, Exercise 6. Chapter one gives the following property

2.4 Property. A sequence $(u_n)_{n=N}^{\infty}$ is bounded if and only if $(\exists C > 0) (\forall n \geq N) (|u_n| \leq C)$.

2.5 Definition. Let $(u_n)_{n=0}^{\infty}$ be a sequence in $\mathbb{R}$ and $\ell \in \mathbb{R}$. We say that (u_n) converges to ℓ and we write $\lim_{n \rightarrow \infty} u_n = \ell$ or $u_n \xrightarrow{n \rightarrow \infty} \ell$ if

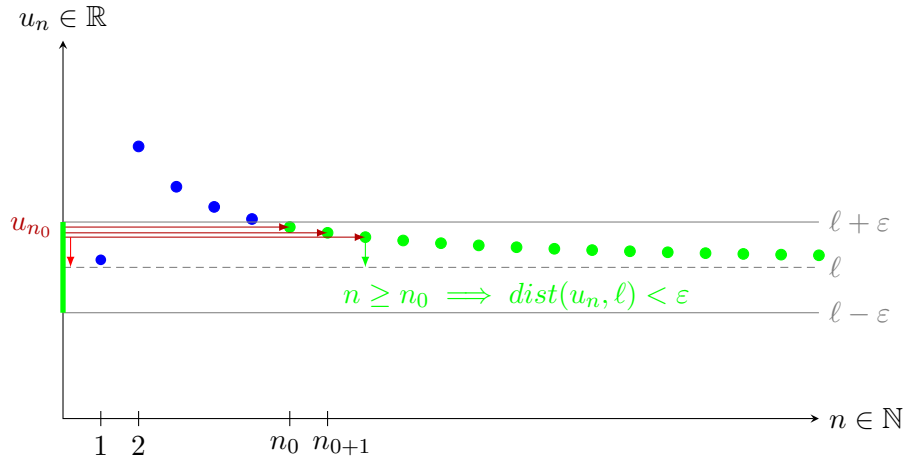
$$(\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |u_n - \ell| < \varepsilon).$$

i.e., for all $\varepsilon > 0$, the sequence (u_n) is eventually in the open interval $(\ell - \varepsilon, \ell + \varepsilon) = I_\varepsilon(\ell)$.

i.e., for all $\varepsilon > 0$, the distance between u_n and ℓ is eventually less than ε .

Hence the limit of a convergent sequence do not depend on the behavior of the beginning of the sequence. Only eventuality matters for the limit.

In the definition above, the number n_0 may depends on ε .



Graph of a convergent sequence: $u_n \xrightarrow{n \rightarrow \infty} \ell$

How to show $\lim_{n \rightarrow \infty} u_n = \ell$?

$$\lim_{n \rightarrow \infty} u_n = \ell \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \xrightarrow{(\star)} |u_n - \ell| < \varepsilon).$$

- We let $\varepsilon > 0$ be arbitrary.
- We search for n_0 which may depends on ε such that the implication $(\star)$ holds true.

2.6 Example. Using the definition of the limit, show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Solution. Let $\varepsilon > 0$ be arbitrary.

Take $n_0 = E(\frac{1}{\varepsilon}) + 1 > \frac{1}{\varepsilon}$. Now if $n \geq n_0$, then we have $n > \frac{1}{\varepsilon}$ and from scratch we obtain $|\frac{1}{n} - 0| = \frac{1}{n} < \varepsilon$.

Scratch. (u_n) is defined for $n \geq 1$.

To find n_0 we start from the R.H.S of the implication $(\star)$:

$$\bullet \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}.$$

2.7 Example. Using the definition of the limit, show that $\lim_{n \rightarrow \infty} \frac{n^2+n-1}{n^2-1} = 1$.

Solution. Let $\varepsilon > 0$ be arbitrary.

Take $n_0 = E(\frac{1}{\varepsilon} + 1) + 1 > \frac{1}{\varepsilon} + 1$. Now if $n \geq n_0$, then we have $n > \frac{1}{\varepsilon} + 1$ and from scratch we obtain $\left| \frac{n^2+n-1}{n^2-1} - 1 \right| \leq \frac{1}{n-1} < \varepsilon$.

Scratch. (u_n) is defined for $n \geq 2$.

To find n_0 we start from the R.H.S of the implication $(\star)$:

$$\bullet \left| \frac{n^2+n-1}{n^2-1} - 1 \right| = \frac{n}{n^2-1} \leq \frac{n+1}{(n-1)(n+1)} \leq \frac{1}{n-1}. \text{ (We drop } |\cdot| \text{ because } \frac{n}{n^2-1} > 0 \text{ for } n \geq 2)$$

$$\bullet \frac{1}{n-1} \leq \varepsilon \iff n > \frac{1}{\varepsilon} + 1$$

2.8 Example. Using the definition of the limit, show that $\lim_{n \rightarrow \infty} \frac{4n^2+3n-6}{n^2-6} = 4$.

Scratch. (u_n) is defined for all $n \in \mathbb{N}$.

Given $\varepsilon > 0$ and $n \geq n_0$ (n_0 to be determined!). We search for some n_0 such that the implication $(\star)$ holds. As usual we start from the R.H.S of $(\star)$:

- $\left| \frac{4n^2+3n-6}{n^2-6} - 4 \right| = \left| \frac{3n+18}{n^2-6} \right| = \frac{3n+18}{n^2-6}$ (we can drop $|\cdot|$ for $n \geq 3$)
- To find the least n_0 such that $(\star)$ holds true, we have to solve $\frac{3n+18}{n^2-6} < \varepsilon$, but we do not need to find the least such n_0 . So we will simplify matters by making estimates. The idea is that $\frac{3n+18}{n^2-6}$ is bounded above by some constant times $\frac{n}{n^2} = \frac{1}{n}$ for sufficiently large n , which simplifies the inequality into easier one.
- $\frac{3n+18}{n^2-6} \leq \frac{3n+18n}{n^2/2} \leq \frac{42}{n}$ provided that; $n^2 - 6 \geq n^2/2 \Leftrightarrow n^2 \geq 12 \Leftrightarrow n \geq 4$.
- $\frac{42}{n} < \varepsilon \iff n > \frac{42}{\varepsilon}$.

Solution. Let $\varepsilon > 0$ be arbitrary. Take $n_0 = \max\{E(\frac{42}{\varepsilon}) + 1, 3, 4\}$. Now if $n \geq n_0$, then we have $n > \frac{42}{\varepsilon}$, $n > 3$ and $n > 4$. So from scratch we obtain $\left| \frac{4n^2+3n-6}{n^2-6} - 4 \right| = \frac{3n+18}{n^2-6} \leq \frac{42}{n} < \varepsilon$.

2.9 Definition. Let $(u_n)_{n=0}^\infty$ be a sequence in $\mathbb{R}$.

1. We say that (u_n) diverges to ∞ and we write $\lim_{n \rightarrow \infty} u_n = \infty$ or $u_n \xrightarrow{n \rightarrow \infty} \infty$ if

$$(\forall A > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies u_n > A).$$

i.e., for all $A > 0$, the sequence (u_n) is eventually in the open interval (A, ∞) .

2. We say that (u_n) diverges to $-\infty$ and we write $\lim_{n \rightarrow \infty} u_n = -\infty$ or $u_n \xrightarrow{n \rightarrow \infty} -\infty$ if

$$(\forall A < 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies u_n < A).$$

i.e., for all $A < 0$, the sequence (u_n) is eventually in the open interval $(-\infty, A)$.

Note that, to show $\lim_{n \rightarrow \infty} u_n = -\infty$ is equivalent to show $\lim_{n \rightarrow \infty} (-u_n) = \infty$.

How to show $\lim_{n \rightarrow \infty} u_n = \infty$?

$$(\forall A > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \xrightarrow{(\star)} u_n > A).$$

- We let $A > 0$ be arbitrary.
- We search for n_0 which may depends on A such that the implication $(\star)$ holds true.

2.10 Example. Using the definition of the limit, show that $\lim_{n \rightarrow \infty} \frac{n^2+(-1)^n}{\sqrt{n^2+n+7}} = \infty$.

Solution.

Let $A > 0$ be arbitrary.

Take $n_0 = \max\{E(9A) + 1, 2\}$. Now if $n \geq n_0$, then we have $n > 9A$, $n \geq 2$ and

$$\frac{n^2+(-1)^n}{\sqrt{n^2+n+7}} \geq \frac{n^2-1}{\sqrt{n^2+n^2+7n^2}} \geq \frac{n^2/3}{3n} \geq \frac{n}{9} > A.$$

Take $n_0 = \max\{E(A+1) + 1, 5\}$. Now if $n \geq n_0$, then we have $n > A+1$ and $n \geq 5$. Consequently,

$$\frac{n^2+(-1)^n}{\sqrt{n^2+n+6}} \geq \frac{n^2-1}{\sqrt{n^2+2n+1}} \geq \frac{(n-1)(n+1)}{|n+1|} \geq n-1 > A.$$

Scratch. (u_n) is defined for $n \geq 0$.

- $\frac{n^2+(-1)^n}{\sqrt{n^2+n+7}} \geq \frac{n^2-1}{\sqrt{n^2+n^2+7n^2}} \geq \frac{n^2/3}{3n} \geq \frac{n}{9}$, provided that: $n^2 - 1 \geq \frac{n^2}{3} \Leftrightarrow n^2 \geq \frac{3}{2} \Leftrightarrow n \geq 2$.
- $\frac{n}{9} > A \iff n > 9A$.

- Another Estimate
- $\frac{n^2+(-1)^n}{\sqrt{n^2+n+6}} \geq \frac{n^2-1}{\sqrt{n^2+2n+1}} \geq \frac{(n-1)(n+1)}{|n+1|} \geq n-1$. Provided that: $n^2 + n + 6 \leq n^2 + 2n + 1 \Leftrightarrow n \geq 5$.
 - $n-1 > A \iff n > A+1$

2.11 Example. Let $\ell \in \mathbb{R}$. Show that $\lim_{n \rightarrow \infty} (-1)^n \neq \ell$ and $\lim_{n \rightarrow \infty} (-1)^n \neq \pm\infty$. (i.e., $\lim_{n \rightarrow \infty} (-1)^n \nexists$ in $\overline{\mathbb{R}}$)

solution.

Negation Of Convergence.

$$\lim_{n \rightarrow \infty} u_n \neq \ell \iff (\exists \varepsilon > 0)(\forall n_0 \in \mathbb{N})(\exists n \in \mathbb{N})(n \geq n_0 \wedge |u_n - \ell| \geq \varepsilon).$$

Choose $\varepsilon = 0.1$ and let $n_0 \in \mathbb{N}$. If $|u_{2n_0} - \ell| = |1 - \ell| < \varepsilon = 0.1$ and $|u_{2n_0+1} - \ell| = |1 + \ell| < \varepsilon = 0.1$, then $2 = |1 - \ell + \ell + 1| \leq |1 - \ell| + |\ell + 1| < 0.1 + 0.1 < 2$ a contradiction. Hence there exists $n \geq n_0$ ($n = 2n_0$ or $n = 2n_0 + 1$) such that $|u_n - \ell| \geq \varepsilon$.

One can show $\forall \ell \in \mathbb{R}, \lim_{n \rightarrow \infty} (-1)^n \neq \ell$ by contradiction:

Assume to the contrary that $\exists \ell \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} (-1)^n = \ell$. Then $\forall \varepsilon > 0, |\pm 1 - \ell| < \varepsilon$. Hence $2 = |1 - \ell + \ell + 1| \leq |1 - \ell| + |\ell + 1| < \varepsilon + \varepsilon = 2\varepsilon$. So $1 < \varepsilon$ for all $\varepsilon > 0$ is a contradiction.

Negation Of Infinite Limit .

$$\lim_{n \rightarrow \infty} u_n \neq \infty \iff (\exists A > 0)(\forall n_0 \in \mathbb{N})(\exists n \in \mathbb{N})(n \geq n_0 \wedge u_n \leq A).$$

Choose $A = 2$ and let $n_0 \in \mathbb{N}$. We have $u_{n_0} = \pm 1 \leq 2$.

One can show that $\lim_{n \rightarrow \infty} u_n \neq \infty$ by contradiction:

Assume to the contrary that $\lim_{n \rightarrow \infty} u_n = \infty$. Then for all $A > 0$, eventually $u_n = \pm 1 > A$ which is a contradiction. Thus $\lim_{n \rightarrow \infty} u_n \neq \infty$.

2.12 Definition. A sequence (u_n) in $\mathbb{R}$ is said to be convergent if $\lim_{n \rightarrow \infty} u_n = \ell$ for some $\ell \in \mathbb{R}$. Otherwise it is divergent.

Note that a sequence (u_n) in $\mathbb{R}$ is said to be divergent if $\lim_{n \rightarrow \infty} u_n = \infty$ or $\lim_{n \rightarrow \infty} u_n = -\infty$ or the limit of (u_n) does not exist (in $\overline{\mathbb{R}}$).

2.13 Property. If (u_n) is a convergent sequence, then its limit is unique.

Proof. Suppose that the sequence (u_n) has two limits ℓ_1 and ℓ_2 and let us show $\ell_1 = \ell_2$. The equality will follow if we show that $\forall \varepsilon > 0, |\ell_1 - \ell_2| < \varepsilon$.

Let $\varepsilon > 0$, then by the definition of the limit:

$$\begin{aligned} \exists n_1 \in \mathbb{N}, \quad n \geq n_1 &\implies |u_n - \ell_1| < \frac{\varepsilon}{2} \\ \exists n_2 \in \mathbb{N}, \quad n \geq n_2 &\implies |u_n - \ell_2| < \frac{\varepsilon}{2}. \end{aligned}$$

For $n_3 > \max\{n_1, n_2\}$ and with the help of triangle inequality we obtain

$$|\ell_1 - \ell_2| = |\ell_1 - u_{n_3} + u_{n_3} - \ell_2| \leq |u_{n_3} - \ell_1| + |u_{n_3} - \ell_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \square$$

2.14 Property. Every convergent sequence is bounded.

Proof. Let (u_n) be a sequence converges to $\ell \in \mathbb{R}$. Then

$$\text{For } \varepsilon = 1, \exists n_0 \in \mathbb{N}, \quad n \geq n_0 \implies |u_n - \ell| < 1.$$

Hence, for $n \geq n_0$ we have $|u_n| = |u_n - \ell + \ell| \leq |u_n - \ell| + |\ell| < 1 + |\ell|$. Define $M := \{1 + |\ell|, |u_0|, |u_1|, \dots, |u_{n_0}|\}$. Then we have $\forall n \in \mathbb{N}, |u_n| < M$ and thus (u_n) is a bounded sequence. $\square$

2.15 Remark. If $u_n \xrightarrow{n \rightarrow \infty} \ell \in \mathbb{R}^*$, then there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $|u_n| > \frac{|\ell|}{2}$. (Hence, if the limit is not zero, then $u_n \neq 0$ eventually)

Proof. We have $\lim_{n \rightarrow \infty} u_n = \ell$. Let N corresponds to $\varepsilon := \frac{|\ell|}{2}$. Then,

$$n \geq N \implies ||u_n| - |\ell|| \leq |u_n - \ell| < \frac{|\ell|}{2} \implies |u_n| > \frac{|\ell|}{2}.$$

□

2.16 Property. Let (u_n) and (v_n) be sequences in $\mathbb{R}$. The following limit properties hold in $\overline{\mathbb{R}}$ in the sense that if the expression on the right side of the equation exists in $\overline{\mathbb{R}}$, then the limit on the left side exists in $\overline{\mathbb{R}}$ and equality holds.

1. $\lim_{n \rightarrow \infty} \lambda u_n = \lambda \lim_{n \rightarrow \infty} u_n, \lambda \in \mathbb{R}.$
2. $\lim_{n \rightarrow \infty} (u_n + v_n) = \lim_{n \rightarrow \infty} u_n + \lim_{n \rightarrow \infty} v_n.$
3. $\lim_{n \rightarrow \infty} (u_n v_n) = \lim_{n \rightarrow \infty} u_n \lim_{n \rightarrow \infty} v_n.$
4. $\lim_{n \rightarrow \infty} (u_n / v_n) = \lim_{n \rightarrow \infty} u_n / \lim_{n \rightarrow \infty} v_n$ provided $\lim_{n \rightarrow \infty} v_n \neq 0.$
5. $\lim_{n \rightarrow \infty} |u_n| = |\lim_{n \rightarrow \infty} u_n|.$ (Hence, $\lim_{n \rightarrow \infty} u_n = \ell_1 \iff \lim_{n \rightarrow \infty} |u_n - \ell_1| = 0.$)

Proof. We prove the theorem for the case $\lim_{n \rightarrow \infty} u_n = \ell_1 \in \mathbb{R}$ and $\lim_{n \rightarrow \infty} v_n = \ell_2 \in \mathbb{R}$.

1. $\lim_{n \rightarrow \infty} \lambda u_n = \lambda \ell_1 \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |\lambda u_n - \lambda \ell_1| < \varepsilon).$

$$\text{Note that: } |\lambda u_n - \lambda \ell_1| = |\lambda| |u_n - \ell_1| < \varepsilon \iff |u_n - \ell_1| < \frac{\varepsilon}{|\lambda|}.$$

We have $\lim_{n \rightarrow \infty} u_n = \ell_1$. Let n_1 corresponds to $\frac{\varepsilon}{|\lambda|} > 0$ and take $n_0 := n_1$. Then,

$$n > n_0 = n_1 \implies |u_n - \ell_1| < \frac{\varepsilon}{|\lambda|} \implies |\lambda u_n - \lambda \ell_1| = |\lambda| |u_n - \ell_1| < \varepsilon.$$

2. $\lim_{n \rightarrow \infty} (u_n + v_n) = \ell_1 + \ell_2 \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |(u_n + v_n) - (\ell_1 + \ell_2)| < \varepsilon).$

$$\text{Note that: } |(u_n + v_n) - (\ell_1 + \ell_2)| = |(u_n - \ell_1) + (v_n - \ell_2)| \leq |u_n - \ell_1| + |v_n - \ell_2| < \varepsilon \text{ if for example } |u_n - \ell_1| < \frac{\varepsilon}{2} \text{ and } |v_n - \ell_2| < \frac{\varepsilon}{2}.$$

We have $\lim_{n \rightarrow \infty} u_n = \ell_1$ and $\lim_{n \rightarrow \infty} v_n = \ell_2$. Then $|u_n - \ell_1| < \frac{\varepsilon}{2}$ for $n \geq n_1$ and $|v_n - \ell_2| < \frac{\varepsilon}{2}$ for $n \geq n_2$. Choose $n_0 := \max\{n_1, n_2\}$. If $n \geq n_0$, then $|(u_n + v_n) - (\ell_1 + \ell_2)| = |(u_n - \ell_1) + (v_n - \ell_2)| \leq |u_n - \ell_1| + |v_n - \ell_2| < \varepsilon$.

3. $\lim_{n \rightarrow \infty} (u_n v_n) = \ell_1 \ell_2 \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |u_n v_n - \ell_1 \ell_2| < \varepsilon).$

Note that:

- Since (u_n) is convergent, then its bounded i.e., $|u_n| \leq A$ for some $A > 0$.
- $|u_n v_n - \ell_1 \ell_2| = |u_n v_n - u_n \ell_2 + u_n \ell_2 - \ell_1 \ell_2| \leq |u_n v_n - u_n \ell_2| + |u_n \ell_2 - \ell_1 \ell_2| = |u_n| |v_n - \ell_2| + |\ell_2| |u_n - \ell_1| \leq A |v_n - \ell_2| + (|\ell_2| + 1) |u_n - \ell_1| < \varepsilon$ if $|v_n - \ell_2| < \frac{\varepsilon}{2A}$ and $|u_n - \ell_1| < \frac{\varepsilon}{2(|\ell_2| + 1)}$.

We have $\lim_{n \rightarrow \infty} u_n = \ell_1$ and $\lim_{n \rightarrow \infty} v_n = \ell_2$. Then $|u_n - \ell_1| < \frac{\varepsilon}{2(|\ell_2| + 1)}$ for $n \geq n_1$ and $|v_n - \ell_2| < \frac{\varepsilon}{2A}$ for $n \geq n_2$. Choose $n_0 := \max\{n_1, n_2\}$. If $n \geq n_0$, then $|u_n v_n - \ell_1 \ell_2| < \varepsilon$.

4. It is sufficient to show:

$$\lim_{n \rightarrow \infty} (1/v_n) = 1/\ell_2 \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |1/v_n - 1/\ell_2| < \varepsilon).$$

Note that:

- Since $\lim_{n \rightarrow \infty} v_n = \ell_2 \neq 0$, then $|v_n| \geq \frac{\ell_2}{2}$ for $n \geq N \in \mathbb{N}$.
- $|1/v_n - 1/\ell_2| = \left| \frac{\ell_2 - v_n}{v_n \ell_2} \right| \leq 2 \left| \frac{v_n - \ell_2}{\ell_2^2} \right| < \varepsilon$ if $|v_n - \ell_2| < \frac{\ell_2^2 \varepsilon}{2}$.

We have $\lim_{n \rightarrow \infty} v_n = \ell_2$. Then $|v_n - \ell_2| < \frac{\ell_2^2 \varepsilon}{2}$ for $n \geq n_2$. Choose $n_0 := \max\{N, n_2\}$. If $n \geq n_0$, then $|1/v_n - 1/\ell_2| < \varepsilon$.

5. $\lim_{n \rightarrow \infty} |u_n| = |\ell_1| \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies ||u_n| - |\ell_1|| < \varepsilon).$

Note that: $||u_n| - |\ell_1|| \leq |u_n - \ell_1|$.

We have $\lim_{n \rightarrow \infty} u_n = \ell_1$. Then $|u_n - \ell_1| < \varepsilon$ for $n \geq n_1$. Choose $n_0 := n_1$. If $n \geq n_0$, then $||u_n| - |\ell_1|| \leq |u_n - \ell_1| < \varepsilon$.

□

2.2 Limits And Ordered Relation

2.17 Property. If $\lim_{n \rightarrow \infty} u_n = \ell \in \mathbb{R}$ and $u_n \geq 0$ for $n \geq N \in \mathbb{N}$, then $\ell \geq 0$.

Proof. Assume to the contrary that $\ell < 0$. We have $\lim_{n \rightarrow \infty} u_n = \ell$. Let n_0 corresponds to $\varepsilon := \frac{-\ell}{2} > 0$. If $n \geq n_0$, then $u_n - \ell < \frac{-\ell}{2}$. Hence $u_n < \frac{\ell}{2} < 0$ eventually which is a contradiction. Thus $\ell \geq 0$. □

2.18 Corollary. Let (u_n) and (v_n) be convergent sequences such that $u_n \leq v_n$ for $n \geq N \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n$.

Proof. Apply the previous property to the sequence $w_n := v_n - u_n$

□

2.19 Remark. Let (u_n) and (v_n) be convergent sequences.

1. $\forall n \geq n_0, u_n < v_n \implies \lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n$.

for example $u_n = -\frac{1}{n} < v_n = \frac{1}{n}$ and $\lim_{n \rightarrow \infty} u_n = 0 = \lim_{n \rightarrow \infty} v_n$.

2. $\forall n \geq n_0, u_n \leq c$ ($c \in \mathbb{R}$) $\implies \lim_{n \rightarrow \infty} u_n \leq c$.

2.20 Property. (Sandwich Rule) Let (u_n) , (v_n) and (w_n) be sequences in $\mathbb{R}$ such that $u_n \leq v_n \leq w_n$ for all $n \geq N$ ($N \in \mathbb{N}$) and $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} w_n = \ell$ ($\ell \in \mathbb{R}$). Then $\lim_{n \rightarrow \infty} v_n = \ell$.

Proof. $\lim_{n \rightarrow \infty} v_n = \ell \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies \ell - \varepsilon < v_n < \ell + \varepsilon).$

Let $\varepsilon > 0$, then by the definition of the limit:

$$\exists n_1 \in \mathbb{N}, \quad n \geq n_1 \implies \ell - \varepsilon < u_n < \ell + \varepsilon$$

$$\exists n_2 \in \mathbb{N}, \quad n \geq n_2 \implies \ell - \varepsilon < w_n < \ell + \varepsilon.$$

Choose $n_0 := \max\{N, n_1, n_2\}$. If $n \geq n_0$, then $\ell - \varepsilon < u_n \leq v_n \leq w_n < \ell + \varepsilon$.

□

2.21 Example. Show that $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$.

Solution.

First Way: $-1 \leq \sin n \leq 1 \implies -1/n \leq (\sin n)/n \leq 1/n \implies \lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$.

Second Way: To show $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$ is equivalent to show $\lim_{n \rightarrow \infty} \left| \frac{\sin n}{n} \right| = 0$. We have $0 \leq \left| \frac{\sin n}{n} \right| \leq \frac{1}{n}$, then by Sandwich rule $\lim_{n \rightarrow \infty} \left| \frac{\sin n}{n} \right| = 0$.

2.22 Property. If (u_n) is a bounded sequence and $\lim_{n \rightarrow \infty} v_n = 0$, then $\lim_{n \rightarrow \infty} u_n v_n = 0$.

Proof. Since (u_n) is a bounded sequence, then there exists $(A > 0)$ such that $|u_n| < A$ for all n . Hence, $0 \leq |u_n v_n| = |u_n| |v_n| \leq A |v_n|$. Then by Sandwich rule $\lim_{n \rightarrow \infty} |u_n v_n| = 0$. Thus $\lim_{n \rightarrow \infty} u_n v_n = 0$ $\square$

2.23 Property. For any real number x there exist sequences (r_n) in $\mathbb{Q}$ and (i_n) in $\mathbb{R} \setminus \mathbb{Q}$ such that $\lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} i_n = x$.

Proof. By density of rationals and irrationals in $\mathbb{R}$, for all $n \in \mathbb{N}^*$, there exist $r_n \in \mathbb{Q}$ and $i_n \in \mathbb{R} \setminus \mathbb{Q}$ such that $x - \frac{1}{n} < r_n < x + \frac{1}{n}$ and $x - \frac{1}{n} < i_n < x + \frac{1}{n}$. The Sandwich rule then implies that $\lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} i_n = x$. $\square$

2.24 Property. Let (u_n) and (v_n) be sequences in $\mathbb{R}$ such that $u_n \leq v_n$ for all $n \geq N$ ($N \in \mathbb{N}$).

1. If $\lim_{n \rightarrow \infty} u_n = \infty$, then $\lim_{n \rightarrow \infty} v_n = \infty$.
2. If $\lim_{n \rightarrow \infty} v_n = -\infty$, then $\lim_{n \rightarrow \infty} u_n = -\infty$.

Proof. 1. $\lim_{n \rightarrow \infty} v_n = \infty \iff (\forall A > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies v_n > A)$.

Let $A > 0$, then by the definition of the limit $\exists n_1 \in \mathbb{N}$, $n \geq n_1 \implies u_n > A$. Choose $n_0 := \max\{N, n_1\}$. If $n \geq n_0$, then $v_n \geq u_n > A$.

2. $\lim_{n \rightarrow \infty} u_n = -\infty \iff (\forall A < 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies u_n < A)$.

Let $A < 0$, then by the definition of the limit $\exists n_1 \in \mathbb{N}$, $n \geq n_1 \implies v_n < A$. Choose $n_0 := \max\{N, n_1\}$. If $n \geq n_0$, then $u_n \leq v_n < A$. $\square$

For example: $u_n = \sum_{k=1}^n \sqrt{k} > \sqrt{n}$. Since $\lim_{n \rightarrow \infty} \sqrt{n} = \infty$, then $\lim_{n \rightarrow \infty} u_n = \infty$.

2.3 Monotone Sequences

2.25 Definition.

- A sequence (u_n) in $\mathbb{R}$ is said to be increasing if $\forall n \in \mathbb{N}, u_n \leq u_{n+1}$.
- A sequence (u_n) in $\mathbb{R}$ is said to be strictly increasing if $\forall n \in \mathbb{N}, u_n < u_{n+1}$.
- A sequence (u_n) in $\mathbb{R}$ is said to be decreasing if $\forall n \in \mathbb{N}, u_n \geq u_{n+1}$.
- A sequence (u_n) in $\mathbb{R}$ is said to be strictly decreasing if $\forall n \in \mathbb{N}, u_n > u_{n+1}$.

A sequence that is either increasing or decreasing is called monotone and a sequence that is either strictly increasing or strictly decreasing is called strictly monotone.

For example $u_n = \frac{n-1}{n+1}$ is strictly increasing because $\forall n \in \mathbb{N}, u_n - u_{n+1} = \frac{-2}{(n+1)(n+2)} < 0$.

2.26 Theorem. (Monotone Sequence Theorem)

Let (u_n) be an increasing sequence. Then we have two possibilities:

1. If (u_n) is bounded above, then it converges to $\sup_{n \in \mathbb{N}} u_n$.
2. If (u_n) is not bounded above, then it diverges to $\sup_{n \in \mathbb{N}} u_n = \infty$.

Proof.

1. Since the set $\{u_n; n \in \mathbb{N}\}$ is nonempty and bounded above, then $\sup_{n \in \mathbb{N}} u_n := \sup\{u_n; n \in \mathbb{N}\}$ exists in $\mathbb{R}$. Claim: $\lim_{n \rightarrow \infty} u_n = \sup_{n \in \mathbb{N}} u_n \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies \sup_{n \in \mathbb{N}} u_n - \varepsilon < u_n < \sup_{n \in \mathbb{N}} u_n + \varepsilon)$.

Let $\varepsilon > 0$, then by ε -characterization of supremum, there exists $n_0 \in \mathbb{N}$ such that $\sup_{n \in \mathbb{N}} u_n - \varepsilon < u_{n_0}$.
If $n \geq n_0$, then by the monotony of the sequence we have

$$\sup_{n \in \mathbb{N}} u_n - \varepsilon < u_{n_0} \leq u_n \leq \sup_{n \in \mathbb{N}} u_n < \sup_{n \in \mathbb{N}} u_n + \varepsilon.$$

2. Claim: $\lim_{n \rightarrow \infty} u_n = \infty \iff (\forall A > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies u_n > A)$.

Let $A > 0$. Since (u_n) is not bounded above, then there exists $n_0 \in \mathbb{N}$ such that $u_{n_0} > A$. If $n \geq n_0$, then by the monotony of the sequence we have $u_n \geq u_{n_0} > A$ □

2.27 Corollary. (Monotone Sequence Theorem)

Let (u_n) be a decreasing sequence. Then we have two possibilities:

1. If (u_n) is bounded below, then it converges to $\inf_{n \in \mathbb{N}} u_n$.
2. If (u_n) is not bounded below, then it diverges to $\inf_{n \in \mathbb{N}} u_n = -\infty$.

Proof. Apply the previous theorem to the sequence $(-u_n)$. □

2.28 Property. Let (u_n) be a sequence in $\mathbb{R}$.

1. If (u_n) is bounded above by ℓ and $\lim_{n \rightarrow \infty} u_n = \ell$, then $\sup u_n = \ell$.
2. If (u_n) is bounded below by ℓ and $\lim_{n \rightarrow \infty} u_n = \ell$, then $\inf u_n = \ell$.

Proof.

1. **First Way:** We have $u_n \leq \sup u_n \leq \ell$. Then by Sandwich rule $\sup u_n = \ell$.

Second Way: (ε -characterization).

• We have ℓ is an upper bound of (u_n) .

• Claim: $(\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\ell - \varepsilon < u_{n_0})$.

Let $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} u_n = \ell$, then there exists $n_1 \in \mathbb{N}$ such that $n \geq n_1 \implies \ell - \varepsilon < u_n < \ell + \varepsilon$.

Choose $n_0 := n_1$, then $\ell - \varepsilon < u_{n_0}$.

2. **First Way:** We have $\ell \leq \inf u_n \leq u_n$. Then by Sandwich rule $\inf u_n = \ell$.

Second Way: (ε -characterization).

• We have ℓ is a lower bound of (u_n) .

• Claim: $(\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(u_{n_0} < \ell + \varepsilon)$.

Let $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} u_n = \ell$, then there exists $n_1 \in \mathbb{N}$ such that $n \geq n_1 \implies \ell - \varepsilon < u_n < \ell + \varepsilon$.

Choose $n_0 := n_1$, then $u_{n_0} < \ell + \varepsilon$.

Third way: Apply 1. to the sequence $(-u_n)$. □

For example: The sequence $u_n = \frac{n^2}{n^2+1}$ is bounded above by 1 and $\lim_{n \rightarrow \infty} u_n = 1$, then $\sup u_n = 1$.

2.29 Property. (Sequential Characterization Of Supremum And Infimum)

1. Let A be a nonempty bounded above subset of $\mathbb{R}$. Then $\ell = \sup A$ if and only if ℓ is an upper bound of A and ℓ is a limit of a sequence (a_n) in A .
2. Let A be a nonempty bounded below subset of $\mathbb{R}$. Then $\ell = \inf A$ if and only if ℓ is a lower bound of A and ℓ is a limit of a sequence (a_n) in A .

Proof. Exercise. □

For example: If A admits a supremum in $\mathbb{R}$, then $\sup A$ is a limit of a sequence (a_n) in A , and if $B \subset (0, \infty)$ admits an infimum, then $\inf B$ is a limit of a sequence (b_n) in B . Hence the set $C := \{a - \sqrt{b} : a \in A, b \in B\}$ admits a supremum and $\sup C = \sup A - \sqrt{\inf B}$ because $\sup A - \sqrt{\inf B}$ is an upper bound of C and $\sup A - \sqrt{\inf B}$ is the limit of the sequence $(a_n - \sqrt{b_n}) \in C$.

2.4 Sub Sequences

2.30 Definition. A sub sequence of a sequence (u_n) "is a part of (u_n) appear in the same order" denoted by $(u_{\varphi(n)})$ where $\varphi : \mathbb{N} \mapsto \mathbb{N}$ is a strictly increasing sequence.

For example (u_{2n}) is a sub sequence of (u_n) because $\varphi(n) = 2n$ is a strictly increasing sequence and (u_{2n+1}) is a sub sequence of (u_n) because $\varphi(n) = 2n + 1$ is a strictly increasing sequence.

2.31 Property. The strictly increasing function $\varphi : \mathbb{N} \mapsto \mathbb{N}$ satisfies $\varphi(n) \geq n$ for all n .

Proof. By induction. □

2.32 Remark. If (u_n) is a bounded sequence, then any sub sequence $(u_{\varphi(n)})$ is also bounded.

2.33 Property. If $u_n \xrightarrow{n \rightarrow \infty} \ell$, then any sub sequence $u_{\varphi(n)} \xrightarrow{n \rightarrow \infty} \ell$.

Proof. $u_{\varphi(n)} \xrightarrow{n \rightarrow \infty} \ell \iff (\forall \varepsilon > 0)(\exists n_0 \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq n_0 \implies |u_{\varphi(n)} - \ell| < \varepsilon)$.

Let $\varepsilon > 0$. Since $u_n \xrightarrow{n \rightarrow \infty} \ell$, then there exists $n_1 \in \mathbb{N}$ such that $n \geq n_1 \xrightarrow{(\star)} |u_n - \ell| < \varepsilon$. Choose $n_0 := n_1$. If $n \geq n_0$ then $\varphi(n) \geq \varphi(n_0) \geq n_0 = n_1$ and by the implication $(\star)$ we obtain $|u_{\varphi(n)} - \ell| < \varepsilon$. □

2.34 Corollary. 1 If (u_n) has a divergent sub sequence, then (u_n) is divergent.

2. If (u_n) has two sub sequences converge to different limits, then (u_n) is divergent.

2.35 Theorem. Every sequence (u_n) has a monotonic sub sequence.

Proof. Let $S := \{n \in \mathbb{N} : u_m < u_n \text{ for } m > n\}$. We discuss three cases:

Case 1: S is finite. Let $M := \max S$. Let $n_1 > M$, then $n_1 \notin S$ and there exists $n_2 > n_1$ such that $u_{n_2} \geq u_{n_1}$. Now, $n_2 > n_1 > M$, then $n_2 \notin S$ and there exists $n_3 > n_2$ such that $u_{n_3} \geq u_{n_2}$. Proceeding in the same way, we obtain an increasing sub sequence $(u_{n_k})_{k=1}^\infty$.

Case 2: S is empty. Similar to case 1, take $n_1 = 1 \notin S$.

Case 3: S is infinite. Let $n_1 = \min S$, $n_2 = \min\{S \setminus \{n_1\}\}$ and so on. We obtain a strictly increasing sequence $(n_k)_{k=1}^\infty \subset S$. Consequently (u_n) has a decreasing sub sequence $(u_{n_k})_{k=1}^\infty$. $\square$

2.36 Theorem. (Bolzano-Weierstrass Theorem)

Every bounded sequence has a convergent sub sequence.

Proof. (u_n) has a monotone sub sequence (u_{n_k}) . Since (u_n) is bounded then (u_{n_k}) is also bounded. Hence (u_{n_k}) is a monotone bounded sequence. Thus (u_{n_k}) is a convergent sub sequence of (u_n) . $\square$
For example: The sequence $(\cos n)$ has a convergent sub sequence.

2.5 Exercises

Exercise 1. Using the definition of the limit of a sequence, prove that

$1. \lim_{n \rightarrow \infty} \sqrt{1 + \frac{\sqrt{n-1}}{n}} = 1.$ $2. \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0.$ $3. \lim_{n \rightarrow \infty} \frac{\sqrt{n^2 - n + 1}}{n + 1} = 1.$ $4. \lim_{n \rightarrow \infty} \frac{n \cos n}{n^2 + 1} = 0.$	$5. \lim_{n \rightarrow \infty} \frac{2n \cos n \sin n - 3n \sin n \cos^2 n}{n^2 + 1} = 0.$ $6. \lim_{n \rightarrow \infty} \frac{2n \cos n \sin n - 3n \sin n \cos^2 n}{n^2 + 1 + \sin n} = 0.$ $7. \lim_{n \rightarrow \infty} \frac{n^2 + (-1)^n}{\sqrt{n^2 + n + 7}} = \infty.$ $8. \lim_{n \rightarrow \infty} \frac{n^2}{4 - n} = -\infty.$
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Exercise 2. Let (u_n) be a sequence in $[0, \infty)$ with limit in $\overline{\mathbb{R}}$. Show that $\lim_{n \rightarrow \infty} \sqrt{u_n} = \sqrt{\lim_{n \rightarrow \infty} u_n}$.

Exercise 3. Let $0 \leq c < 1$ and $A := \{u_n = c^n : n \in \mathbb{N}^*\}$.

1. Show that the sequence (u_n) is a decreasing sequence.
2. Show that $0 \leq c^n < 1$ for all $n \in \mathbb{N}^*$.
3. Show that $\lim_{n \rightarrow \infty} u_n$ exists in $\mathbb{R}$ and $\lim_{n \rightarrow \infty} u_n = 0$. (Hint: $u_{n+1} = cu_n$.)
4. Deduce the value of $\inf A$.
5. Assume $-1 < c < 1$. Show that $\lim_{n \rightarrow \infty} c^n = 0$.

Exercise 4. Let $0 < u_0 < v_0$ and define $u_{n+1} = \sqrt{u_n v_n}$ and $v_{n+1} = \frac{u_n + v_n}{2}$. Show that $0 < u_n < u_{n+1} < v_{n+1} < v_n$ and that $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} v_n$.

Exercise 5. Let (u_n) be a sequence in $\mathbb{R}$.

1. If (u_n) is bounded above by ℓ and $\exists u_{\varphi(n)}$ sub sequence s.t $\lim_{n \rightarrow \infty} u_{\varphi(n)} = \ell$, then $\sup u_n = \ell$.
2. If (u_n) is bounded below by ℓ and $\exists u_{\varphi(n)}$ sub sequence s.t $\lim_{n \rightarrow \infty} u_{\varphi(n)} = \ell$, then $\inf u_n = \ell$.
3. Let $A := \{u_n = (-1)^n + \frac{1}{2n} : n \in \mathbb{N}^*\}$. Determine $\sup A$ and $\inf A$.

Exercise 6. Show that the limit of the sequence $(\cos n)$ does not exist in $\mathbb{R}$. (Use $\cos(2n) = 2\cos^2 n - 1$ and $\cos(n+1) + \cos(n-1) = 2\cos n \cos 1$ or $\exists m_k, n_k \in \mathbb{N}$ s.t $m_k \in [2k\pi - \frac{\pi}{6}, 2k\pi + \frac{\pi}{6}]$, $n_k \in [2k\pi + \frac{\pi}{4}, 2k\pi + \frac{3\pi}{4}]$.)

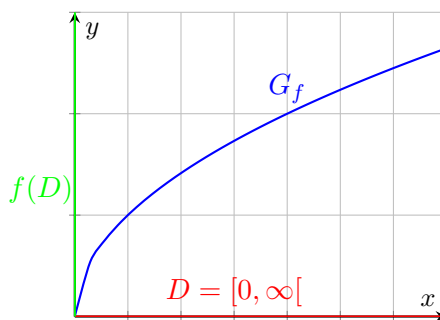
3.1 Functions

3.1 Definition. Let $D \subseteq \mathbb{R}$. A **function** f from D into $\mathbb{R}$ denoted by $f : D \mapsto \mathbb{R}$ or $D \xrightarrow{f} \mathbb{R}$, is a rule that assigns to each $x \in D$ a unique image in $\mathbb{R}$ denoted by $f(x)$.

- D is called the domain of f .
- $f(D) := \{f(x) : x \in D\}$ is called the image or range of f .
- $G_f := \{(x, f(x)) : x \in D\}$ is called the graph of f .

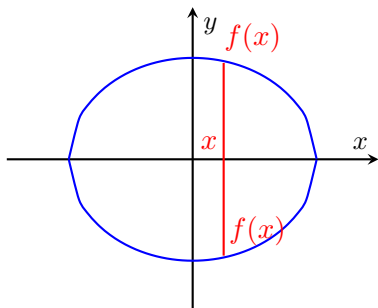
For example: $f : [0, \infty[\mapsto \mathbb{R}$ defined by $f(x) = \sqrt{x}$ is a function (by the uniqueness of the square root).

- The domain of f is $D = [0, \infty[$.
- The range of f is $f(D) = [0, \infty[$.
- The graph of f is $G_f := \{(x, \sqrt{x}) : x \in [0, \infty[\}$.

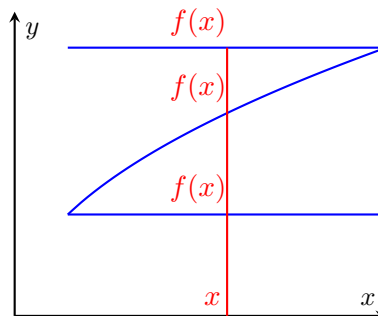


The function $f(x) = \sqrt{x}$.

Here are some graphs of non functions.



Graph of a non function. $(x^2 + y^2 = 4)$



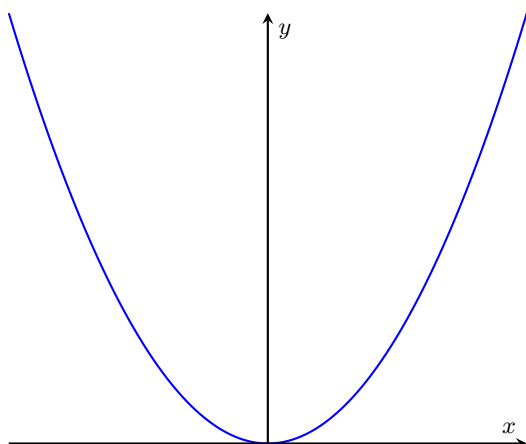
Graph of a non function

3.2 Definition. Let $f : D \mapsto \mathbb{R}$ be a function. We say that

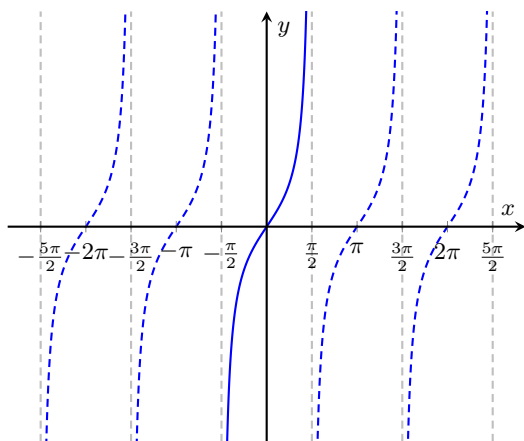
1. f is **even** on D if $\begin{cases} \bullet D \text{ is symmetric with respect to the origin i.e.; } x \in D \implies -x \in D. \\ \bullet f(-x) = f(x) \forall x \in D. \end{cases}$
2. f is **odd** on D if $\begin{cases} \bullet D \text{ is symmetric with respect to the origin i.e.; } x \in D \implies -x \in D. \\ \bullet f(-x) = -f(x) \forall x \in D. \end{cases}$
3. f is **periodic** on D if $\exists T > 0$ s.t. $\begin{cases} \bullet D \text{ is invariant under translations by } \pm T \text{ i.e.,} \\ \quad (x \in D \implies x \pm T \in D) \\ \bullet f(x+T) = f(x) \forall x \in D. \end{cases}$

The **period** of f is the smallest positive number T (if exists) satisfying $f(x+T) = f(x) \forall x \in D$. If the period of f is T , then we say that f is T -**periodic** on D .

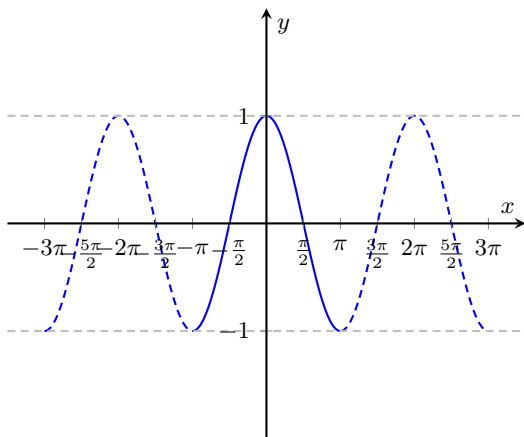
Note that if $D = \mathbb{R}$, then the first condition in the above three definitions is always satisfied. Also, a constant function on $\mathbb{R}$ is periodic of any period $T > 0$ and thus has no period. Here are some examples about even, odd and periodic functions.



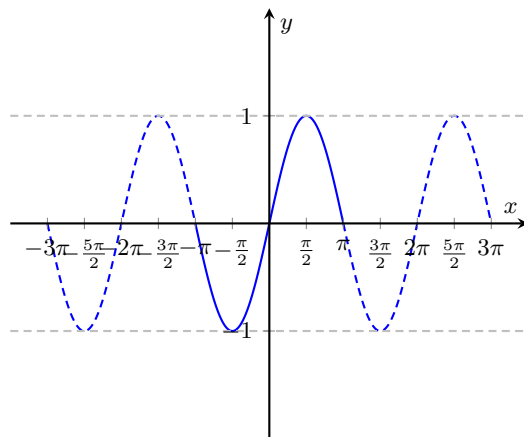
$x \mapsto x^2$ is an even function on $D = \mathbb{R}$ or $D = [-1, 1]$ but not $D = [-1, 1[$.



$x \mapsto \tan x$ is π -periodic and odd on $D = \bigcup_{k \in \mathbb{Z}}]-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi[$.



$x \mapsto \cos x$ is 2π -periodic and even on $D = \mathbb{R}$.



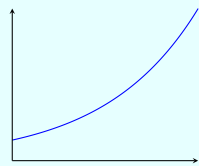
$x \mapsto \sin x$ is 2π -periodic and odd on $D = \mathbb{R}$.

3.3 Property. If f is T -periodic on D , then for all $x \in D$ we have $f(x + nT) = f(x) \forall n \in \mathbb{Z}$.

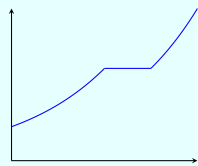
proof. Exercise

3.4 Definition. Let $f : D \mapsto \mathbb{R}$ be a function. We say that

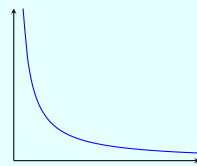
1. f is **increasing on D** if; $\forall x_1, x_2 \in D, x_1 < x_2 \implies f(x_1) \leq f(x_2).$
2. f is **strictly increasing on D** if; $\forall x_1, x_2 \in D, x_1 < x_2 \implies f(x_1) < f(x_2).$
3. f is **decreasing on D** if; $\forall x_1, x_2 \in D, x_1 < x_2 \implies f(x_1) \geq f(x_2).$
4. f is **strictly decreasing on D** if; $\forall x_1, x_2 \in D, x_1 < x_2 \implies f(x_1) > f(x_2).$
5. f is **monotone on D** if it is either increasing or decreasing on D .
6. f is **strictly monotone on D** if it is either strictly increasing or strictly decreasing on D .



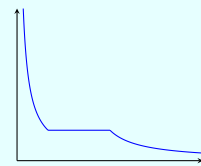
St. inc. function.



Inc. function.



St. dec. function.



Dec. function.

Note that, if a function is strictly increasing then it is increasing and if a function is strictly decreasing then it is decreasing as well.

3.5 Definition. Let $f : D \mapsto \mathbb{R}$ be a function. We say that

1. f is **bounded above on D** if; $\exists M \in \mathbb{R}, \forall x \in D, f(x) \leq M.$
2. f is **bounded below on D** if; $\exists m \in \mathbb{R}, \forall x \in D, m \leq f(x).$
3. f is **bounded on D** if; $\exists m, M \in \mathbb{R}, \forall x \in D, m \leq f(x) \leq M.$

Note that f is bounded on D if and only if it is bounded above and below on D .

For example:

- ★ $f(x) = \sin x$ is bounded below on $D = \mathbb{R}$ by $m = -1$ and bounded above by $M = 1$.
- ★ $f(x) = \cos x$ is bounded below on $D = \mathbb{R}$ by $m = -1$ and bounded above by $M = 1$.
- ★ $f(x) = e^x$ is bounded below on $\mathbb{R}$ by $m = 0$ and it is not bounded above on $\mathbb{R}$.
- ★ $f(x) = e^x$ is bounded below on $D =]-\infty, 3]$ by $m = 0$ and bounded above by $M = e^3$.
- ★ $f(x) = \tan x$ is neither bounded below nor bounded above on $D =]-\frac{\pi}{2}, \frac{\pi}{2}[$.
- ★ $f(x) = |\cos x|$ is bounded below on $D = \mathbb{R}$ by $m = 0$ and bounded above by $M = 1$.

If we let $A := \{f(x) : x \in D\}$, then

- f is bounded above on D if and only if A is bounded above.
- f is bounded below on D if and only if A is bounded below.
- f is bounded on D if and only if A is bounded.

All properties of bounded sets can be restated for bounded functions. In particular, Exercise 6. Chapter one gives the following property

3.6 Property. A function $f : D \mapsto \mathbb{R}$ is **bounded on D** if and only if $\exists C > 0, \forall x \in D, |f(x)| \leq C$.

3.2 Limits

The notion of limit is conveniently described in terms of the notion of neighborhoods.

3.7 Definition. (*Neighborhoods*) Let $r > 0$.

1. Let $a \in \mathbb{R}$, the set

$I_r(a) :=]a - r, a + r[= \{x \in \mathbb{R} : |x - a| < r\}$ is called an **r -neighborhood** of a .

$I_r(a^+) := [a, a + r[= \{x \in \mathbb{R} : 0 \leq x - a < r\}$ is called a **right neighborhood** of a .

$I_r(a^-) :=]a - r, a] = \{x \in \mathbb{R} : 0 \leq a - x < r\}$ is called a **left neighborhood** of a .

$I_r(a) \setminus \{a\} :=]a - r, a[\cup]a, a + r[= \{x \in \mathbb{R} : 0 < |x - a| < r\}$ is called a **deleted neighborhood** of a .

$I_r(a^+) \setminus \{a\} :=]a, a + r[= \{x \in \mathbb{R} : 0 < x - a < r\}$.

$I_r(a^-) \setminus \{a\} :=]a - r, a[= \{x \in \mathbb{R} : 0 < a - x < r\}$.

2. A **neighborhood** of ∞ is an interval of the form $I_r(\infty) := (r, \infty)$.

3. A **neighborhood** of $-\infty$ is an interval of the form $I_r(-\infty) := (-\infty, -r)$.

Using the conventions $I_r(\infty) \setminus \{\infty\} = I_r(\infty)$, $I(a) = I_r(a)$ and $I(a^+) \setminus \{a^+\} = I(a^+) \setminus \{a\}$, the following definition of limit is sufficiently general to include the usual limits encountered in calculus: two-sided limits, one-sided limits and limits at infinity.

The symbol α will denote any of $a, a^+, a^-, \infty, -\infty$.

3.8 Definition. Let f be a real valued function defined on some $I(\alpha) \setminus \{\alpha\}$ and $\ell \in \overline{\mathbb{R}}$. We say that $f(x)$ tends to ℓ as x tends to α and we write $\lim_{x \rightarrow \alpha} f(x) = \ell$ if

$$\forall I(\ell), \quad \exists I(\alpha), \quad \forall x \in D, \quad x \in I(\alpha) \setminus \{\alpha\} \implies f(x) \in I(\ell).$$

We tabulate $I(\ell)$ in various cases:

ℓ	$\ell \in \mathbb{R}$	$\ell = \infty$	$\ell = -\infty$
$I(\ell)$	$(\ell - \varepsilon, \ell + \varepsilon)$	$(M, \infty), M > 0$	$(-\infty, M), M < 0$
$f(x) \in I(\ell)$	$ f(x) - \ell < \varepsilon$	$f(x) > M$	$f(x) < M$

In analogy to sequences we start with the case $\alpha = \infty$.

a. The case $\alpha = \infty$.

$I(\alpha) \setminus \{\alpha\} = (A, \infty), A > 0$
$x \in I(\alpha) \setminus \{\alpha\} \iff x > A$

$$1. \lim_{x \rightarrow \infty} f(x) = \ell \iff \left(\forall \varepsilon > 0, \exists A > 0, \forall x \in D, x > A \implies |f(x) - \ell| < \varepsilon \right).$$

$$2. \lim_{x \rightarrow \infty} f(x) = \infty \iff \left(\forall M > 0, \exists A > 0, \forall x \in D, x > A \implies f(x) > M \right).$$

$$3. \lim_{x \rightarrow \infty} f(x) = -\infty \iff \left(\forall M < 0, \exists A > 0, \forall x \in D, x > A \implies f(x) < M \right).$$

b. The case $\alpha = -\infty$.

$I(\alpha) \setminus \{\alpha\} = (-\infty, A), A < 0$
$x \in I(\alpha) \setminus \{\alpha\} \iff x < A$

1. $\lim_{x \rightarrow -\infty} f(x) = \ell \iff (\forall \varepsilon > 0, \exists A < 0, \forall x \in D, x < A \implies |f(x) - \ell| < \varepsilon).$
2. $\lim_{x \rightarrow -\infty} f(x) = \infty \iff (\forall M > 0, \exists A < 0, \forall x \in D, x < A \implies f(x) > M).$
3. $\lim_{x \rightarrow -\infty} f(x) = -\infty \iff (\forall M < 0, \exists A < 0, \forall x \in D, x < A \implies f(x) < M).$

c. The case $\alpha = a \in \mathbb{R}$.

$I(a) \setminus \{a\} = (a - \delta, a) \cup (a, a + \delta), \delta > 0$
$x \in I(a) \setminus \{a\} \iff 0 < x - a < \delta \iff (x \neq a \wedge x - a < \delta)$

1. $\lim_{x \rightarrow a} f(x) = \ell \iff (\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D, 0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon).$
2. $\lim_{x \rightarrow a} f(x) = \infty \iff (\forall M > 0, \exists \delta > 0, \forall x \in D, 0 < |x - a| < \delta \implies f(x) > M).$
3. $\lim_{x \rightarrow a} f(x) = -\infty \iff (\forall M < 0, \exists \delta > 0, \forall x \in D, 0 < |x - a| < \delta \implies f(x) < M).$

d. The case $\alpha = a^+$. (right-hand limit)

$I(a^+) \setminus \{a\} = (a, a + \delta), \delta > 0$
$x \in I(a^+) \setminus \{a\} \iff 0 < x - a < \delta$

1. $\lim_{x \rightarrow a^+} f(x) = \ell \iff (\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D, 0 < x - a < \delta \implies |f(x) - \ell| < \varepsilon).$
2. $\lim_{x \rightarrow a^+} f(x) = \infty \iff (\forall M > 0, \exists \delta > 0, \forall x \in D, 0 < x - a < \delta \implies f(x) > M).$
3. $\lim_{x \rightarrow a^+} f(x) = -\infty \iff (\forall M < 0, \exists \delta > 0, \forall x \in D, 0 < x - a < \delta \implies f(x) < M).$

e. The case $\alpha = a^-$. (left-hand limit)

$I(a^-) \setminus \{a\} = (a - \delta, a), \delta > 0$
$x \in I(a^-) \setminus \{a\} \iff 0 < a - x < \delta$

1. $\lim_{x \rightarrow a^-} f(x) = \ell \iff (\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D, 0 < a - x < \delta \implies |f(x) - \ell| < \varepsilon).$
2. $\lim_{x \rightarrow a^-} f(x) = \infty \iff (\forall M > 0, \exists \delta > 0, \forall x \in D, 0 < a - x < \delta \implies f(x) > M).$
3. $\lim_{x \rightarrow a^-} f(x) = -\infty \iff (\forall M < 0, \exists \delta > 0, \forall x \in D, 0 < a - x < \delta \implies f(x) < M).$

How to show $\lim_{x \rightarrow a} f(x) = \ell$?

$$\lim_{x \rightarrow a} f(x) = \ell \iff (\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D \setminus \{a\}, |x - a| < \delta \xrightarrow{(\star)} |f(x) - \ell| < \varepsilon).$$

• We let $\varepsilon > 0$ be arbitrary.

• We search for $\delta = \delta(\varepsilon)$ which may depends on ε such that the implication $(\star)$ holds true.

How to find δ s.t $(\star)$ holds true? (How to find at least one δ -neighborhood of a s.t $(\star)$ holds true?)

We will simplify matters by making estimates: For $|x - a| < \delta$ given,

$$|f(x) - \ell| \leq \dots \leq |x - a|^{\frac{1}{2}} \text{ (or } \alpha > 0 \text{) (expression in } x \text{ bounded above by } c \text{) } \leq c\delta^{\frac{1}{2}} < \varepsilon$$

If $\delta < \frac{\varepsilon^2}{c^2}$. So we choose $\delta = \frac{1}{2} \frac{\varepsilon^2}{c^2} < \frac{\varepsilon^2}{c^2}$.

3.9 Example. Let $f(x) = x^2 + 3$. Using the definition of the limit, show that $\lim_{x \rightarrow 1} f(x) = 4$.

Solution. $\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D \setminus \{1\}, |x - 1| < \delta \implies |f(x) - 4| < \varepsilon$.

Let $\varepsilon > 0$. We search for $\delta > 0$ such that;

for $x \in D = \mathbb{R}, x \neq 1$ the implication $|x - 1| < \delta \implies |f(x) - 4| < \varepsilon$ holds true.

For $|x - 1| < \delta$ given i.e., $1 - \delta < x < 1 + \delta$,

$$\begin{aligned} |f(x) - 4| &= |x^2 - 1| = |x - 1||x + 1| && (1 - \delta < x < 1 + \delta \implies 2 - \delta < x + 1 < 2 + \delta) \\ &= |x - 1|(x + 1) && (\text{we drop absolute value for } 0 < 2 - \delta \text{ i.e., } \delta < 2) \\ &\leq \delta 4 && (\delta < 2 \implies x + 1 < 2 + \delta < 4) \\ &< \varepsilon && (4\delta < \varepsilon \text{ if } \delta < \frac{\varepsilon}{4}). \end{aligned}$$

Choose $0 < \delta < \min\{2, \frac{\varepsilon}{4}\}$, for example $\delta = \frac{1}{2} \times \min\{2, \frac{\varepsilon}{4}\}$.

Here another estimate:

$$\begin{aligned} |f(x) - 4| &= |x^2 - 1| = |x - 1||x + 1| = |x - 1||x - 1 + 2| \\ &\leq |x - 1|(|x - 1| + |2|) && (\text{using triangle inequality}) \\ &< \delta(\delta + 2) && (\because |x - 1| < \delta) \\ &< 3\delta && (\text{assume } \delta < 1) \\ &< \varepsilon && (3\delta < \varepsilon \text{ if } \delta < \frac{\varepsilon}{3}). \end{aligned}$$

Choose $0 < \delta < \min\{1, \frac{\varepsilon}{3}\}$, for example $\delta = \frac{1}{2} \times \min\{1, \frac{\varepsilon}{3}\}$.

3.10 Example. Let $f(x) = \frac{x - \sqrt{2x - 1}}{x - 1}$. Using the definition of the limit, show that $\lim_{x \rightarrow 1} f(x) = 0$.

Solution. $\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D \setminus \{1\}, |x - 1| < \delta \implies |f(x) - 0| < \varepsilon$.

Let $\varepsilon > 0$. We search for $\delta > 0$ such that;

for $x \in D = [\frac{1}{2}, 1[\cup]1, \infty[$, $x \neq 1$ the implication $|x - 1| < \delta \implies |f(x)| < \varepsilon$ holds true.

For $|x - 1| < \delta$ given i.e., $1 - \delta < x < 1 + \delta$,

$$\begin{aligned} |f(x)| &= \left| \frac{x - \sqrt{2x - 1}}{x - 1} \right| = \left| \frac{x - \sqrt{2x - 1}}{x - 1} \times \frac{x + \sqrt{2x - 1}}{x + \sqrt{2x - 1}} \right| && (\text{try to get } |x - 1| \text{ in numerator}) \\ &= \left| \frac{(x - 1)^2}{(x - 1)(x + \sqrt{2x - 1})} \right| && (\text{cancel by } |x - 1|) \\ &= |x - 1| \times \frac{1}{x + \sqrt{2x - 1}} && x \geq \frac{1}{2} \implies x + \sqrt{2x - 1} \geq \frac{1}{2} \\ &\leq 2\delta < \varepsilon && 2\delta < \varepsilon \text{ if } \delta < \frac{\varepsilon}{2}. \end{aligned}$$

Choose $0 < \delta < \frac{\varepsilon}{2}$, for example $\delta = \frac{1}{2} \times \frac{\varepsilon}{2} = \frac{\varepsilon}{4}$.

Here is another estimate independent of the fact that $x \geq \frac{1}{2}$:

$$\begin{aligned} |f(x)| &= \left| \frac{x - \sqrt{2x - 1}}{x - 1} \right| = \left| \frac{x - \sqrt{2x - 1}}{x - 1} \times \frac{x + \sqrt{2x - 1}}{x + \sqrt{2x - 1}} \right| && (\text{try to get } |x - 1| \text{ in numerator}) \\ &= \left| \frac{(x - 1)^2}{(x - 1)(x + \sqrt{2x - 1})} \right| && (\text{cancel by } |x - 1|) \\ &= |x - 1| \times \frac{1}{x + \sqrt{2x - 1}} && \frac{1}{2} < 1 - \delta < x < 1 + \delta \text{ for } \delta < \frac{1}{2} \\ &\leq 2\delta < \varepsilon && 2\delta < \varepsilon \text{ if } \delta < \frac{\varepsilon}{2}. \end{aligned}$$

Choose $0 < \delta < \min\{\frac{1}{2}, \frac{\varepsilon}{2}\}$, for example $\delta = \frac{1}{2} \times \min\{\frac{1}{2}, \frac{\varepsilon}{2}\}$.

How to show $\lim_{x \rightarrow \infty} f(x) = \ell$?

$$\lim_{x \rightarrow \infty} f(x) = \ell \iff \left(\forall \varepsilon > 0, \quad \exists A > 0, \quad \forall x \in D, \quad x > A \xrightarrow{(\star)} |f(x) - \ell| < \varepsilon \right)$$

• We let $\varepsilon > 0$ be arbitrary.

• We search for $A = A(\varepsilon)$ which may depends on ε such that the implication $(\star)$ holds true.

How to find A s.t $(\star)$ holds true?(How to find at least one nbhd $]A, \infty[$ of ∞ s.t $(\star)$ holds true?)

We will simplify matters by making estimates: For $x > A$ given,

$$|f(x) - \ell| \leq \dots \leq \theta(x) < \varepsilon$$

where $\theta(x) \xrightarrow{x \rightarrow \infty} 0$ and the inequality $\theta(x) < \varepsilon$ is easy to solve.

For example, $\theta(x) = \frac{1}{\sqrt{x}} < \varepsilon$ if $x > \frac{1}{\varepsilon^2}$. So we choose $A > \frac{1}{\varepsilon^2}$. Hence $x > A > \frac{1}{\varepsilon^2}$ implies $\theta(x) < \varepsilon$.

3.11 Example. Let $f(x) = \frac{4x^2+3x-6}{x^2-4}$. Using the definition of the limit, show that $\lim_{x \rightarrow \infty} f(x) = 4$.

Solution. $\forall \varepsilon > 0, \quad \exists A > 0, \quad \forall x \in D =]-\infty, -2[\cup]-2, 2[\cup]2, \infty[, \quad x > A \xrightarrow{(\star)} |f(x) - \ell| < \varepsilon$.
Let $\varepsilon > 0$, we search for A such that the implication $(\star)$ holds true.

$$\begin{aligned} |f(x) - \ell| &= \left| \frac{4x^2 + 3x - 6}{x^2 - 4} - 4 \right| = \left| \frac{3x + 10}{x^2 - 4} \right| \\ &= \frac{3x + 10}{x^2 - 4} && \text{(we drop the absolute value for } x > 2 \text{)} \\ &\leq \frac{3x + 10}{x^2/2} && (x^2 - 4 > x^2/2 \text{ if } x > \sqrt{8}) \\ &\leq \frac{3x + 10x}{x^2/2} && (10 < 10x \text{ for } x > 1) \\ &\leq \frac{26}{x} && (\text{cancel by } x) \\ &< \varepsilon && (\text{for } x > \frac{26}{\varepsilon}). \end{aligned}$$

So we choose $A > \max\{\frac{26}{\varepsilon}, 1, 2, \sqrt{8}\}$.

Hence $x > A$ implies $x > 1, x > 2, x > \sqrt{8}, x > \frac{26}{\varepsilon}$ and $|f(x) - \ell| < \varepsilon$.

How to show $\lim_{x \rightarrow \infty} f(x) = \infty$?

$$\lim_{x \rightarrow \infty} f(x) = \infty \iff \left(\forall M > 0, \quad \exists A > 0, \quad \forall x \in D, \quad x > A \xrightarrow{(\star)} f(x) > M \right)$$

• We let $M > 0$.

• We search for $A = A(M)$ which may depends on M such that the implication $(\star)$ holds true.

How to find A s.t $(\star)$ holds true?(How to find at least one nbhd $]A, \infty[$ of ∞ s.t $(\star)$ holds true?)

We will simplify matters by making estimates: For $x > A$ given,

$$f(x) \geq \dots \geq \theta(x) > M$$

where $\theta(x) \xrightarrow{x \rightarrow \infty} \infty$ and the inequality $\theta(x) > M$ is easy to solve.

3.12 Example. Let $f(x) = \frac{x^2+(-1)^{E(x)}}{\sqrt{x^2+x+7}}$. Using the definition of the limit, show that $\lim_{x \rightarrow \infty} f(x) = \infty$.

Solution. $\forall M > 0, \quad \exists A > 0, \quad \forall x \in D = \mathbb{R}, \quad x > A \xrightarrow{(\star)} f(x) > M$.

Let $M > 0$, we search for A such that the implication $(\star)$ holds true.

$$\begin{aligned}
 f(x) = \frac{x^2 + (-1)^{E(x)}}{\sqrt{x^2 + x + 7}} &\geq \frac{x^2 + -1}{\sqrt{x^2 + x + 7}} && ((-1)^{E(x)} = \pm 1 \leq 1) \\
 &\geq \frac{x^2/2 + x^2/2 - 1}{\sqrt{x^2 + x^2 + 7x^2}} && (x \leq x^2, 7 \leq 7x^2 \text{ for } 1 < x) \\
 &\geq \frac{x^2/2}{3|x|} && (x^2/2 - 1 > 0 \text{ for } 2 < x) \\
 &\geq \frac{x}{6} && (|x| = x \text{ for } 0 < x) \\
 &> M && (\text{ for } 6M < x).
 \end{aligned}$$

So we choose $A > \max\{6M, 0, 1, 2\} = \max\{6M, 2\}$.

Hence $x > A$ implies $x > 0, x > 1, x > 2, x > 6M$ and $f(x) > M$.

How to show $\lim_{x \rightarrow a} f(x) = \infty$?

$$\lim_{x \rightarrow a} f(x) = \infty \iff \left(\forall M > 0, \quad \exists \delta > 0, \quad \forall x \in D \setminus \{a\}, \quad |x - a| < \delta \xRightarrow{(\star)} f(x) > M \right).$$

• We let $M > 0$.

• We search for $\delta = \delta(\varepsilon)$ which may depends on ε such that the implication $(\star)$ holds true.

How to find δ s.t $(\star)$ holds true? (How to find at least one δ -neighborhood of a s.t $(\star)$ holds true?)

We will simplify matters by making estimates: For $|x - a| < \delta$ given,

$$f(x) \geq \dots \geq \frac{(\text{expression in } x \text{ bounded below by positive } c)}{|x - a|^{\frac{1}{2}} (\text{ or } \alpha > 0)} \geq \frac{c}{\delta^{\frac{1}{2}}} > M$$

If $\delta < \frac{c^2}{M^2}$. So we choose $\delta = \frac{1}{2} \frac{c^2}{M^2} < \frac{c^2}{M^2}$.

3.13 Example. Let $f(x) = \frac{|x-1|}{x-\sqrt{2x-1}}$. Using the definition of the limit, show that $\lim_{x \rightarrow 1} f(x) = \infty$.

Solution. $\forall M > 0, \quad \exists \delta > 0, \quad \forall x \in D \setminus \{1\} = [\frac{1}{2}, 1[\cup]1, \infty[, \quad |x - 1| < \delta \xRightarrow{(\star)} f(x) > M$.

Let $M > 0$. We search for $\delta > 0$ such that the implication $(\star)$ holds true.

For $|x - 1| < \delta$ given i.e., $1 - \delta < x < 1 + \delta$,

$$\begin{aligned}
 f(x) &= \frac{|x - 1|}{x - \sqrt{2x - 1}} = \frac{|x - 1|}{x - \sqrt{2x - 1}} \times \frac{x + \sqrt{2x - 1}}{x + \sqrt{2x - 1}} && (\text{try to get } |x - 1|^{\alpha > 0} \text{ in denominator}) \\
 &= \frac{|x - 1|(x + \sqrt{2x - 1})}{(x - 1)^2} && (\text{cancel by } |x - 1|) \\
 &= \frac{1}{|x - 1|} \times (x + \sqrt{2x - 1}) && x + \sqrt{2x - 1} \geq 1 - \delta + \sqrt{1 - 2\delta} \\
 &\geq \frac{1}{\delta} (1 - \delta + \sqrt{1 - 2\delta}) && 1 - \delta + \sqrt{1 - 2\delta} > \frac{1}{2} \text{ if } \delta < \frac{1}{2} \\
 &\geq \frac{1}{\delta} \frac{1}{2} > M && \frac{1}{2\delta} > M \text{ if } \delta < \frac{1}{2M}.
 \end{aligned}$$

Choose $0 < \delta < \frac{1}{2M}$, for example $\delta = \frac{1}{2} \times \frac{1}{2M} = \frac{1}{4M}$.

3.14 Remark. We discussed four cases, the other five cases can be transformed into these cases by the change of variable $t = -x$ in the case $x \rightarrow -\infty$ or by computing $\lim(-f(x))$ in case $f(x) \rightarrow -\infty$.

3.15 Example. Let $f(x) = \frac{x-1}{x-\sqrt{2x-1}}$. Using the definition of the limit, show that $\lim_{x \rightarrow 1^+} f(x) = \infty$.

Solution. $\forall M > 0, \exists \delta > 0, \forall x \in D \setminus \{1\} = [\frac{1}{2}, 1[\cup]1, \infty[, 0 < x - 1 < \delta \xRightarrow{(*)} f(x) > M.$

Let $M > 0$. We search for $\delta > 0$ such that the implication $(*)$ holds true.

For $0 < x - 1 < \delta$ given i.e., $1 < x < 1 + \delta$,

$$\begin{aligned} f(x) &= \frac{x-1}{x-\sqrt{2x-1}} = \frac{x-1}{x-\sqrt{2x-1}} \times \frac{x+\sqrt{2x-1}}{x+\sqrt{2x-1}} && \text{(try to get } (x-1)^{\alpha>0} \text{ in denominator)} \\ &= \frac{(x-1)(x+\sqrt{2x-1})}{(x-1)^2} && \text{(cancel by } (x-1)) \\ &= \frac{1}{x-1} \times (x+\sqrt{2x-1}) && x > 1 \implies x+\sqrt{2x-1} > 2 \\ &\geq \frac{2}{\delta} > M && \frac{2}{\delta} > M \text{ if } \delta < \frac{2}{M}. \end{aligned}$$

Choose $0 < \delta < \frac{2}{M}$.

3.16 Remark. Infact, all cases can be transformed to $\lim_{x \rightarrow 0^\pm} f(x) = 0$

3.17 Property. (Uniqueness of the limit)

Let $\ell \in \overline{\mathbb{R}}$. If $\lim_{x \rightarrow \alpha} f(x) = \ell$, then f admits no other limit for x tends to α .

Proof. We prove the property for the case $\alpha = a \in \mathbb{R}$ and leave the other cases for the reader. Suppose that f has two limits ℓ_1 and ℓ_2 and let us show $\ell_1 = \ell_2$. We consider only the case where ℓ_1 and ℓ_2 are both finite. The equality will follow if we show that $\forall \varepsilon > 0, |\ell_1 - \ell_2| < \varepsilon$. Let $\varepsilon > 0$, then by the definition of the limit:

$$\begin{aligned} \exists \delta_1 > 0, \quad \forall x \in D, \quad 0 < |x - a| < \delta_1 &\implies |f(x) - \ell_1| < \frac{\varepsilon}{2} \\ \exists \delta_2 > 0, \quad \forall x \in D, \quad 0 < |x - a| < \delta_2 &\implies |f(x) - \ell_2| < \frac{\varepsilon}{2} \end{aligned}$$

For $\delta_3 < \min\{\delta_1, \delta_2\}$, $x_0 \in D$ and $0 < |x_0 - a| < \delta_3$ and with the help of triangle inequality we obtain $|\ell_1 - \ell_2| = |\ell_1 - f(x_0) + f(x_0) - \ell_2| \leq |f(x_0) - \ell_1| + |f(x_0) - \ell_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. $\square$

3.18 Property. Let f be a function defined on $I \setminus \{a\}$ for some open interval I containing a and $\ell \in \overline{\mathbb{R}}$. Then

$$\lim_{x \rightarrow a} f(x) = \ell \iff \lim_{x \rightarrow a^-} f(x) = \ell = \lim_{x \rightarrow a^+} f(x).$$

Proof. Exercise. $\square$

3.19 Property. (Sequential Characterization of Limit.)

Let $\ell \in \overline{\mathbb{R}}$ and f defined on $I(\alpha) \setminus \{\alpha\}$. Then

$$\lim_{x \rightarrow \alpha} f(x) = \ell \iff \left(\text{for every sequence } (x_n) \text{ in } I(\alpha) \setminus \{\alpha\}, \quad x_n \xrightarrow{n \rightarrow \infty} \alpha \implies f(x_n) \xrightarrow{n \rightarrow \infty} \ell \right).$$

Proof. We prove the property for the case $\alpha = a \in \mathbb{R}$ and $\ell \in \mathbb{R}$. The proofs for the other cases are similar, adapting the same argument.

$\Rightarrow$) Assume $\lim_{x \rightarrow a} f(x) = \ell$ and let (x_n) be a sequence in $I(a) \setminus \{a\}$ s.t. $x_n \xrightarrow{n \rightarrow \infty} a$. Claim $f(x_n) \xrightarrow{n \rightarrow \infty} \ell$:
Let $\varepsilon > 0$. Since $\lim_{x \rightarrow a} f(x) = \ell$, then

$$\exists \delta > 0, \quad \forall x \in D, \quad 0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon. \quad (\star)$$

Since $x_n \xrightarrow{n \rightarrow \infty} a$ and $\delta > 0$, then

$$\exists n_0 \in \mathbb{N}, \quad n \geq n_0 \implies |x_n - a| < \delta.$$

Hence,

$$\exists n_0 \in \mathbb{N}, \quad n \geq n_0 \implies 0 < |x_n - a| < \delta \xrightarrow{\text{using } (\star)} |f(x_n) - \ell| < \varepsilon.$$

Thus $f(x_n) \xrightarrow{n \rightarrow \infty} \ell$.

$\Leftarrow$) Assume the seq condition; for every sequence (x_n) in $I(a) \setminus \{a\}$, $x_n \xrightarrow{n \rightarrow \infty} a \implies f(x_n) \xrightarrow{n \rightarrow \infty} \ell$.

Suppose to the contrary that $\lim_{x \rightarrow a} f(x) \neq \ell$ i.e.,

$$\exists \varepsilon > 0, \quad \forall \delta > 0, \quad \exists x \in D = I(a) \setminus \{a\}, \quad |x - a| < \delta \wedge |f(x) - \ell| \geq \varepsilon.$$

Let $\delta = \frac{1}{n}$. Then $\exists x_n \in D = I(a) \setminus \{a\}$, $|x_n - a| < \frac{1}{n} \wedge |f(x_n) - \ell| \geq \varepsilon$. Thus $x_n \xrightarrow{n \rightarrow \infty} a$ and $f(x_n) \not\xrightarrow{n \rightarrow \infty} \ell$ which contradicts the sequential condition. $\square$

3.20 Remark. If there exist two sequences (x_n) and (y_n) in $D \setminus \{a\}$ s.t. $x_n \xrightarrow{n \rightarrow \infty} a \wedge y_n \xrightarrow{n \rightarrow \infty} a$ but $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(y_n)$, then $\lim_{x \rightarrow a} f(x)$ does not exist (in $\overline{\mathbb{R}}$).

3.21 Examples.

1. Show that $\lim_{x \rightarrow \infty} \sin x$ does not exist.

Solution. $x_n = 2\pi n \xrightarrow{n \rightarrow \infty} \infty \wedge y_n = 2\pi n + \frac{\pi}{2} \xrightarrow{n \rightarrow \infty} \infty$ but $\lim_{n \rightarrow \infty} \sin(x_n) = 0 \neq \lim_{n \rightarrow \infty} \sin(y_n) = 1$.

2. Show that $\lim_{x \rightarrow \infty} \cos x$ does not exist.

Solution. $x_n = 2\pi n \xrightarrow{n \rightarrow \infty} \infty \wedge y_n = 2\pi n + \frac{\pi}{2} \xrightarrow{n \rightarrow \infty} \infty$ but $\lim_{n \rightarrow \infty} \cos(x_n) = 1 \neq \lim_{n \rightarrow \infty} \cos(y_n) = 0$.

3. Show that $\lim_{x \rightarrow 0^+} \cos \frac{1}{x}$ does not exist.

Solution. $x_n = \frac{1}{2\pi n} \xrightarrow{n \rightarrow \infty} 0^+ \wedge y_n = \frac{1}{2\pi n + \frac{\pi}{2}} \xrightarrow{n \rightarrow \infty} 0^+$ but $\lim_{n \rightarrow \infty} \cos(\frac{1}{x_n}) = 1 \neq \lim_{n \rightarrow \infty} \cos(\frac{1}{y_n}) = 0$.

4. Show that $\lim_{x \rightarrow \infty} \cos(\ln x)$ does not exist.

Solution. $x_n = e^{2\pi n} \xrightarrow{n \rightarrow \infty} \infty \wedge y_n = e^{2\pi n + \frac{\pi}{2}} \xrightarrow{n \rightarrow \infty} \infty$ but $\lim_{n \rightarrow \infty} \cos(\ln x_n) = 1 \neq \lim_{n \rightarrow \infty} \cos(\ln y_n)$.

3.22 Property. Assume $\lim_{x \rightarrow \alpha} f(x) = \ell$.

1. If ℓ is finite, then f is bounded on some deleted neighborhood $I_r(\alpha) \setminus \{\alpha\}$.

2. If ℓ is finite and $\ell \neq 0$, then $|f(x)| \geq \frac{|\ell|}{2}$ on some deleted neighborhood $I_r(\alpha) \setminus \{\alpha\}$.

Proof. Exercise. $\square$

3.23 Property. Let f and g be functions defined on $I(\alpha) \setminus \{\alpha\}$. The following limit properties hold in $\overline{\mathbb{R}}$ in the sense that if the expression on the right side of the equation exists in $\overline{\mathbb{R}}$, then the limit on the left side exists in $\overline{\mathbb{R}}$ and the equality holds.

1. $\lim_{x \rightarrow \alpha} \lambda f(x) = \lambda \lim_{x \rightarrow \alpha} f(x), \lambda \in \mathbb{R}.$
2. $\lim_{x \rightarrow \alpha} (f(x) + g(x)) = \lim_{x \rightarrow \alpha} f(x) + \lim_{x \rightarrow \alpha} g(x).$
3. $\lim_{x \rightarrow \alpha} (f(x)g(x)) = \lim_{x \rightarrow \alpha} f(x) \lim_{x \rightarrow \alpha} g(x).$
4. $\lim_{x \rightarrow \alpha} (f(x)/g(x)) = \lim_{x \rightarrow \alpha} f(x) / \lim_{x \rightarrow \alpha} g(x)$ provided $\lim_{x \rightarrow \alpha} g(x) \neq 0.$
5. $\lim_{x \rightarrow \alpha} |f(x)| = |\lim_{x \rightarrow \alpha} f(x)|.$ (Hence, $\lim_{x \rightarrow \alpha} f(x) = \ell_1 \iff \lim_{x \rightarrow \alpha} |f(x) - \ell_1| = 0.$)

Proof. The properties follows immediately from the sequential characterization of limit property (3.19) and the algebra of limits of sequences (2.16). However, it is instructive to formulate direct proofs. We do the two methods for part (4).

4. Using the sequential characterization of limit:

Let (x_n) be a sequence in $I(\alpha) \setminus \{\alpha\}$ s.t $x_n \xrightarrow{n \rightarrow \infty} \alpha$. By hypotheses $\lim_{n \rightarrow \infty} g(x_n) \neq 0$. Hence,

$$\lim_{x \rightarrow \alpha} (f(x)/g(x)) = \lim_{n \rightarrow \infty} (f(x_n)/g(x_n)) = \lim_{n \rightarrow \infty} f(x_n) / \lim_{n \rightarrow \infty} g(x_n) = \lim_{x \rightarrow \alpha} f(x) / \lim_{x \rightarrow \alpha} g(x).$$

Using the direct method for the case $\lim_{x \rightarrow a} f(x) = \ell_1 \in \mathbb{R}$ and $\lim_{x \rightarrow a} g(x) = \ell_2 \in \mathbb{R}^*$: Let $\varepsilon > 0$.

Then $|f(x) - \ell_1| < k\varepsilon$ on $I_{\delta_1}(a) \setminus \{a\}$ and $|g(x) - \ell_2| < k\varepsilon$ on $I_{\delta_2}(a) \setminus \{a\}$ for $k > 0$. Now,

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{\ell_1}{\ell_2} \right| &= \left| \frac{\ell_2 f(x) - \ell_1 g(x)}{\ell_2 g(x)} \right| = \frac{|\ell_2 f(x) - \ell_2 \ell_1 + \ell_2 \ell_1 - \ell_1 g(x)|}{|\ell_2 g(x)|} \quad (\text{ we add and subtract } \ell_2 \ell_1) \\ &\leq \frac{|\ell_2 f(x) - \ell_2 \ell_1| + |\ell_2 \ell_1 - \ell_1 g(x)|}{|\ell_2 g(x)|} \quad (\text{ using triangle inequality }) \\ &\leq \frac{|\ell_2| |f(x) - \ell_1| + |\ell_1| |\ell_2 - g(x)|}{|\ell_2| |\ell_2/2|} \quad (\text{ on } I_r(a) \setminus \{a\}, \text{ property(3.22)}) \\ &\leq \frac{|\ell_2| k\varepsilon + |\ell_1| k\varepsilon}{|\ell_2| |\ell_2/2|} = k \left(\frac{2[|\ell_2| + |\ell_1|]}{\ell_2^2} \right) \varepsilon \quad (\text{ on } I_{\min\{r, \delta_1, \delta_2\}}(a) \setminus \{a\}) \\ &\leq \varepsilon. \quad (\text{ take } k = \frac{\ell_2^2}{2[|\ell_2| + |\ell_1|]}). \end{aligned}$$

The proofs for the other cases are similar.

5. and 3. Exercises.

□

3.24 Property. Let f and g be functions defined on $I(\alpha) \setminus \{\alpha\}$ s.t $f(x) \leq g(x) \forall x \in I(\alpha) \setminus \{\alpha\}$.

1. If $\lim_{x \rightarrow \alpha} f(x) = \ell_1$ and $\lim_{x \rightarrow \alpha} g(x) = \ell_2$, then $\ell_1 \leq \ell_2$.
2. If $\lim_{x \rightarrow \alpha} f(x) = \infty$, then $\lim_{x \rightarrow \alpha} g(x) = \infty$.
3. If $\lim_{x \rightarrow \alpha} g(x) = -\infty$, then $\lim_{x \rightarrow \alpha} f(x) = -\infty$.

Proof. Using the sequential characterization of limit:

Let (x_n) be a sequence in $I(\alpha) \setminus \{\alpha\}$ s.t $x_n \xrightarrow{n \rightarrow \infty} \alpha$. By hypotheses we have $f(x_n) \leq g(x_n)$. Then the three items follows from the properties of limits and ordered relation on sequences.

Using the direct method:

1. The case $\alpha = a$.

Require to show; $\ell_1 \leq \ell_2 \iff \ell_1 - \ell_2 \leq 0 \iff \forall \varepsilon > 0, \ell_1 - \ell_2 < \varepsilon$.

Let $\varepsilon > 0$.

Then $|f(x) - \ell_1| < k\varepsilon$ on $I_{\delta_1}(a) \setminus \{a\}$ and $|g(x) - \ell_2| < k\varepsilon$ on $I_{\delta_2}(a) \setminus \{a\}$ for $k > 0$. Now,

$$\begin{aligned} \ell_1 - \ell_2 &= \ell_1 - \cancel{f(x)} + \cancel{f(x)} - \cancel{g(x)} + \cancel{g(x)} - \ell_2 \\ &\leq (\ell_1 - f(x)) + (g(x) - \ell_2) && (\because f(x) - g(x) \leq 0) \\ &\leq |\ell_1 - f(x)| + |g(x) - \ell_2| && (\because \forall X, X \leq |X|) \\ &< 2k\varepsilon && (\text{ on } I_{\min\{\delta_1, \delta_2\}}(a) \setminus \{a\}) \\ &< \varepsilon. && (\text{ take } k = \frac{1}{2}). \end{aligned}$$

The proofs for the other cases are similar.

2. The case $\alpha = \infty$.

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) = \infty &\implies \left(\forall M > 0, \exists A > 0, \forall x \in D, x > A \implies f(x) > M \right) \\ &\implies \left(\forall M > 0, \exists A > 0, \forall x \in D, x > A \implies g(x) \geq f(x) > M \right) \\ &\implies \lim_{x \rightarrow \infty} g(x) = \infty. \end{aligned}$$

The proofs for the other cases are similar.

3. The case $\alpha = -\infty$.

$$\begin{aligned} \lim_{x \rightarrow -\infty} g(x) = -\infty &\implies \left(\forall M < 0, \exists A < 0, \forall x \in D, x < A \implies g(x) < M \right) \\ &\implies \left(\forall M < 0, \exists A < 0, \forall x \in D, x < A \implies f(x) \leq g(x) < M \right) \\ &\implies \lim_{x \rightarrow -\infty} f(x) = -\infty. \end{aligned}$$

The proofs for the other cases are similar.

□

3.25 Property. (Sandwich Rule) Let f, g and h be functions defined on $I(\alpha) \setminus \{\alpha\}$ s.t $f(x) \leq g(x) \leq h(x) \forall x \in I(\alpha) \setminus \{\alpha\}$. If $\lim_{x \rightarrow \alpha} f(x) = \ell$ and $\lim_{x \rightarrow \alpha} h(x) = \ell$ then $\lim_{x \rightarrow \alpha} g(x) = \ell$.

Proof. The cases $\ell = \pm\infty$ done (previous property).

Using the sequential characterization of limit:

Let (x_n) be a sequence in $I(\alpha) \setminus \{\alpha\}$ s.t $x_n \xrightarrow{n \rightarrow \infty} \alpha$. By hypotheses we have $f(x_n) \leq g(x_n) \leq h(x_n)$.

Then by the Sandwich Rule for sequences, we obtain $\lim_{n \rightarrow \infty} g(x_n) = \ell$. Thus $\lim_{x \rightarrow \alpha} g(x) = \ell$.

Using the direct method: The case $\alpha = a$.

Let $\varepsilon > 0$. Then $\ell - \varepsilon < f(x) < \ell + \varepsilon$ for $0 < |x - a| < \delta_1$ and $\ell - \varepsilon < h(x) < \ell + \varepsilon$ for $0 < |x - a| < \delta_2$.

Let $\delta = \min\{\delta_1, \delta_2\}$. Then for $0 < |x - a| < \delta$ we have $\ell - \varepsilon < f(x) \leq g(x) \leq h(x) < \ell + \varepsilon$.

The proofs for the other cases are similar.

□

3.26 Remark. Let f and g be functions defined on $I(\alpha) \setminus \{\alpha\}$. If $f(x)$ is bounded and $\lim_{x \rightarrow \alpha} g(x) = 0$,

then $\lim_{x \rightarrow \alpha} (f(x)g(x)) = 0$. For example $\lim_{x \rightarrow 0} (x \sin \frac{1}{x}) = 0$.

Solution. Since f is bounded on a deleted neighborhood $I(\alpha) \setminus \{\alpha\}$, then $\exists M > 0$ s.t. $\forall x \in I(\alpha) \setminus \{\alpha\}$, $|f(x)| \leq M$. Hence $0 \leq |f(x)g(x)| \leq M|g(x)| \quad \forall x \in I(\alpha) \setminus \{\alpha\}$. By Sandwich rule we obtain $\lim_{x \rightarrow \alpha} |f(x)g(x)| = 0$. Consequently, $\lim_{x \rightarrow \alpha} (f(x)g(x)) = 0$.

3.27 Theorem. (Monotone Function Theorem) Let $a, b \in \overline{\mathbb{R}}$ and $f :]a, b[\mapsto \mathbb{R}$.

1. If f is increasing, then

$$\lim_{x \rightarrow a^+} f(x) = \inf\{f(x) : a < x < b\} \in \overline{\mathbb{R}} \text{ and } \lim_{x \rightarrow b^-} f(x) = \sup\{f(x) : a < x < b\} \in \overline{\mathbb{R}}.$$

2. If f is decreasing, then

$$\lim_{x \rightarrow a^+} f(x) = \sup\{f(x) : a < x < b\} \in \overline{\mathbb{R}} \text{ and } \lim_{x \rightarrow b^-} f(x) = \inf\{f(x) : a < x < b\} \in \overline{\mathbb{R}}.$$

Proof. Exercise. □

3.28 Corollary. Let $a, b \in \overline{\mathbb{R}}$, $a < c < b$ and $f :]a, b[\mapsto \mathbb{R}$. If f is monotone on $]a, b[$, then $\lim_{x \rightarrow c^\pm} f(x)$ exist in $\mathbb{R}$.

Proof. Exercise. □

3.3 Continuous Functions

3.29 Definition. Let $I \subseteq \mathbb{R}$ be open interval, $a \in I$ and $f : I \mapsto \mathbb{R}$ be a function. We say that:

1. f is **continuous at** a if $\lim_{x \rightarrow a} f(x) = f(a)$. Using quantifiers,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in I, \quad |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

2. f is **continuous on** I if f is continuous at each point of I . Using quantifiers,

$$\forall a \in I, \quad \forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in I, \quad |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

3.30 Examples. Using the definition of limit,

1. Show that $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ is continuous at $x = 0$.

Solution.

$$\lim_{x \rightarrow 0} f(x) = f(0) \iff \left(\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in \mathbb{R}, \quad |x - 0| < \delta \implies |f(x) - f(0)| < \varepsilon \right).$$

Let $\varepsilon > 0$. For $|x| < \delta$ given,

$|f(x) - f(0)| = |x| \left| \sin \frac{1}{x} \right| \leq |x| < \delta < \varepsilon$ if we chose $0 < \delta < \varepsilon$. Thus f is continuous at $x = 0$.

2. Show that $f(x) = x^2$ is continuous on $\mathbb{R}$.

Solution.

f is continuous on $\mathbb{R} \iff (\forall a \in \mathbb{R}, \forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon)$.

Let $a \in \mathbb{R}$ and $\varepsilon > 0$. For $|x - a| < \delta$ given,

$$\begin{aligned}
 |f(x) - f(a)| &= |x^2 - a^2| = |x - a||x + a| \\
 &< \delta|x - a + 2a| && \text{(we add } a \text{ and subtract } a) \\
 &< \delta(|x - a| + |2a|) && \text{(using triangle inequality)} \\
 &< \delta(\delta + |2a|) && (\because |x - a| < \delta) \\
 &< \delta(1 + |2a|) && \text{(assume } \delta < 1) \\
 &< \varepsilon && (\delta(1 + |2a|) < \varepsilon \text{ if } \delta < \frac{\varepsilon}{1 + |2a|})
 \end{aligned}$$

Choose $0 < \delta < \min\{1, \frac{\varepsilon}{1 + |2a|}\}$.

3. Show that $f(x) = |x|$ is continuous on $\mathbb{R}$.

Solution.

f is continuous on $\mathbb{R} \iff (\forall a \in \mathbb{R}, \forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon)$.

Let $a \in \mathbb{R}$ and $\varepsilon > 0$. For $|x - a| < \delta$ given,

$$|f(x) - f(a)| = ||x| - |a|| \leq |x - a| < \delta < \varepsilon \quad (\text{if } \delta < \varepsilon)$$

Choose $0 < \delta < \varepsilon$.

3.31 Definition. Let $I \subseteq \mathbb{R}$ be an interval of any type, $a \in I$ and $f : I \mapsto \mathbb{R}$. We say that:

1. f is **continuous from right at** a if $\lim_{x \rightarrow a^+} f(x) = \lim_{\substack{x \rightarrow a \\ x > a}} f(x) = f(a)$. Using quantifiers,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in I, \quad 0 < x - a < \delta \implies |f(x) - f(a)| < \varepsilon.$$

2. f is **continuous from left at** a if $\lim_{x \rightarrow a^-} f(x) = \lim_{\substack{x \rightarrow a \\ x < a}} f(x) = f(a)$. Using quantifiers,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in I, \quad 0 < a - x < \delta \implies |f(x) - f(a)| < \varepsilon.$$

All properties of limit of functions can be restated for continuous functions. By continuity at a^+ we mean continuity from right at a and by continuity at a^- we mean continuity from left at a . Also $x_n \xrightarrow[n \rightarrow \infty]{} a^+$ means $x_n \xrightarrow[n \rightarrow \infty]{} a$ and $x_n > a$ eventually. Analogously, $x_n \xrightarrow[n \rightarrow \infty]{} a^-$ means $x_n \xrightarrow[n \rightarrow \infty]{} a$ and $x_n < a$ eventually.

3.32 Property. Assume $\alpha \in \{a, a^+, a^-\}$.

1. Let f, g be continuous at α and $\lambda \in \mathbb{R}$. Then

(a) λf is continuous at α .

(b) $f + g$ is continuous at α .

(c) fg is continuous at α .

(d) $\frac{f}{g}$ is continuous at α provided that $g(\alpha) \neq 0$.

2. f is continuous at $a \iff f$ is continuous from right and from left at a .

3. f is continuous at $\alpha \iff \left(\text{for every sequence } (x_n) \text{ in } I, x_n \xrightarrow{n \rightarrow \infty} \alpha \implies f(x_n) \xrightarrow{n \rightarrow \infty} f(\alpha) \right)$.

Constant functions and the function $f(x) = x$ are clearly continuous on $\mathbb{R}$. It follows from previous property that polynomials are continuous on $\mathbb{R}$ and rational functions are continuous on their domain of definition.

3.33 Property. Let $f : I \mapsto \mathbb{R}$ be continuous at a and g defined in a neighborhood of $f(a)$ and continuous at $f(a)$. Then the composite $g \circ f$ is continuous at a .

Proof. $g \circ f$ is continuous at $a \iff \left(\forall (x_n) \text{ in } I, \lim_{n \rightarrow \infty} x_n = a \implies \lim_{n \rightarrow \infty} g(f(x_n)) = g(f(a)) \right)$.
Let (x_n) be a sequence in I s.t $x_n \xrightarrow{n \rightarrow \infty} a$.

$$\begin{aligned} x_n \xrightarrow{n \rightarrow \infty} a &\implies y_n = f(x_n) \xrightarrow{n \rightarrow \infty} f(a) && (\because f \text{ is continuous at } a) \\ &\implies g(y_n) = g(f(x_n)) \xrightarrow{n \rightarrow \infty} g(f(a)) && (\because g \text{ is continuous at } f(a)) \end{aligned}$$

Using $\varepsilon - \delta$: Claim $g \circ f(x) \xrightarrow{x \rightarrow a} g(f(a))$.

Let $\varepsilon > 0$. Since g is continuous at $f(a)$, then there exist $\delta_1 > 0$ such that

$$|y - f(a)| < \delta_1 \xrightarrow{(\star)} |g(y) - g(f(a))| < \varepsilon.$$

Since $\delta_1 > 0$ and f is continuous at a , then there exists $\delta > 0$ such that

$$|x - a| < \delta \xrightarrow{(\star\star)} |f(x) - f(a)| < \delta_1.$$

Hence, there exists $\delta > 0$ such that

$$\forall x \in D_{g \circ f}, \quad |x - a| < \delta \xrightarrow{(\star\star)} |f(x) - f(a)| < \delta_1 \xrightarrow{(\star)} |g(f(x)) - g(f(a))| < \varepsilon.$$

□

Note that; $x \in D_{g \circ f} \iff (x \in D_f \wedge f(x) \in D_g)$. The following property will be frequently used.

3.34 Property. All elementary functions "polynomials, rational functions, power functions, trigonometric functions, exponential functions, hyperbolic functions" and their inverses are continuous over their entire domains. (Chapter Five)

3.35 Example. Show that $f(x) = \begin{cases} \sin x + e^x & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ x^3 + x^2 + |\cos x| & \text{if } x > 0. \end{cases}$ is continuous on $\mathbb{R}$.

Solution.

- f is continuous $] -\infty, 0[$ as sum of continuous functions.
- f is continuous $]0, \infty[$ as composite, product and sum of continuous functions.
- Continuity of f at 0:

$\lim_{x \rightarrow 0^-} f(x) = \lim_{\substack{x \rightarrow 0 \\ x < 0}} f(x) = \lim_{x \rightarrow 0} (\sin x + e^x) = 1 = f(0)$. Hence f is continuous from left at 0.

$\lim_{x \rightarrow 0^+} f(x) = \lim_{\substack{x \rightarrow 0 \\ x > 0}} f(x) = \lim_{x \rightarrow 0} (x^3 + x^2 + |\cos x|) = 1 = f(0)$. Hence f is continuous from right at 0.

Therefore f is continuous at 0 and thus f is continuous on $\mathbb{R}$.

3.36 Definition. A function $f : [a, b] \mapsto \mathbb{R}$ is continuous on $[a, b]$ closed if the following holds

- f is continuous on $]a, b[$.
- f is continuous from right at a .
- f is continuous from left at b .

Similarly, f is continuous on $[a, b[$ if f is continuous on $]a, b[$ and f is continuous from right at a . Also f is continuous on $]a, b]$ if f is continuous on $]a, b[$ and f is continuous from left at b .

3.37 Definition. Let I be an interval, $a \in I$ and $f : I \setminus \{a\} \mapsto \mathbb{R}$. We say that f has an extension by continuity at $x = a \in \{a, a^+, a^-\}$ if $\lim_{x \rightarrow a} f(x) = \ell$ with ℓ finite and we define its extension by

continuity $\tilde{f} : I \mapsto \mathbb{R}$ by $\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in I \setminus \{a\} \\ \ell & \text{if } x = a. \end{cases}$

Moreover if f is continuous on $I \setminus \{a\}$, then $\tilde{f}$ is continuous on I .

For example, the function $f(x) = 1 + x \sin(\frac{1}{x})$ defined on $\mathbb{R}^*$ has an extension by continuity at $x = 0$ because $\lim_{x \rightarrow 0} f(x) = 1$ is a finite number and its extension $\tilde{f}(x) = \begin{cases} 1 + x \sin(\frac{1}{x}) & \text{if } x \in \mathbb{R} \setminus \{0\} \\ 1 & \text{if } x = 0. \end{cases}$

3.4 Properties of Continuous Functions

3.38 Property. If $f : D \rightarrow \mathbb{R}$ is strictly monotone on its domain, then it is injective.

Proof. Let $x, x' \in D$ s.t $x \neq x'$.

$$\begin{aligned} x \neq x' &\implies x > x' \vee x < x' \\ &\implies f(x) > f(x') \vee f(x) < f(x') && (f \text{ is strictly monotone}) \\ &\implies f(x) \neq f(x'). \end{aligned}$$

Thus f is injective. □

3.39 Theorem. (Intermediate Value Theorem- First Form)[I.V.T]

Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function on the closed bounded interval $[a, b]$. If $f(a)f(b) < 0$ i.e., $f(a) < 0 < f(b) \vee f(b) < 0 < f(a)$, then there exists $c \in]a, b[$ such that $f(c) = 0$.

Moreover; if f is strictly monotone on $[a, b]$, then c is unique.

Proof. Assume $f(a) < 0$ and $f(b) > 0$ and let $A := \{x \in [a, b] : f(x) \leq 0\}$.

- Since $a \in A$ and A bounded above by b , then $\sup A$ exists in $\mathbb{R}$. Let $c := \sup A \in [a, b]$. Then there exists a sequence (x_n) in A s.t $x_n \xrightarrow{n \rightarrow \infty} c$.
- Claim $f(c) \leq 0$:

$$\begin{aligned} x_n \xrightarrow{n \rightarrow \infty} c \text{ and } f \text{ continuous at } c &\implies \lim_{n \rightarrow \infty} f(x_n) = f(c) \\ &\implies f(c) \leq 0 \quad ((x_n) \text{ in } A \text{ implies } f(x_n) \leq 0) \end{aligned}$$

- Claim $f(c) \geq 0$:

Since $f(b) > 0$, then $c \neq b$. By the definition of c we have $\forall x \in]c, b[$, $f(x) > 0$. Since f is continuous at c and by applying the limit both sides as $x \rightarrow c^+$ to the previous equality, we obtain $f(c) \geq 0$. Thus there exists $c \in]a, b[$ such that $f(c) = 0$.

Uniqueness of c : Assume that there exists $c' \in]a, b[$ s.t $f(c') = 0$.

$$f(c) = f(c') \implies c = c' \quad (f \text{ is injective, property 3.38}).$$

□

Note that, if we replace $f(a)f(b) < 0$ by $f(a)f(b) \leq 0$, then $c \in [a, b]$. Also if we replace $f(a) < 0 < f(b)$ by $f(a) \leq 0 < f(b)$, then $c \in [a, b[$.

3.40 Example. Show that $f(x) = x^{25} - \frac{1}{2}$ has a unique root in $]0, 1[$.

Solution. We have:

- f is continuous on $[0, 1]$. (f is a polynomial, continuous on $\mathbb{R}$ hence continuous on $[0, 1] \subset \mathbb{R}$)
- $f(0)f(1) = (-\frac{1}{2})(\frac{1}{2}) < 0$.

Then by I.V.T there exist $c \in]0, 1[$ s.t $f(c) = 0$. Since f is strictly increasing, then c is unique.

3.41 Theorem. (Intermediate Value Theorem- Second Form)[I.V.T]

Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function on the closed bounded interval $[a, b]$. If $f(a) < T < f(b) \vee f(b) < T < f(a)$, then there exists $c \in]a, b[$ such that $f(c) = T$.

Moreover; if f is strictly monotone on $[a, b]$, then c is unique.

Proof. Apply the previous theorem to the function $g : [a, b] \mapsto \mathbb{R}$ defined by $g(x) = f(x) - T$. □

3.42 Corollary. Let I be an interval and $f : I \mapsto \mathbb{R}$. If f is continuous, then $f(I)$ is an interval.

Proof. Let $y_1, y_2 \in f(I)$ and $y_1 < z < y_2$. Claim $z \in f(I)$.

Since $y_1, y_2 \in f(I)$, then there exists $x_1, x_2 \in I$ s.t $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Since $y_1 < y_2$, then $x_1 \neq x_2$. Assume $x_1 < x_2$ and let us try I.V.T for f on $[x_1, x_2]$.

- f is continuous on $[x_1, x_2]$. (f is continuous on I hence continuous on $[x_1, x_2] \subset I$)
- $f(x_1) < z < f(x_2)$.

Then by I.V.T there exist $c \in]x_1, x_2[\subset I$ s.t $f(c) = z$. Thus $z \in f(I)$. □

We will see in the next theorem that if f is continuous on $[a, b]$, then $f([a, b]) = [m, M]$ where $m = \min_{x \in [a, b]} f(x)$ and $M = \max_{x \in [a, b]} f(x)$. In case I is open or half-open interval, its image $f(I)$ can be an interval of any kind. For example, take $f(x) = \sin x$: If $I =]0, 2\pi[$, then $f(I) =]-1, 1[$. If $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$, then $f(I) =]-1, 1[$. If $I =]-\frac{\pi}{2}, \frac{3\pi}{2}[$, then $f(I) =]-1, 1[$. Take $g(x) = \tan x$: if $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$, then $g(I) =]-\infty, \infty[$.

3.43 Theorem. (*Extreme Value Theorem*)

If f is continuous on a closed bounded interval $[a, b]$, then f is bounded and attains its maximum and minimum i.e., $\exists c, d \in [a, b]$ s.t. $\min_{x \in [a, b]} f(x) = f(c)$ and $\max_{x \in [a, b]} f(x) = f(d)$. Moreover $f([a, b]) = [f(c), f(d)]$ i.e.,

$$f : [a, b] \mapsto f([a, b]) = [f(c), f(d)].$$

Proof.

Claim f is bounded: Suppose to the contrary that f is not bounded. Then for each $n \in \mathbb{N}$ there exists $u_n \in [a, b]$ s.t. $|f(u_n)| > n$. Since u_n is bounded, then by Bolzano–Weierstrass theorem, there exists a sub sequence u_{n_k} converging to $x_0 \in [a, b]$. By the continuity of $|f|$, we get $|f(u_{n_k})|$ converges to $|f(x_0)|$. Hence $|f(u_{n_k})|$ is a bounded sequence, a contradiction.

Claim f attains its maximum and minimum: Since f is bounded, then $\sup_{x \in [a, b]} f(x)$ is finite. Let

$M := \sup_{x \in [a, b]} f(x)$. By the sequential characterization of supremum, there exists a sequence $x_n \in [a, b]$

s.t. $f(x_n) \xrightarrow{n \rightarrow \infty} M$. By the Bolzano–Weierstrass theorem, there exists a sub sequence x_{n_k} converging to some $d \in [a, b]$. Since f is continuous, then $M = f(d)$. Therefore $\max_{x \in [a, b]} f(x) = f(d)$. The proof

for the minimum is similar.

Claim $f([a, b]) = [f(c), f(d)]$: By the previous corollary, $f([a, b])$ is an interval. Since $f(c)$ is the lower end point and $f(d)$ is the upper end point, then $f([a, b]) = [f(c), f(d)]$. $\square$

3.44 Remark. A function $f : D \mapsto f(D)$ is surjective (onto).

3.45 Theorem. (*Inverse Function Theorem-1*)

Let I be an interval of $\mathbb{R}$ and $f : I \mapsto f(I) = J$. Suppose that ① f is continuous on I and ② f is strictly monotone on I , then ① $J = f(I)$ is an interval, ② $f : I \mapsto J$ is bijective (hence invertible) and ③ $f^{-1} : J \mapsto I$ is continuous and strictly monotone as f . Moreover, the representative curve of f^{-1} is obtained from the representative curve of f by symmetry with respect to the first bisector $y = x$.

Proof. ① The image of an interval by a continuous function is an interval.

② Since f is strictly monotone, then it is injective. But $f : I \mapsto f(I) = J$ is surjective. Thus f defines a bijective map from I onto J .

③ Assume $f : I \mapsto J$ is strictly increasing and let us show that $f^{-1} : J \mapsto I$ is strictly increasing. Let $y_1, y_2 \in J$ s.t. $y_1 < y_2$, then there exist $x_1, x_2 \in I$ s.t. $f(x_1) = y_1$ and $f(x_2) = y_2$. Claim that $f^{-1}(y_1) < f^{-1}(y_2)$: Suppose to the contrary that $f^{-1}(y_1) \geq f^{-1}(y_2)$ i.e., $x_1 \geq x_2$. Since f is strictly increasing, then $f(x_1) \geq f(x_2)$ i.e., $y_1 \geq y_2$, a contradiction. For continuity of f^{-1} on $f(I)$: Suppose that $y_1 = f(x_1) \in f(I)$ and $\varepsilon > 0$ are given. Assume that x_1 is not an end point of I ; a slight change will deal with an end point. Choose $x_0, x_2 \in I$ s.t. $x_1 - \varepsilon < x_0 < x_1 < x_2 < x_1 + \varepsilon$. Set $y_j = f(x_j)$ and take $0 < \delta < \min\{y_2 - y_1, y_1 - y_0\}$. Then $|y - y_1| < \delta$ implies $y_0 < y < y_2$ and $x_1 - \varepsilon < x_0 < f^{-1}(y) < x_2 < x_1 + \varepsilon$. $\square$

Note that if f is continuous and strictly monotone on an interval I , then $f(I)$ is also an interval; namely,

If f is strictly increasing	$2. I =]a, b] \implies f(I) =] \lim_{x \rightarrow a^+} f(x), f(b)].$
1. $I = [a, b] \implies f(I) = [f(a), f(b)]$.	$3. I = [a, b[\implies f(I) = [f(a), \lim_{x \rightarrow b^-} f(x)[.$

4. $I =]a, b[\implies f(I) =] \lim_{x \rightarrow a^+} f(x), \lim_{x \rightarrow b^-} f(x)[.$ If f is strictly decreasing 1. $I = [a, b] \implies f(I) = [f(b), f(a)].$	2. $I =]a, b] \implies f(I) = [f(b), \lim_{x \rightarrow a^+} f(x)[.$ 3. $I = [a, b[\implies f(I) =] \lim_{x \rightarrow b^-} f(x), f(a)].$ 4. $I =]a, b[\implies f(I) =] \lim_{x \rightarrow b^-} f(x), \lim_{x \rightarrow a^+} f(x)[.$
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3.5 Substitution Theorem

3.46 Theorem. (Substitution Theorem)

Suppose $\lim_{x \rightarrow \alpha} f(x) = \ell \in \overline{\mathbb{R}}$ and g defined in a neighborhood of ℓ excluding possibly the point ℓ such that one of the following three conditions satisfied,

- ① $\ell \in \mathbb{R}$ and g is continuous at ℓ ,
- ② $\ell = \pm\infty$ and $\lim_{y \rightarrow \ell} g(y) \in \mathbb{R}$.
- ③ $\ell \in \mathbb{R}, \quad f(x) \neq \ell \quad \forall x \in I(\alpha) \setminus \{\alpha\}$ and $\lim_{y \rightarrow \ell} g(y) \in \overline{\mathbb{R}}$.

Then $\lim_{x \rightarrow \alpha} g(f(x)) = \lim_{y \rightarrow \ell} g(y)$.

Proof. Exercise.” You may do it by two ways as property 3.33” □

- In case ①, $\lim_{x \rightarrow \alpha} g(f(x)) = \lim_{y \rightarrow \ell} g(y) = g(\ell) = g(\lim_{x \rightarrow \alpha} f(x))$.
- The theorem fails if $\ell \in \mathbb{R}$ and ① and ② do not hold i.e. g is not continuous at ℓ and $f(x) = \ell$ for infinitely many times in every deleted neighborhood of α . A counterexample is given by the functions $f(x) = 0$ and $g(y) = \begin{cases} 1 & \text{if } y \neq 0 \\ 3 & \text{if } y = 0. \end{cases}$ We have $\lim_{x \rightarrow 0} g(f(x)) = 3 \neq 1 = \lim_{y \rightarrow 0} g(y)$.

3.6 More About Limit

In this section we give a definition of the limit including the limits encountered in section 3.2.

- An accumulation point of a nonempty set D of real numbers is an extended real number α such that every neighborhood of α contains a point of D not equal to α .
- A point of D that is not an accumulation point of D is called an isolated point of D .
- α is an isolated point of D if $\alpha \in D$ and there is a neighborhood of α that contains no other point of D .
- Let f be a real valued function defined on a non empty subset D of $\mathbb{R}$. Let $\alpha, \ell \in \overline{\mathbb{R}}$ where either α is an accumulation point of D or α is an isolated point of D . We say that $f(x)$ tends to ℓ as x tends to α along D and we write $\lim_{\substack{x \rightarrow \alpha \\ x \in D}} f(x) = \ell$ if

$$\forall I(\ell), \quad \exists \mathbb{I}(\alpha), \quad \forall x \in D, \quad x \in \mathbb{I}(\alpha) \implies f(x) \in I(\ell).$$

- the limit ℓ is unique.
- The definition of limit as $x \rightarrow \alpha$ where $\alpha \in \{a, a^+, a^-, \infty, -\infty\}$ in section 3.2 is a particular case of this definition as follows: $\lim_{x \rightarrow \alpha} f(x) = \lim_{\substack{x \rightarrow \alpha \\ x \in I_\alpha \setminus \{\alpha\}}} f(x)$.
- Let f be a real valued function defined on a non empty subset D and $D_1 \cup D_2 = D$. Suppose that one of the following four possibilities hold: a is an accumulation point (or isolated point) of D_1 and a is an accumulation point (or isolated point) of D_2 . Then, $\lim_{\substack{x \rightarrow a \\ x \in D}} f(x)$ exists in $\overline{\mathbb{R}}$ iff both limits $\lim_{\substack{x \rightarrow a \\ x \in D_1}} f(x)$ and $\lim_{\substack{x \rightarrow a \\ x \in D_2}} f(x)$ exist in $\overline{\mathbb{R}}$ and are equal. In this case all three limits are equal.

- ★ For example: Let $D_1 = \mathbb{N}$ and $D_2 =]0, 1[$. Then 1 is an isolated point of D_1 and an accumulation point of D_2 , and $\lim_{\substack{x \rightarrow 1 \\ x \in D_1}} f(x) = f(1)$. Therefore $\lim_{x \rightarrow 1} f(x)$ exists iff $\lim_{\substack{x \rightarrow 1 \\ x \in D_1}} f(x) = \lim_{\substack{x \rightarrow 1 \\ x \in D_2}} f(x) = f(1)$.
- ★ In particular: $\lim_{x \rightarrow a} f(x)$ exists in $\overline{\mathbb{R}}$ iff $\lim_{x \rightarrow a^\pm} f(x)$ exist and are equal. In this case all three limits are equal.
- All other properties, like sequential characterization, algebra of limits, Sandwich Rule can be proved analogously as the ones proved in section 3.2.

3.7 Exercises

Exercise 1. Let $f(x) = \frac{E(\frac{1}{x})+x}{E(\frac{1}{x})-x}$.

1. Find the domain of definition of f .
2. Show that $\forall x \in]0, 1/2[, \quad 1 < f(x) \leq \frac{1+x^2}{1-x^2-x}$.
3. Using the definition of the limit of functions, show that $\lim_{x \rightarrow 0^+} f(x) = 1$.

Exercise 2. Let $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$. Using the density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$, show that f is discontinuous at each real number $a \in \mathbb{R}$.

Exercise 3. Show that if f is continuous at a and $f(a) \neq 0$, then f is different from zero in a neighborhood of a .

Exercise 4. Let $f : \mathbb{R} \mapsto \mathbb{R}$ be periodic function with period T .

1. Assume f is continuous on $\mathbb{R}$.
 - (a) Let $x \in \mathbb{R}$. Show that $f(x + nT) = f(x) \quad \forall n \in \mathbb{Z}$.
 - (b) Show that there exists $c, d \in [0, T]$ such that $f(c) \leq f(x) \leq f(d)$ for all $x \in [0, T]$.
 - (c) Let $x \in \mathbb{R}$ and $n_0 = E\left(\frac{x}{T}\right)$. Show that $0 \leq x - n_0T < T$.
 - (d) Deduce that f is bounded and attains its maximum and minimum on $\mathbb{R}$.
2. Assume f is a non constant function and let $a, b \in \mathbb{R}$ such that $f(a) \neq f(b)$. Take $x_n = a + nT$ and $y_n = b + nT$. Using the sequential characterization of limit, show that $\lim_{x \rightarrow \infty} f(x) \nexists$.

Exercise 5. Let the function $f(x) = \frac{x - \sqrt{2x - 1}}{x - 1}$. Consider the two sets

$$A = \{f(x), \frac{1}{2} \leq x < 1\} \quad \text{and} \quad B = \{f(x), x > 1\}.$$

1. Find the domain of definition of the function f .
2. Using the definition of the limit of a function, show that $\lim_{x \rightarrow 1} f(x) = 0$.
3. Show that $f(x) \leq 0$ for all $\frac{1}{2} \leq x < 1$ and $0 \leq f(x)$ for $x > 1$.
4. Deduce that $\sup A$ and $\inf B$ exist.
5. Deduce that $f(x) \leq \sup A \leq 0$ for all $\frac{1}{2} \leq x < 1$ and $0 \leq \inf B \leq f(x)$ for $x > 1$.
6. Determine the values of $\sup A$ and $\inf B$.
7. Do $\max A$ and $\min B$ exist? Justify your answer.

8. Using the ε - characterization of sup and inf, show that $\sup A = 0$ and $\inf B = 0$.

Exercise 6. Let $f : [a, b] \mapsto [a, b]$ be a continuous function.

1. Show that f has a fixed point $c \in [a, b]$ i.e., $f(c) = c$.

2. Show that c is unique in each of the following cases:

(a) f is decreasing.

(b) There exists $k \in [0, 1[$ s.t $\forall x, y \in [a, b], \quad |f(x) - f(y)| \leq k|x - y|$.

Exercise 7.(Generalization of the I.V.T- First Form)

1. Let I be an interval and $f : I \mapsto \mathbb{R}$ be a continuous function. Assume that the limits (finite $\neq 0$ or infinite) of $f(x)$ as x tends to the end points of I is of different signs.

(a) Show that there exist $c \in I$ s.t $f(c) = 0$.

(b) Deduce that if f is strictly monotone on I , then c is unique.

2. Deduce that $f(x) = x + \sin x + \ln x$ has exactly one zero in $]0, \infty[$.

3. Deduce that a polynomial of odd degree has a real root.

Exercise 8. (Generalization of the I.V.T- Second Form)

Let f and g be two continuous functions on the closed bounded interval $[a, b]$ s.t $f(a) < g(a)$ and $f(b) > g(b)$.

1. Show that there exists at least one point $c \in]a, b[$ s.t $f(c) = g(c)$.

2. Deduce the I.V.T- Second Form.

3. Consider g or f to be the identity function. What you may deduce.

Exercise 9. Let $f, g : [a, b] \mapsto \mathbb{R}$ be continuous functions s.t $f(x) < g(x), \quad \forall x \in [a, b]$. Show that $\exists \alpha > 0$ s.t $f(x) \leq g(x) - \alpha, \quad \forall x \in [a, b]$.

Exercise 10. Let $f : [0, 1] \mapsto \mathbb{R}$ be continuous function and $\alpha, \beta > 0$. Show that there exists $c \in [0, 1]$ s.t $\alpha f(0) + \beta f(1) = (\alpha + \beta)f(c)$.

Exercise 11. (Existence of the nth root)

Let $a > 0, n \in \mathbb{N}^*$ and $f(x) = x^n$.

1. Show that there exists $b > 0$ s.t $f(b) > a$. (Hint: You may use $\lim_{x \rightarrow \infty} f(x) = \infty$)

2. Deduce that there exists a unique $c \in]0, b[$ (c positive) s.t $f(c) = a$.

Exercise 12. Let f be continuous and positive on $\mathbb{R}$ with $\lim_{x \rightarrow \pm\infty} f(x) = 0$. Show that f attains its maximum on $\mathbb{R}$.

Exercise 13. Let f be continuous on $\mathbb{R}$ with $\lim_{x \rightarrow \pm\infty} f(x) = \infty$. Show that f attains its minimum on $\mathbb{R}$.

Exercise 14. Let $f :]2, \infty[\mapsto \mathbb{R}$ be a function defined by $f(x) = \frac{x-3}{x-2}$. Show that f has an inverse function f^{-1} to be determined.

Exercise 15. Let $f : [1, \infty[\mapsto \mathbb{R}$ be a function defined by $f(x) = \sqrt{x-1}$. Show that f has an inverse function f^{-1} to be determined.

Exercise 16. Let $f : [2, \infty[\mapsto \mathbb{R}$ be a function defined by $f(x) = \frac{1}{\sqrt{x^2-4x+8}}$. Show that f has an inverse function f^{-1} to be determined.

Exercise 17. Let $f, g : [0, 1] \mapsto [0, \infty[$ be continuous functions satisfying $\sup_{0 \leq x \leq 1} f(x) = \sup_{0 \leq x \leq 1} g(x)$.

Show that there exists $x_0 \in [0, 1]$ s.t $f(x_0) = g(x_0)$.

Exercise 18. Prove property 3.3, property 3.22, property 3.23, theorem 3.27 and corollary 3.28.

Exercise 19. Let $u_n = \frac{n^2+n+\sin n}{n^2+n+1}$ and $f(x) = \begin{cases} \frac{1+x+x^2 \sin \frac{1}{x} \cos(\frac{2\pi}{x})}{1+x+x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

1. Show that $\lim_{n \rightarrow \infty} u_n = 1$ and $\lim_{n \rightarrow \infty} f(\frac{1}{n}) = 1$.

2. Deduce that f is not continuous at 0.

Exercise 20.(Continuity of sin and cos on $\mathbb{R}$).

1. Show that $|\sin x| \leq |x|$ for all $x \in \mathbb{R}$. (Hint: Graphically consider the unit circle for the case $0 < x \leq \frac{\pi}{2}$.)

2. Using the definition of the limit of functions, show that $x \mapsto \sin x$ is continuous on $\mathbb{R}$. (Hint: Use the identity $\sin x - \sin x_0 = 2 \sin \frac{x-x_0}{2} \cos \frac{x+x_0}{2}$.)

3. Deduce that $x \mapsto \cos x$ is continuous on $\mathbb{R}$.

Exercise 21. (Fundamental limit)

1. Show that $\sin x < x < \tan x$ for all $0 < x \leq \frac{\pi}{2}$. (Hint: Graphically consider the unit circle.)

2. Deduce that $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

Exercise 22. (Fundamental limits)

1. Let $a_n = (1 + \frac{1}{n})^n$, $n \in \mathbb{N}^*$.

(a) Show that $a_n = \sum_{k=0}^n \frac{1}{k!} \cdot 1 \cdot (1 - \frac{1}{n}) \cdots (1 - \frac{k-1}{n})$. Deduce that (a_n) is strictly increasing and $2 < a_n$ for $n \geq 2$.

(b) Show that $a_n < 3$ for all $n \in \mathbb{N}^*$. (Hint: Use $k! \geq 2^{k-1}$.)

(c) Show that the sequence (a_n) is convergent and $2 \leq \lim_{n \rightarrow \infty} a_n \leq 3$.

2. Let $e := \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$.

(a) Let $n = E(x)$ for $x > 0$. Show that $(1 + \frac{1}{n+1})^n \leq (1 + \frac{1}{x})^x \leq (1 + \frac{1}{n})^{n+1}$.

(b) Deduce that $\lim_{x \rightarrow \pm\infty} (1 + \frac{1}{x})^x = e$. (Hint: $x = -|x|$ for $x < 0$.)

(c) Deduce that $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$. (Hint: Use $\frac{\ln(1+x)}{x} = \ln(1+x)^{\frac{1}{x}}$ and the substitution theorem.)

(d) Deduce that $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$. (Hint: put $e^x - 1 = y$.)

(e) Deduce that $\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$ ($\alpha \in \mathbb{R}$). (Hint: put $1+x = e^y$.)

4.1 First Derivative: Properties And Examples

4.1 Definition. Let I be an open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$. We say that:

1. f is differentiable at x_0 if $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exists and finite. In this case the real number

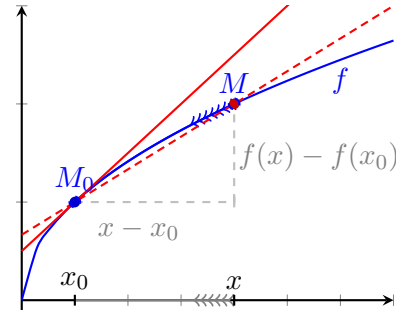
$$f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

is called the (first) derivative of f at x_0 .

2. f is differentiable on I if f is differentiable at each point of I .

Graphical interpretation of derivative:

- $\frac{f(x) - f(x_0)}{x - x_0}$ represents the slope of the line passing through the points $M_0(x_0, f(x_0))$ and $M(x, f(x))$.
- As x approaches x_0 , M approaches M_0 .
- Hence $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ represents the slope of the tangent line to the representative curve of f at the point $M_0(x_0, f(x_0))$.



Graphical interpretation of derivative.

4.2 Property. (Differentiability $\implies$ Continuity)

Let I be open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$. If f is differentiable at x_0 , then f is continuous at x_0 .

Proof. $\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \right) = f'(x_0) \cdot 0 = 0.$ □

The converse of this property is not true. For example, the function $f(x) = |x|$ is continuous but not differentiable at $x = 0$. ($\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x} = \pm 1$, hence does not exist.)

The contrapositive of this property: If f is not continuous at x_0 , then it is not differentiable at x_0 .

4.3 Examples. Differentiability for some usual functions.

1. A constant function $f(x) = c$ is differentiable on $\mathbb{R}$ and $f'(x) = 0$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary. $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{c - c}{x - x_0} = 0$ exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = 0$ for all $x_0 \in \mathbb{R}$.

2. An affine function $f(x) = ax + b$ is differentiable on $\mathbb{R}$ and $f'(x) = a$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary. $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{ax - ax_0}{x - x_0} = a$ exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = a$ for all $x_0 \in \mathbb{R}$.

3. The function $f(x) = x^2$ is differentiable on $\mathbb{R}$ and $f'(x) = 2x$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary.

• $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{x^2 - x_0^2}{x - x_0} = \lim_{x \rightarrow x_0} \frac{(x - x_0)(x + x_0)}{x - x_0} = \lim_{x \rightarrow x_0} (x + x_0) = 2x_0$ exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = 2x_0$ for all $x_0 \in \mathbb{R}$.

• $\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0} \frac{(x_0 + h)^2 - x_0^2}{h} = \lim_{h \rightarrow 0} \frac{2x_0h + h^2}{h} = \lim_{h \rightarrow 0} (2x_0 + h) = 2x_0$ exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = 2x_0$ for all $x_0 \in \mathbb{R}$.

4. The function $f(x) = x^n$, $n \in \mathbb{N}^*$, is differentiable on $\mathbb{R}$ and $f'(x) = nx^{n-1}$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{(x_0 + h)^n - x_0^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} h^k x_0^{n-k} - x_0^n}{h} && \text{(binomial formula)} \\ &= \lim_{h \rightarrow 0} \frac{x_0^n + (n-1)hx_0^{n-1} + \sum_{k=2}^n \binom{n}{k} h^k x_0^{n-k} - x_0^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{(n-1)hx_0^{n-1} + \sum_{k=2}^n \binom{n}{k} h^k x_0^{n-k}}{h} \\ &= (n-1)x_0^{n-1} \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = (n-1)x_0^{n-1}$ for all $x_0 \in \mathbb{R}$.

5. The function $f(x) = x^\alpha$, $\alpha \in \mathbb{R}$, is differentiable on its domain of definition and $f'(x) = \alpha x^{\alpha-1}$.

Note that $D_f = \mathbb{R} \vee \mathbb{R}^* \vee \mathbb{R}_+^*$ depending on $\alpha \in \mathbb{N} \vee \mathbb{Z} \vee \mathbb{R}$.

Solution. Let $x_0 \neq 0$ be in D_f .

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{(x_0 + h)^\alpha - x_0^\alpha}{h} \\ &= \lim_{h \rightarrow 0} \frac{x_0^\alpha (1 + h/x_0)^\alpha - x_0^\alpha}{h} \\ &= x_0^{\alpha-1} \lim_{h \rightarrow 0} \frac{(1 + h/x_0)^\alpha - 1}{h/x_0} && \text{(fundamental limit)} \\ &= \alpha x_0^{\alpha-1} \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = \alpha x_0^{\alpha-1}$ for all $x_0 \in D_f$.

6. The function $f(x) = \sin x$ is differentiable on $\mathbb{R}$ and $f'(x) = \cos x$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{\sin(x_0 + h) - \sin x_0}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \sin(h/2) \cos(x_0 + h/2)}{h} \quad (\sin x - \sin x_0 = 2 \sin \frac{x - x_0}{2} \cos \frac{x + x_0}{2}) \\ &= \lim_{h \rightarrow 0} \frac{\sin(h/2) \cos(x_0 + h/2)}{h/2} \\ &= \cos x_0 \quad (\text{fundamental limit \& continuity of } \cos) \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = \cos x_0$ for all $x_0 \in \mathbb{R}$.

7. The function $f(x) = \cos x$ is differentiable on $\mathbb{R}$ and $f'(x) = -\sin x$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{\cos(x_0 + h) - \cos x_0}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2 \sin(h/2) \sin(x_0 + h/2)}{h} \quad (\cos x - \cos x_0 = -2 \sin \frac{x - x_0}{2} \sin \frac{x + x_0}{2}) \\ &= - \lim_{h \rightarrow 0} \frac{\sin(h/2) \sin(x_0 + h/2)}{h/2} \\ &= -\sin x_0 \quad (\text{fundamental limit \& continuity of } \sin) \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = -\sin x_0$ for all $x_0 \in \mathbb{R}$.

8. The function $f(x) = e^x$ is differentiable on $\mathbb{R}$ and $f'(x) = e^x$ for all $x \in \mathbb{R}$.

Solution. Let $x_0 \in \mathbb{R}$ arbitrary.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{e^{(x_0 + h)} - e^{x_0}}{h} \\ &= e^{x_0} \lim_{h \rightarrow 0} \frac{e^h - 1}{h} \\ &= e^{x_0} \quad (\text{fundamental limit}) \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = e^{x_0}$ for all $x_0 \in \mathbb{R}$.

9. The function $f(x) = \ln x$ is differentiable on $]0, \infty[$ and $f'(x) = \frac{1}{x}$ for all $x \in]0, \infty[$.

Solution. Let $x_0 \in]0, \infty[$ arbitrary.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{\ln(x_0 + h) - \ln x_0}{h} \\ &= \frac{1}{x_0} \lim_{h \rightarrow 0} \frac{\ln(1 + h/x_0)}{h/x_0} \\ &= \frac{1}{x_0} \quad (\text{fundamental limit}) \end{aligned}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = \frac{1}{x_0}$ for all $x_0 \in]0, \infty[$.

4.4 Definition. (Right & Left Derivatives)

1. Suppose f is defined on a right neighborhood $I(x_0^+)$ of x_0 . We say that f is differentiable from right at x_0 if $\lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$ exists and finite. In this case the real number

$$f'_r(x_0) = f'_+(x_0) := \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h}$$

is called the right derivative of f at x_0 .

2. Suppose f is defined on a left neighborhood $I(x_0^-)$ of x_0 . We say that f is differentiable from left at x_0 if $\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}$ exists and finite. In this case the real number

$$f'_\ell(x_0) = f'_-(x_0) := \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h}$$

is called the left derivative of f at x_0 .

4.5 Property. Let I be open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$. Then

$$f \text{ is differentiable at } x_0 \iff \begin{cases} \textcircled{1} f \text{ is differentiable from right at } x_0, \\ \textcircled{2} f \text{ is differentiable from left at } x_0, \\ \textcircled{3} f'_r(x_0) = f'_\ell(x_0). \end{cases}$$

Note that the right derivative of f at x_0 represents the slope of the right tangent to the graph of f at the point $(x_0, f(x_0))$. Similarly, the left derivative of f at x_0 represents the slope of the left tangent to the graph of f at the point $(x_0, f(x_0))$. By the previous property, f is differentiable at x_0 if the right and left tangents coincides (one tangent) and not vertical. Equivalently, f is not differentiable at x_0 if the right and left tangents do not coincide (they form an angle) or atleast one is vertical.

4.6 Examples.

1. $f(x) = |x|$ is not differentiable at $x = 0$.

Solution.

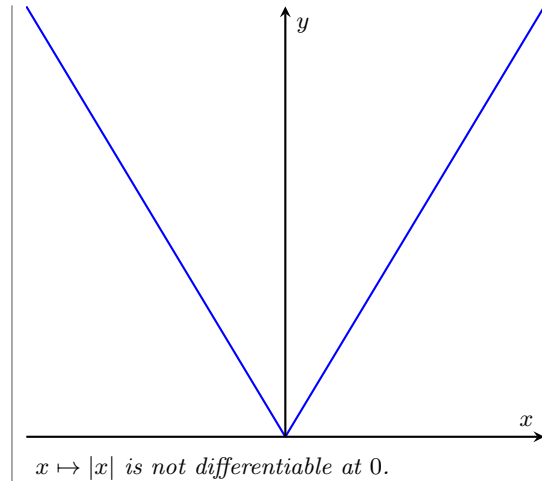
$$\bullet \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{|x|}{x} = \lim_{x \rightarrow 0} \frac{x}{x} = 1 =$$

$f'_r(0)$. So f is differentiable from right at 0.

$$\bullet \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{|x|}{x} = \lim_{x \rightarrow 0} \frac{-x}{x} =$$

$-1 = f'_\ell(0)$. So f is differentiable from left at 0.

$\bullet f'_r(0) \neq f'_\ell(0)$. Thus f is not differentiable at $x = 0$.



Note that:

★ If $f'_r(x_0) \neq f'_\ell(x_0)$ and at least one of them is finite, then x_0 is called a corner point for f .

For example $x = 0$ is a corner point for the function $f(x) = |x|$.

★ If $f'_r(x_0)$ and $f'_\ell(x_0)$ are finite and have different sign, then x_0 is called a cusp point of f .

For example $x = 0$ is a cusp point for the function $f(x) = |x|$.

2. $f(x) = \sqrt{|x|}$ is not differentiable at $x = 0$.

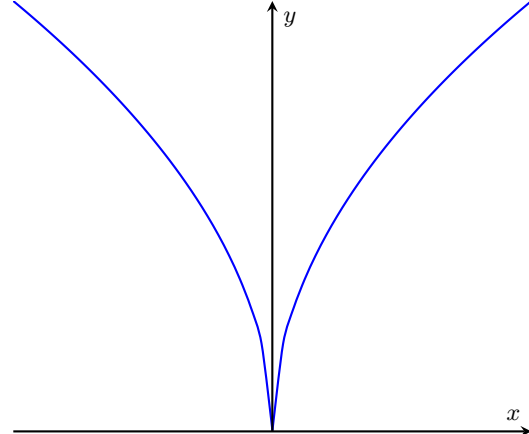
Solution.

- $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{\sqrt{|x|}}{|x|} = \infty$. So f is

not differentiable from right at $x = 0$. Thus f is not differentiable at $x = 0$. Or,

- $\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{\sqrt{|x|}}{-|x|} = -\infty$. So

f is not differentiable from left at $x = 0$. Thus f is not differentiable at $x = 0$.



$x \mapsto \sqrt{|x|}$ is not differentiable at 0.

3. $f(x) = \frac{1}{1+|x-1|}$ is not differentiable at $x = 1$.

Solution.

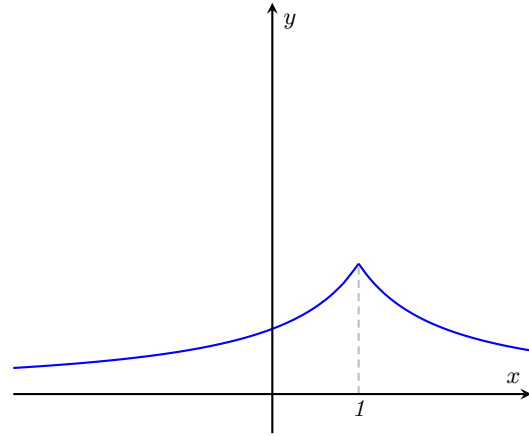
- $\lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1} = \lim_{\substack{x \rightarrow 1 \\ x > 1}} \frac{1/x - 1}{x - 1} =$

$$\lim_{x \rightarrow 1} \frac{-1}{x} = -1 = f'_r(1). \text{ So } f \text{ is differentiable from right at 1.}$$

- $\lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{\substack{x \rightarrow 1 \\ x < 1}} \frac{1/(2-x) - 1}{x - 1} =$

$$\lim_{x \rightarrow 1} \frac{1}{2-x} = 1 = f'_\ell(1) \text{ So } f \text{ is differentiable from left at 1.}$$

- $f'_r(1) \neq f'_\ell(1)$. Thus f is not differentiable at 1.



$x \mapsto 1/(1+|x-1|)$ is not differentiable at 1.

4. $f(x) = \begin{cases} x^2 & \text{if } x > 0 \\ x^3 & \text{if } x \leq 0. \end{cases}$ is differentiable at 0.

Solution.

- $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{x^2}{x} = \lim_{x \rightarrow 0} x = 0 =$

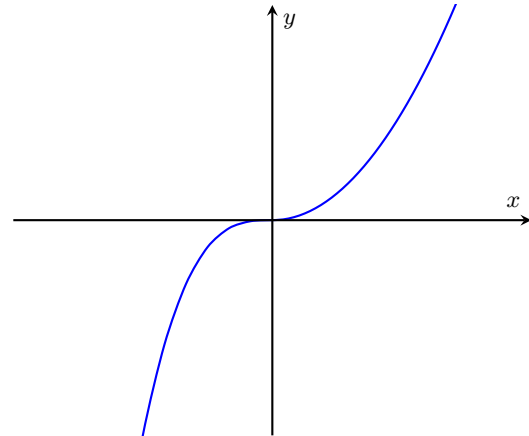
$f'_r(0)$. So f is differentiable from right at 0.

- $\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{x^3}{x} = \lim_{x \rightarrow 0} x^2 = 0 =$

$f'_\ell(0)$. So f is differentiable from left at 0.

- $f'_r(0) = f'_\ell(0)$.

Thus f is differentiable at $x = 0$.



$x \mapsto f(x)$ is differentiable at 0.

4.7 Definition. A function $f : [a, b] \mapsto \mathbb{R}$ is differentiable on $[a, b]$ closed if the following holds

- f is differentiable on $]a, b[$.
- f is differentiable from right at a .
- f is differentiable from left at b .

Similarly, f is differentiable on $[a, b[$ if f is differentiable on $]a, b[$ and f is differentiable from right at a . Also f is differentiable on $]a, b]$ if f is differentiable on $]a, b[$ and f is differentiable from left at b .

4.8 Example. Study the differentiability of the function $f(x) = \sqrt{x}$. Solution.

- $D_f = [0, \infty[$.
- Differentiability of f on $]0, \infty[$: Let $x_0 \in]0, \infty[$ arbitrary.

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\sqrt{x} - \sqrt{x_0}}{x - x_0} = \lim_{x \rightarrow x_0} \left(\frac{\sqrt{x} - \sqrt{x_0}}{x - x_0} \cdot \frac{\sqrt{x} + \sqrt{x_0}}{\sqrt{x} + \sqrt{x_0}} \right) = \lim_{x \rightarrow x_0} \frac{1}{\sqrt{x} + \sqrt{x_0}} = \frac{1}{2\sqrt{x_0}}$$

exist and finite. Hence f is differentiable at x_0 and $f'(x_0) = \frac{1}{2\sqrt{x_0}}$ for all $x_0 \in]0, \infty[$.

- Differentiability of f at 0 from right:

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{x} = \infty.$$

So f is not differentiable from right at 0. Thus $f(x) = \sqrt{x}$ is differentiable on $]0, \infty[$ and $f'(x) = \frac{1}{2\sqrt{x}}$.

4.9 Remark. In the same way as property 4.2 one can show:

- ★ If f is differentiable from right at x_0 , then f is continuous from right at x_0 .
- ★ If f is differentiable from left at x_0 , then f is continuous from left at x_0 .
- ★ If f is differentiable from right and left at x_0 (even if different), then f is continuous at x_0 .
- ★ If f is differentiable on an interval I , then f is continuous on I .

4.10 Property. (Finite expansion up to order one for differentiable function)

Let I be open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$. The function f is differentiable at x_0 if and only if there exists $\varepsilon : I \mapsto \mathbb{R}$ s.t $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$ and there exist $c \in \mathbb{R}$ s.t $f(x) = f(x_0) + (x - x_0)c + (x - x_0)\varepsilon(x)$ for all $x \in I$. In this case $c = f'(x_0)$.

Proof.

$\Rightarrow$) Suppose that f is differentiable at x_0 . Let $\varepsilon(x) = \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0)$. Since f is differentiable at x_0 , then $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$. Hence ε extendable by continuity at x_0 with $\varepsilon(x_0) = 0$. Thus we have

$$\forall x \in I, \quad f(x) = f(x_0) + (x - x_0)c + (x - x_0)\varepsilon(x) \text{ with } c = f'(x_0).$$

$\Leftarrow$) Suppose that $\forall x \in I, \quad f(x) = f(x_0) + (x - x_0)c + (x - x_0)\varepsilon(x)$ with $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$. Then

for $x \in I \setminus \{x_0\}$, we have $\frac{f(x) - f(x_0)}{x - x_0} = c + \varepsilon(x) \xrightarrow{x \rightarrow x_0} c \in \mathbb{R}$. Thus f is differentiable at x_0 and $f'(x_0) = c$. $\square$

Note that:

★ The function $\varepsilon : I \mapsto \mathbb{R}$ is continuous at x_0 . Moreover if f is continuous on I then so ε .

★ $f(x) = \underbrace{f(x_0) + (x - x_0)f'(x_0)}_{Y_T} + (x - x_0)\varepsilon(x)$, gives that a differentiable function f at x_0 can be

approximated in a neighborhood of x_0 by the tangent Y_T to the curve at the point $(x_0, f(x_0))$.

4.11 Property. Let I be open interval, $f, g : I \mapsto \mathbb{R}$ and $\alpha, \beta \in \mathbb{R}$. If f and g are differentiable at $x_0 \in I$, then

1. $\alpha f + \beta g$ is differentiable at x_0 and $(\alpha f + \beta g)'(x_0) = \alpha f'(x_0) + \beta g'(x_0)$.
2. $f \cdot g$ is differentiable at x_0 and $(f \cdot g)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.
3. If $g(x_0) \neq 0$, then $\frac{1}{g}$ is defined in a neighborhood of x_0 and differentiable at x_0 and

$$\left(\frac{1}{g}\right)'(x_0) = \frac{-g'(x_0)}{g^2(x_0)}.$$

4. If $g(x_0) \neq 0$, then $\frac{f}{g}$ is defined in a neighborhood of x_0 and differentiable at x_0 and

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}.$$

Proof.

$$1. \frac{(\alpha f + \beta g)(x) - (\alpha f + \beta g)(x_0)}{x - x_0} = \frac{\alpha f(x) - \alpha f(x_0)}{x - x_0} + \frac{\beta g(x) - \beta g(x_0)}{x - x_0} \xrightarrow{x \rightarrow x_0} \alpha f'(x_0) + \beta g'(x_0).$$

$$\begin{aligned} 2. \frac{f g(x) - f g(x_0)}{x - x_0} &= \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0} \\ &= \frac{f(x) - f(x_0)}{x - x_0} \cdot g(x) + \frac{g(x) - g(x_0)}{x - x_0} \cdot f(x_0) \quad (g \text{ is continuous at } x_0) \\ &\xrightarrow{x \rightarrow x_0} f'(x_0)g(x_0) + f(x_0)g'(x_0). \end{aligned}$$

3. We have g is differentiable at x_0 and $g(x_0) \neq 0$, then g is continuous at x_0 and $g(x_0) \neq 0$. So $g(x) \neq 0$ in some neighborhood of x_0 and thus $\frac{1}{g}$ is defined and continuous in some neighborhood of x_0 . Now

$$\frac{\frac{1}{g(x)} - \frac{1}{g(x_0)}}{x - x_0} = -\frac{\frac{g(x) - g(x_0)}{(x - x_0)g(x)g(x_0)}}{x - x_0} \xrightarrow{x \rightarrow x_0} \frac{-g'(x_0)}{g^2(x_0)}. \quad (g \text{ is continuous at } x_0)$$

4. Write $\frac{f}{g} = f \cdot \frac{1}{g}$ and use the second and third items. □

4.12 Property. (Chain Rule)

Let f be differentiable at x_0 and g be differentiable at $y_0 = f(x_0)$. Then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(f(x_0))f'(x_0)$.

Proof. Claim $g \circ f$ is defined in a neighborhood of x_0 :

Since g is differentiable at y_0 , then its defined in a neighborhood $J = J(y_0)$ of $y_0 = f(x_0)$. Since f is differentiable at x_0 , then its continuous at x_0 and so there exists $I = I(x_0)$ a neighborhood of x_0 s.t $f(I) \subset J$. Thus $g \circ f$ is defined on I a neighborhood of x_0 . Now,
 g is differentiable at y_0 implies $\forall y \in J$, $g(y) = g(y_0) + (y - y_0)g'(y_0) + (y - y_0)\varepsilon(y)$ with $\varepsilon(y) \xrightarrow{y \rightarrow y_0} 0$.

For $x \in I \setminus \{x_0\}$, $y_0 = f(x_0)$ and $y = f(x) \in J$ we have

$$\begin{aligned} \frac{g \circ f(x) - g \circ f(x_0)}{x - x_0} &= \frac{g(y) - g(y_0)}{x - x_0} = \frac{(y - y_0)g'(y_0) + (y - y_0)\varepsilon(y)}{x - x_0} \\ &= \frac{f(x) - f(x_0)}{x - x_0} \cdot g'(f(x_0)) + \frac{f(x) - f(x_0)}{x - x_0} \cdot \varepsilon(f(x)) \\ &\xrightarrow{x \rightarrow x_0} g'(f(x_0))f'(x_0). \quad (\text{substitution theorem}) \end{aligned}$$

Second way: Let $h(y) = \begin{cases} \frac{g(y) - g(y_0)}{y - y_0} & \text{if } y \neq y_0 \\ g'(y_0) & \text{if } y = y_0. \end{cases}$ Then h is continuous at y_0 and

$$\frac{g \circ f(x) - g \circ f(x_0)}{x - x_0} = h(f(x)) \frac{f(x) - f(x_0)}{x - x_0} \xrightarrow{x \rightarrow x_0} g'(f(x_0))f'(x_0). \quad (\text{substitution theorem})$$

□

4.13 Remark.

$(f(u(x)))' = f'(u(x))u'(x)$ provided that u is differentiable at x and f is differentiable at $u(x)$. For example $(f(\sin x))' = f'(\sin x) \cos x$ provided that f is differentiable at $\sin x$. In particular $(\cos(\sin x))' = -\sin(\sin x) \cos x$.

4.2 Higher Order Derivatives

4.14 Definition. Let I be an open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$ be a function.

1. We say that f is twice (or two times) differentiable at x_0 if
 - ① f is differentiable on $I(x_0)$ a neighborhood of x_0 i.e., $f' : I(x_0) \mapsto \mathbb{R}$ exists and
 - ② f' is differentiable at x_0 .
 We denote by $f''(x_0)$ the second derivative of f at x_0 and we have $f''(x_0) = (f')'(x_0)$.
2. We define by recurrence the n th derivative (derivative of order n) of f at x_0 by

$$f^{(n)}(x_0) = (f^{(n-1)})'(x_0).$$

Some notations commonly used: $f^{(0)} = f$, $f^{(1)} = f' = \frac{df}{dx}$, $f^{(2)} = f'' = \frac{d^2f}{dx^2}$ and $f^{(3)} = f''' = \frac{d^3f}{dx^3}$.

4.15 Examples.

1. Let $f(x) = \sin x$. Show that $f^{(n)}(x) = \sin(x + n\frac{\pi}{2}) \quad \forall n \in \mathbb{N}^*$.

Solution. By induction:

$$f^{(n)}(x) = \sin(x + n\frac{\pi}{2}) \quad \forall n \in \mathbb{N}^* \quad (P(n)).$$

- For $n = 1$ we have $f^{(1)}(x) = \cos x = \sin(x + 1\frac{\pi}{2})$. Hence $P(1)$ is true.
- Assume $P(n)$ is true i.e., $f^{(n)}(x) = \sin(x + n\frac{\pi}{2})$. Now

$$f^{(n+1)}(x) = (f^{(n)})'(x) = (\sin(x + n\frac{\pi}{2}))' = \cos(x + n\frac{\pi}{2}) = \sin(x + n\frac{\pi}{2} + \frac{\pi}{2}) = \sin(x + (n+1)\frac{\pi}{2}).$$

Hence $P(n+1)$ is true. Thus $P(n)$ is true for all $n \in \mathbb{N}^*$.

2. Let $g(x) = \cos x$. Show that $g^{(n)}(x) = \cos(x + n\frac{\pi}{2}) \quad \forall n \in \mathbb{N}^*$.

Solution. We have $g(x) = \cos x = \sin(x + \frac{\pi}{2}) = f(x + \frac{\pi}{2})$. Hence by chain rule n times we have $g^{(n)}(x) = f^{(n)}(x + \frac{\pi}{2}) = \sin(x + \frac{\pi}{2} + n\frac{\pi}{2}) = \cos(x + n\frac{\pi}{2})$. Or one can do it by induction as 1.

4.16 Definition. Let I be an open interval and $f : I \mapsto \mathbb{R}$ be a function.

1. We say that f is twice (or two times) differentiable on I if $f' : I \mapsto \mathbb{R}$ is differentiable on I and its derivative is called the second derivative function of f and denoted by f''
2. We define by recurrence the n th derivative of f on I by $f^{(n)} = (f^{(n-1)})'$. If $f^{(n)}$ exists on I , we say that f is n -times differentiable on I .
3. We say that f is of class C^1 (or continuously differentiable) on I and we write $f \in C^1(I)$ if
 - ① f is differentiable on I i.e., $f' : I \mapsto \mathbb{R}$ exists and
 - ② f' is continuous on I .
4. We say that f is of class C^n on I and we write $f \in C^n(I)$ if
 - ① f is differentiable n -times on I (i.e., derivatives up to order n exists on I) and
 - ② $f^{(n)} : I \mapsto \mathbb{R}$ is continuous on I .
5. We say that f is of class C^∞ on I and we write $f \in C^\infty(I)$ if $f \in C^n(I) \quad \forall n \in \mathbb{N}^*$ i.e., $C^\infty(I) := \bigcap_{n \in \mathbb{N}^*} C^n(I)$.

Note that;

- ① $f \in C^n(I) \implies$ derivatives up to order n exist and continuous on I .
- ② $f \in C^\infty(I) \implies$ derivatives up to any order exist and continuous on I .
- ③ $C(I) = C^0(I) := \{f; f : I \mapsto \mathbb{R} \text{ is continuous on } I\}$.
- ④ $\mathcal{F}(I) := \{f; f : I \mapsto \mathbb{R} \text{ is a function } \}$.
- ⑤ $C^\infty(I) \subsetneq \dots \subsetneq C^n(I) \subsetneq \dots \subsetneq C^2(I) \subsetneq C^1(I) \subsetneq C^0(I) \subsetneq \mathcal{F}(I)$. One can check that $f(x) = \begin{cases} x^{n+2} \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ is of class C^n but not of class C^{n+1} on $\mathbb{R}$.

4.3 The Mean Value Theorem

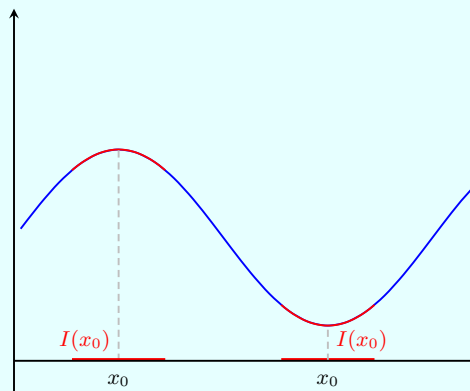
The mean value theorem is one of the most useful theorems in analysis. Under some assumptions, It relates f and f' without involving limits explicitly. So it allow us to get more information about the function from its derivative. The proof of the M.V.T is based on the existence of local extrema.

4.17 Definition. (Local Extrema)

★ A function f is said to have a **local maximum** at x_0 if f is defined on $I(x_0)$ a neighborhood of x_0 and $f(x) \leq f(x_0)$ for all $x \in I(x_0)$.

★ A function f is said to have a **local minimum** at x_0 if f is defined on $I(x_0)$ a neighborhood of x_0 and $f(x) \geq f(x_0)$ for all $x \in I(x_0)$.

★ A function f is said to have a **local extremum** at x_0 if it has either local maximum or local minimum at x_0 .



Local Extrema Of f .

The extremum points of a differentiable function defined on an open interval I should be searched for among the points where the derivative vanish.

4.18 Theorem. (*Local Extremum Theorem*)

If f has a local extremum at x_0 and if f is differentiable at x_0 , then $f'(x_0) = 0$.

Proof. Suppose that f has a local maximum at x_0 . Then there exists I a neighborhood of x_0 such that $f(x) \leq f(x_0)$ for all $x \in I$. Hence,

$$\star \frac{f(x)-f(x_0)}{x-x_0} \geq 0 \quad \text{if } x \in I \wedge x < x_0,$$

$$\star \frac{f(x)-f(x_0)}{x-x_0} \leq 0 \quad \text{if } x \in I \wedge x > x_0.$$

So $f'_\ell(x_0) \geq 0$ and $f'_r(x_0) \leq 0$. Thus $f'(x_0) = f'_\ell(x_0) = f'_r(x_0) = 0$. The proof for the local minimum case is similar. $\square$

4.19 Theorem. (*Rolle's Theorem*)

Let f be continuous on $[a, b]$ and differentiable on $]a, b[$. If $f(a) = f(b)$, then there exists a point $c \in]a, b[$ such that $f'(c) = 0$.

Proof. By the extreme value theorem there exist $c, d \in [a, b]$ s.t. $\min_{x \in [a, b]} f(x) = f(c)$ and $\max_{x \in [a, b]} f(x) = f(d)$. Hence $f(c) \leq f(x) \leq f(d)$, $\forall x \in [a, b]$. If $f(c) = f(d)$, then f is a constant function and $f'(x) = 0 \quad \forall x \in]a, b[$. If $f(c) \neq f(d)$, then either $c \in]a, b[$ or $d \in]a, b[$. If $c \in]a, b[$, then f has a local minimum at c and by the local extremum theorem we get $f'(c) = 0$. Similarly, if $d \in]a, b[$, then f has a local maximum at d and by the local extremum theorem we get $f'(d) = 0$. $\square$

4.20 Theorem. (*Mean Value Theorem*)

If f be continuous on $[a, b]$ and differentiable on $]a, b[$ then there exists a point $c \in]a, b[$ such that $\frac{f(b)-f(a)}{b-a} = f'(c)$.

Proof. The function $h(x) := (f(b) - f(a))x - (b - a)f(x)$ is continuous on $[a, b]$ and differentiable on $]a, b[$ and satisfies $h(a) = h(b)$. By Rolle's theorem, there exists $c \in]a, b[$ such that $h'(c) = 0$. But $h'(c) = f(b) - f(a) - (b - a)f'(c)$. Thus there exists $c \in]a, b[$ such that $\frac{f(b)-f(a)}{b-a} = f'(c)$. $\square$

Here are some applications of the mean value theorem.

4.21 Corollary. If f is differentiable on an open interval I and $f'(x) = 0$ for all $x \in I$, then there exists a constant c such that $f(x) = c$ for all $x \in I$.

Proof. Fix $x_0 \in I$ and claim $f(x) = f(x_0)$ for all $x \in I \setminus \{x_0\}$. Applying the M.V.T for f on $[x, x_0] \subset I$ (or $[x_0, x]$):

$\star$ Since f is continuous on I and $[x, x_0] \subset I$, then f is continuous on $[x, x_0]$.

$\star$ Since f is differentiable on I and $]x, x_0[\subset I$, then f is differentiable on $]x, x_0[$.

By M.V.T, $\exists c \in]x, x_0[$ such that $\frac{f(x)-f(x_0)}{x-x_0} = f'(c) = 0$. Thus $f(x) = f(x_0)$ for all $x \in I$. $\square$

4.22 Corollary. If f and g are differentiable on an open interval I and $f'(x) = g'(x)$ for all $x \in I$, then there exists a constant c such that $f(x) = g(x) + c$ for all $x \in I$.

Proof. Apply the previous corollary to the function $h := f - g$. □

4.23 Corollary. Let f be differentiable on an open interval I .

1. If $f'(x) \geq 0 \quad \forall x \in I$, then f is increasing on I .
2. If $f'(x) \leq 0 \quad \forall x \in I$, then f is decreasing on I .
3. If $f'(x) > 0 \quad \forall x \in I$, then f is strictly increasing on I .
4. If $f'(x) < 0 \quad \forall x \in I$, then f is strictly decreasing on I .

Proof. Let $x, y \in I$ such that $x < y$. Applying the M.V.T for f on $[x, y] \subset I$:

★ Since f is continuous on I and $[x, y] \subset I$, then f is continuous on $[x, y]$.

★ Since f is differentiable on I and $]x, y[\subset I$, then f is differentiable on $]x, y[$.

By M.V.T, $\exists c \in]x, y[$ such that $\frac{f(y)-f(x)}{y-x} = f'(c)$. We discuss four cases:

1. If $f'(x) \geq 0 \quad \forall x \in I$, then $\frac{f(y)-f(x)}{y-x} = f'(c) \geq 0$. So $f(x) \leq f(y)$ and thus f is inc on I .
2. If $f'(x) \leq 0 \quad \forall x \in I$, then $\frac{f(y)-f(x)}{y-x} = f'(c) \leq 0$. So $f(x) \geq f(y)$ and thus f is dec on I .
3. If $f'(x) > 0 \quad \forall x \in I$, then $\frac{f(y)-f(x)}{y-x} = f'(c) > 0$. So $f(x) < f(y)$ and thus f is st. inc on I .
4. If $f'(x) < 0 \quad \forall x \in I$, then $\frac{f(y)-f(x)}{y-x} = f'(c) < 0$. So $f(x) > f(y)$ and thus f is st. dec on I . □

4.24 Remark.

1. If f is strictly increasing (resp increasing) on the intervals $]a, b[$ and $]b, c[$ and f is continuous at b then f is strictly increasing (resp increasing) on $]a, c[$. For example $f(x) = x^3$. Its derivative $f'(x) = 3x^2 > 0$ on $\mathbb{R}^*$. Since f is continuous at zero, then f is strictly increasing on $\mathbb{R}$.
2. If f is strictly decreasing (resp decreasing) on the intervals $]a, b[$ and $]b, c[$ and f is continuous at b then f is strictly decreasing (resp decreasing) on $]a, c[$.
3. Corollary 4.23 does not work if I is disjoint union of open intervals. For example $f(x) = \frac{1}{x}$. Its derivative $f'(x) = -\frac{1}{x^2} < 0$ on $]-\infty, 0[\cup]0, \infty[$ but f is not strictly decreasing on $]-\infty, 0[\cup]0, \infty[$.
4. Corollary 4.21 does not work if I is disjoint union of open intervals. For example $f(x) = \begin{cases} 1 & \text{if } x \in]-\infty, 0[\\ 2 & \text{if }]0, \infty[\end{cases}$ Its derivative $f'(x) = 0$ on $I =]-\infty, 0[\cup]0, \infty[$ but f is not a constant function on I .

4.25 Example. Show that $\frac{1}{x+1} < \ln(x+1) - \ln x < \frac{1}{x} \quad \forall x \in]0, \infty[$.

Solution. Let $x \in]0, \infty[$ and $f(t) = \ln t$. Let us try MVT on $[x, x+1]$:

① Since f is continuous on $]0, \infty[$, then f is continuous on $[x, x+1] \subseteq]0, \infty[$.

② Since f is differentiable on $]0, \infty[$, then f is differentiable on $]x, x+1[\subseteq]0, \infty[$.

By the MVT, there exists $c \in]x, x+1[$ such that $\frac{f(x+1)-f(x)}{(x+1)-x} = f'(c) = \frac{1}{c}$. Hence

$$\frac{1}{x+1} < \ln(x+1) - \ln x = \frac{1}{c} < \frac{1}{x}.$$

4.4 Taylor's Formula

4.26 Theorem. (Taylor-Lagrange Formula)

Let $f : [a, b] \mapsto \mathbb{R}$ (or $f : [b, a] \mapsto \mathbb{R}$) and $n \in \mathbb{N}$. If

① $f \in C^n([a, b])$ and

② $f^{(n+1)}$ exists on $]a, b[$,

then there exist $c \in]a, b[$, not necessarily unique, such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \cdots + \frac{(b-a)^n}{n!}f^{(n)}(a) + \frac{(b-a)^{n+1}}{(n+1)!}f^{(n+1)}(c).$$

Proof. For simplicity we proof the theorem for $n = 1$.

Given that $f \in C^1([a, b])$ and $f^{(2)} = f''$ exists on $]a, b[$. Claim there exist $c \in]a, b[$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(c) :$$

Let $h(x) := f(x) + (b-x)f'(x) + \frac{(b-x)^2}{2!}M$ where M is chosen so that $h(a) = h(b)$. Hence

$$M = \frac{2}{(b-a)^2}[f(b) - f(a) - (b-a)f'(a)].$$

Since $h(b) = f(b) = h(a)$, then $f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}M$.

The result follows if we show $M = f''(c)$. Let us try Roll's theorem for h on $[a, b]$:

- $h(a) = h(b)$ (by the choice of M).
- $h \in C([a, b])$ (sum of continuous functions).
- h is differentiable on $]a, b[$ as sum of differentiable functions ($f'' = (f')'$ exists on $]a, b[$ means f' is differentiable on $]a, b[$).

Then by Roll's theorem, there exists $c \in]a, b[$ such that $h'(c) = 0$. This gives $M = f''(c)$. $\square$

Note that Taylor-Lagrange with $n = 0$ is the mean value theorem.

4.27 Theorem. (Mac-Laurin Formula $a = 0$, $b = x \in \mathbb{R}$)

Let $f : [0, x] \mapsto \mathbb{R}$ (or $f : [x, 0] \mapsto \mathbb{R}$) and $n \in \mathbb{N}$. If

① $f \in C^n([0, x])$ and

② $f^{(n+1)}$ exists on $]0, x[$,

then there exist $c \in]0, x[$, not necessarily unique, such that

$$\begin{aligned} f(x) &= f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \frac{x^{n+1}}{(n+1)!}f^{(n+1)}(c) \\ &= \sum_{k=0}^n \frac{x^k}{k!}f^{(k)}(0) + \frac{x^{n+1}}{(n+1)!}f^{(n+1)}(c). \end{aligned}$$

4.28 Example. Give the Mac-Laurin expansion up to order n for the functions $f(x) = e^x$.

Solution. Note that $f \in C^n([0, x])$ and $f^{(n+1)}$ exists on $]0, x[$. Also $f^{(n)}(x) = e^x$ and $f^{(n)}(0) = 1$ for all $n \in \mathbb{N}^*$. Thus

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!}e^c = \sum_{k=0}^n \frac{x^k}{k!} + \frac{x^{n+1}}{(n+1)!}e^c.$$

Similarly, the formula holds for $x < 0$.

4.5 Inverse Function Theorem

4.29 Theorem. (Inverse Function Theorem-2)

Let I be an interval and $f : I \mapsto \mathbb{R}$ be continuous and injective on I . If f is differentiable at $x_0 \in I$ and $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and $(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}$.

Proof. Let $y = f(x) \in J = f(I)$ such that $y \neq y_0$. Since f is bijective, then $x \neq x_0$ and

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}} = g(f(x)).$$

Since f is continuous at x_0 and f injective, then $x \rightarrow x_0$ implies $f(x) \rightarrow f(x_0)$ and $f(x) \neq f(x_0)$. Now, f is differentiable at x_0 and $f'(x_0) \neq 0$ gives with the help of substitution theorem that

$$\lim_{y \rightarrow y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{y \rightarrow y_0} g(y) = \lim_{x \rightarrow x_0} g(f(x)) = \frac{1}{f'(x_0)}.$$

Second way: We have f is injective, let $h(x) = \begin{cases} \frac{x - x_0}{f(x) - f(x_0)} & \text{if } x \neq x_0 \\ \frac{1}{f'(x_0)} & \text{if } x = x_0. \end{cases}$ Since f is differentiable at x_0 and $f'(x_0) \neq 0$, then h is continuous at x_0 and

$$\lim_{y \rightarrow y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{y \rightarrow y_0} h(f^{-1}(y)) = \frac{1}{f'(x_0)}. \quad (\text{substitution theorem})$$

Third way: Let (y_n) be a sequence in $f(I) \setminus \{y_0\}$ converging to y_0 and $x_n = f^{-1}(y_n)$. Since f^{-1} is continuous at y_0 , then

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} f^{-1}(y_n) = f^{-1}(y_0) = x_0.$$

Since f is differentiable at x_0 , $f'(x_0) \neq 0$ and f is bijective then $x_n \neq x_0$ and

$$\lim_{n \rightarrow \infty} \frac{f^{-1}(y_n) - f^{-1}(y_0)}{y_n - y_0} = \lim_{n \rightarrow \infty} \frac{x_n - x_0}{f(x_n) - f(x_0)} = \frac{1}{f'(x_0)}.$$

□

4.30 Theorem. (Inverse Function Theorem-3)

Let I be an interval and $f : I \mapsto \mathbb{R}$ be differentiable on I and $f'(x) \neq 0$ for all $x \in I$. Then
 ① f is injective, ② $J = f(I)$ is an interval and ③ f^{-1} is differentiable on J and we have
 $(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad \forall x \in I.$

Proof. The result follows from IFT-1 and IFT-2 if we show f is injective.

Assume that f is not injective, then there exist $a, b \in I$ such that $a \neq b$ and $f(a) = f(b)$. Since $[a, b] \subseteq I$, then f is continuous on $[a, b]$ and differentiable on $]a, b[$. Hence by Roll's theorem there exists $c \in]a, b[\subseteq I$ such that $f'(c) = 0$ which is a contradiction. Thus f is injective. □

Note that;

① If $f \in C^1(I)$ and $f'(x) \neq 0 \quad \forall x \in I$, then $g = \frac{1}{f'} \in C^1(I)$.

② More general, If $f \in C^n(I)$ and $f'(x) \neq 0 \quad \forall x \in I$, then $g = \frac{1}{f'} \in C^n(I)$.

4.31 Theorem. (Inverse Function Theorem-4)

Let I be an interval and $f : I \mapsto \mathbb{R}$ be of class C^n on I and $f'(x) \neq 0$ for all $x \in I$. Then ① f is injective, ② $J = f(I)$ is an interval and ③ f^{-1} is of class C^n on J .

Exercise 1. Let $a, b \in \mathbb{R}$ and $f(x) = \begin{cases} \sqrt{x} & \text{if } 0 \leq x \leq 1 \\ a(x^2 - 1) + b & \text{if } x > 1. \end{cases}$ Determine a and b so that f is of class C^1 on $]0, \infty[$.

Exercise 2. Let f be a differentiable function on $\mathbb{R}$ such that $f(0) = 1$ and $g(x) = \begin{cases} f(\frac{x}{2}) & \text{if } x > 0 \\ x + 1 & \text{if } x \leq 0. \end{cases}$

1. Show that g is continuous on $\mathbb{R}$.

2. Assume that $f'_r(0) = 2$. Show that g is differentiable on $\mathbb{R}$ and determine $g'(x)$.

Exercise 3. Let $f :]-a, a[\mapsto \mathbb{R}$ be a differentiable function. Prove that if f is an even function then f' is an odd function and if f is an odd function then f' is an even function.

Exercise 4. Let $f : \mathbb{R} \mapsto \mathbb{R}$ be differentiable function on $\mathbb{R}$ such that $f(0) = 0$. Show that $\lim_{x \rightarrow 0} \frac{f(x)f(-x)}{x^2} = -(f'(0))^2$. (Hint: Use the equivalent definition of differentiation at a point i.e., finite expansion to order one.)

Exercise 5. Let $f \in C^1([a, b])$. Show that there exists $k \geq 0$ such that

$$|f(x) - f(y)| \leq k|x - y| \quad \forall x, y \in [a, b].$$

Exercise 6. (Extension by differentiability)

1. Let I be an open interval and $a \in I$. Suppose that f is continuous on I , differentiable on $I \setminus \{a\}$ and $\lim_{x \rightarrow a} f'(x) = \ell$ ($\ell \in \mathbb{R}$). Prove that f is differentiable at a , $f'(a) = \ell$ and f' is continuous at a .

2. Show that $f(x) = x\sqrt{|x|}$ is differentiable on $\mathbb{R}$.

3. Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ Show that f is differentiable on $\mathbb{R}$ but f is not of class C^1 on $\mathbb{R}$. (Check that $f'(0)$ exists but $\lim_{x \rightarrow 0} f'(x)$ does not exist.)

Exercise 7. (Extension by differentiability)

1. Let $I = [a, b[$. Suppose that f is continuous on $I = [a, b[$, differentiable on $I \setminus \{a\} =]a, b[$ and $\lim_{x \rightarrow a^+} f'(x) = \ell$ ($\ell \in \mathbb{R}$). Prove that f is differentiable from right at a , $f'_r(a) = \ell$ and f' is continuous from right at a .

2. Compute $\lim_{x \rightarrow 0^+} \frac{\cos \sqrt{x} - 1}{x}$.

Exercise 8. (No extension by differentiability)

Let I be an open interval and $a \in I$. Suppose that f is continuous on I , differentiable on $I \setminus \{a\}$ and $\lim_{x \rightarrow a} f'(x) = \pm\infty$. Prove that $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \pm\infty$. (i.e., f is not differentiable at a and the tangent to the representative curve of f at the point $(a, f(a))$ is vertical.)

Exercise 9. Let f be differentiable on $\mathbb{R}$ such that $f(0) = 1$ and $g(x) = \begin{cases} f(x^2 \ln x + \frac{x}{2}) & \text{if } x > 0 \\ x + 1 & \text{if } x \leq 0. \end{cases}$

1. Show that g is continuous on $\mathbb{R}$.

2. Assume that $\lim_{x \rightarrow 0^+} f'(x) = 2$. Show that g is differentiable on $\mathbb{R}$ and determine $g'(x)$.

Exercise 10. Let f be differentiable on $\mathbb{R}$ such that $f'(x) \neq 0$ for all $x \in \mathbb{R}$. Show that f is not a periodic function.

Exercise 11. Let I be an open interval and $f : I \mapsto \mathbb{R}$ satisfying $\forall x, y \in I, |f(x) - f(y)| \leq K|x - y|^2$ for some $k > 0$. Show that f is a constant function.

Exercise 12. Let $f(x) = \begin{cases} x^2 \ln x^2 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ Show that f is of class C^1 on $\mathbb{R}$ but f is not twice differentiable on $\mathbb{R}$.

Exercise 13. Let $a < b < c$ and $f : [a, c] \mapsto \mathbb{R}$ be continuous on $[a, c]$, twice differentiable on $]a, c[$ and $f(a) = f(b) = f(c)$.

1. Show that there exist $\xi_1 \in]a, b[$, $\xi_2 \in]b, c[$ and such that $f'(\xi_1) = f'(\xi_2)$.

2. Deduce that there exists $\xi \in]a, c[$ such that $f''(\xi) = 0$.

Exercise 14.

1. Let $f : [0, 1] \mapsto \mathbb{R}$ be continuous on $[0, 1]$ with $f(0) = f(1) = 0$ and twice differentiable on $]0, 1[$ with $f'' < 0$ on $]0, 1[$.

(a) Show that there exists $c \in]0, 1[$ such that $f'(c) = 0$.

(b) Show that $f' > 0$ on $]0, c[$ and $f' < 0$ on $]c, 1[$.

(c) Deduce that $f(x) > 0$, for all $x \in]0, 1[$.

2. Let $a, b \in \mathbb{R}$ such that $a > b > 1$. consider the function f defined by

$$f(x) = \ln \ln(xa + (1 - x)b) - x \ln \ln a - (1 - x) \ln \ln b.$$

(a) Show that f is continuous on $[0, 1]$ with $f(0) = f(1) = 0$ and twice differentiable on $]0, 1[$ with $f'' < 0$ on $]0, 1[$.

(b) Deduce the following inequality :

$$\ln \ln(xa + (1 - x)b) > x \ln \ln a + (1 - x) \ln \ln b, \quad \forall x \in]0, 1[.$$

(c) Show that $\ln(\frac{a+b}{2}) > \sqrt{\ln a \ln b}$.

Exercise 15. Show that the following inequalities:

$$1 + x - \frac{x^2}{2} \leq \sqrt{1 + 2x} < 1 + x - \frac{x^2}{2} + \frac{x^3}{2} \quad \forall x \in [0, \infty[.$$

Exercise 16. Let $f : [\frac{\pi}{2}, \pi[\mapsto \mathbb{R}$ defined by $f(x) = \frac{1}{\sin x}$.

1. Show that f defines a bijection from $I = [\frac{\pi}{2}, \pi[$ onto some interval J to be determined.

2. Find the domain of differentiability of f^{-1} and calculate $(f^{-1})'(y)$ (without determining f^{-1}).

Exercise 17. Let $f :]-\frac{\pi}{2}, \frac{\pi}{2}[\mapsto \mathbb{R}$ defined by $f(x) = \tan x - x$.

1. Show that f defines a bijection from $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$ onto some interval J to be determined.

2. Let $g = f^{-1}$.

(a) Study the continuity and monotony of g .

(b) Show that g is of class C^2 on J except at one point a to be determined.

(c) Let $y_0 = 1 - \frac{\pi}{4}$. Calculate $g(y_0)$, $g'(y_0)$ and $g''(y)$ then deduce Taylor up to $n = 1$ for $g = f^{-1}$.

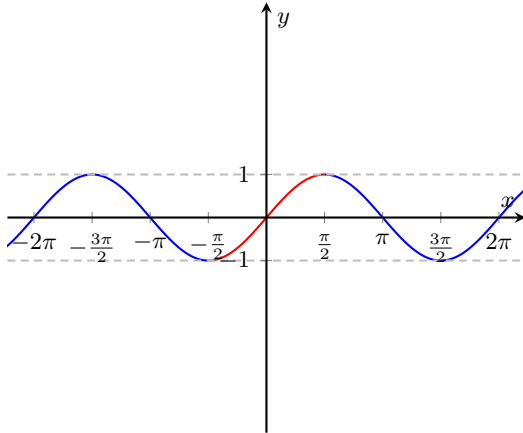
5.1 Trigonometric Functions And Inverses

We consider the standard definitions of sine and cosine as coordinates of points on the unit circle. From this definition one can derive the usual trigonometric identities.

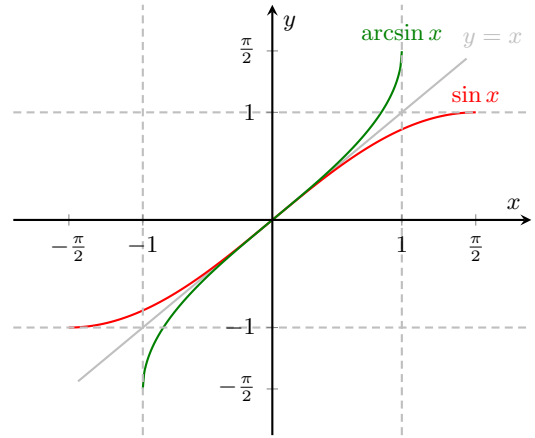
5.1.1 Sine And Arcsine Functions

The function $\sin : \mathbb{R} \mapsto [-1, 1]$ has the following properties:

- $\text{dom}(\sin) = \mathbb{R}$ and $-1 \leq \sin x \leq 1 \quad \forall x \in \mathbb{R}$.
- $\sin \in C^\infty(\mathbb{R})$ and $\sin^{(n)}(x) = \sin(x + n\frac{\pi}{2}) \quad \forall x \in \mathbb{R} \quad \forall n \in \mathbb{N}$. In particular $\sin'(x) = \cos x \quad \forall x \in \mathbb{R}$.
- $\sin$ is odd. $\begin{cases} \text{dom}(\sin) = \mathbb{R} \text{ is centered at } 0. \\ \sin(-x) = -\sin(x) \quad \forall x \in \mathbb{R}. \end{cases}$
- $\sin$ is 2π -periodic. $\begin{cases} \sin(x + 2\pi) = \sin x \quad \forall x \in \mathbb{R}. \\ \sin(x + 2k\pi) = \sin x \quad \forall x \in \mathbb{R}, \quad \forall k \in \mathbb{Z}. \end{cases}$



$x \mapsto \sin x$ is 2π -periodic and odd on $D = \mathbb{R}$.



$\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \mapsto [-1, 1], \quad \arcsin : [-1, 1] \mapsto [-\frac{\pi}{2}, \frac{\pi}{2}].$
 $x \mapsto \sin x \quad \quad \quad x \mapsto \arcsin x$

The restriction of the sine function, $\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \mapsto [-1, 1]$ has the following properties:

- Continuous on $I = [-\frac{\pi}{2}, \frac{\pi}{2}]$.
- Strictly increasing on $I = [-\frac{\pi}{2}, \frac{\pi}{2}]$:

$$\begin{aligned} \sin'(x) = \cos x > 0 \text{ on }]-\frac{\pi}{2}, \frac{\pi}{2}[&\implies \sin \text{ is st inc on }]-\frac{\pi}{2}, \frac{\pi}{2}[\\ &\implies \sin \text{ is st inc on } [-\frac{\pi}{2}, \frac{\pi}{2}]. \quad (\because \sin \text{ is continuous at } \pm \frac{\pi}{2}) \end{aligned}$$

Then by the inverse function theorem;

★ The function sine defines a bijection from the interval $I = [-\frac{\pi}{2}, \frac{\pi}{2}]$ onto the interval $J = [\sin(-\frac{\pi}{2}), \sin(\frac{\pi}{2})] = [-1, 1]$.

$$\begin{array}{ccc} \sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \mapsto [-1, 1], & \arcsin : [-1, 1] \mapsto [-\frac{\pi}{2}, \frac{\pi}{2}]. \\ x \rightarrow y = \sin x & y \rightarrow x = \arcsin y \end{array}$$

$$y = \sin x, \quad x \in [-\frac{\pi}{2}, \frac{\pi}{2}] \iff x = \arcsin y, \quad y \in [-1, 1].$$

★ Since $\sin$ is differentiable on $I = [-\frac{\pi}{2}, \frac{\pi}{2}]$, then $\arcsin$ is differentiable on

$$J \setminus \{y = \sin x : \sin'(x) = 0\} = J \setminus \{y = \sin x : x = \pm \frac{\pi}{2}\} =]-1, 1[.$$

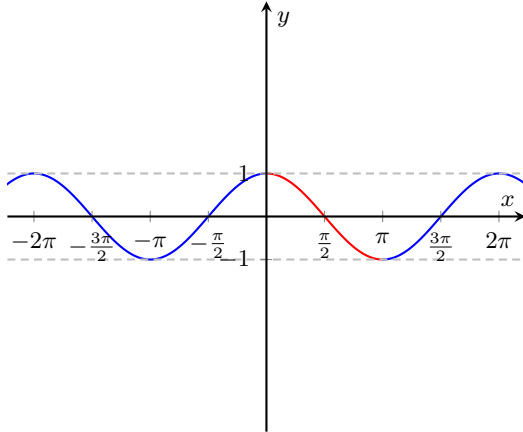
★ Since $\sin$ is of class C^∞ on I , then $\arcsin$ is of class C^∞ on the domain of differentiation $] -1, 1[$.

$$\arcsin'(y) = (\sin^{-1})'|_{y=\sin x} = \frac{1}{\sin'(x)} = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}} \quad \forall y \in]-1, 1[.$$

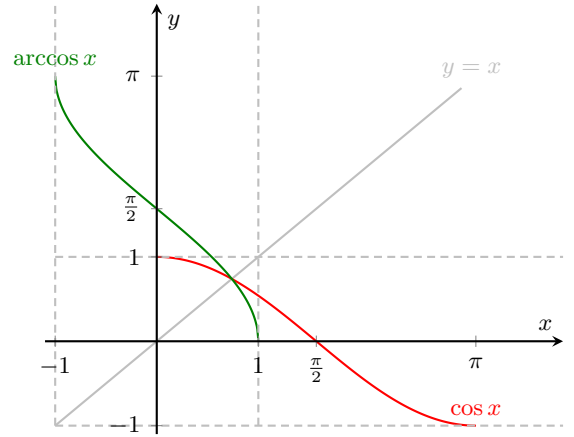
5.1.2 Cosine And Arccosine Functions

The function $\cos : \mathbb{R} \mapsto [-1, 1]$ has the following properties:

- $\text{dom}(\cos) = \mathbb{R}$ and $-1 \leq \cos x \leq 1 \quad \forall x \in \mathbb{R}$.
- $\cos \in C^\infty(\mathbb{R})$ and $\cos^{(n)}(x) = \cos(x + n\frac{\pi}{2}) \quad \forall x \in \mathbb{R} \quad \forall n \in \mathbb{N}$. In particular $\cos'(x) = -\sin x$.
- $\cos$ is even. $\begin{cases} \text{dom}(\cos) = \mathbb{R} \text{ is centered at } 0. \\ \cos(-x) = \cos(x) \quad \forall x \in \mathbb{R}. \end{cases}$
- $\cos$ is 2π -periodic. $\begin{cases} \cos(x + 2\pi) = \cos x \quad \forall x \in \mathbb{R}. \\ \cos(x + 2k\pi) = \cos x \quad \forall x \in \mathbb{R}, \quad \forall k \in \mathbb{Z}. \end{cases}$



$x \mapsto \cos x$ is 2π -periodic and even on $D = \mathbb{R}$.



$\cos : [0, \pi] \mapsto [-1, 1], \quad \arccos : [-1, 1] \mapsto [0, \pi].$
 $x \rightarrow \sin x \quad x \rightarrow \arcsin x$

The restriction of the cosine function, $\cos : [0, \pi] \mapsto [-1, 1]$ has the following properties:

- Continuous on $I = [0, \pi]$.
- Strictly decreasing on $I = [0, \pi]$:

$$\begin{aligned} \cos'(x) = -\sin x < 0 \text{ on }]0, \pi[&\implies \cos \text{ is st dec on }]0, \pi[\\ &\implies \cos \text{ is st dec on } [0, \pi]. \quad (\because \cos \text{ is continuous at } 0 \text{ and } \pi) \end{aligned}$$

Then by the inverse function theorem;

★ The function cosine defines a bijection from the interval $I = [0, \pi]$ onto the interval $J = [\cos(\pi), \cos(0)] = [-1, 1]$.

$$\cos : [0, \pi] \mapsto [-1, 1], \quad \arccos : [-1, 1] \mapsto [0, \pi].$$

$$x \rightarrow y = \cos x \quad y \rightarrow x = \arccos y$$

$$y = \cos x, \quad x \in [0, \pi] \iff x = \arccos y, \quad y \in [-1, 1].$$

★ Since $\cos$ is differentiable on $I = [0, \pi]$, then $\arccos$ is differentiable on

$$J \setminus \{y = \cos x : \cos'(x) = 0\} = J \setminus \{y = \cos x : x = 0, x = \pi\} =]-1, 1[.$$

★ Since $\cos$ is of class C^∞ on I , then $\arccos$ is of class C^∞ on the domain of differentiation $] -1, 1[$.

$$\arccos'(y) = (\cos^{-1})'|_{y=\cos x} = \frac{1}{\cos'(x)} = \frac{1}{-\sin x} = \frac{-1}{\sqrt{1-y^2}} \quad \forall y \in]-1, 1[.$$

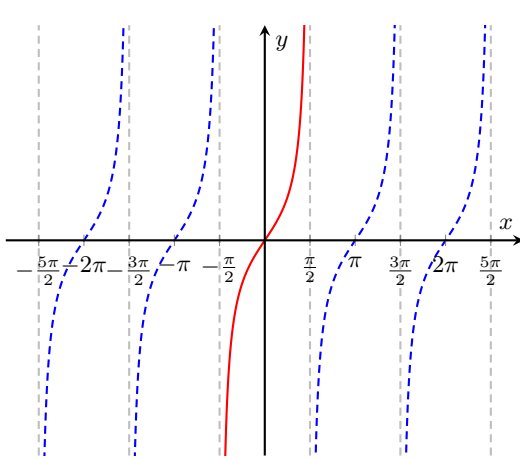
5.1.3 Tangent And Arctangent Functions

The tangent function is defined by $\tan x = \frac{\sin x}{\cos x}$ on $D = \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\}$. The function $\tan : D \mapsto \mathbb{R}$ has the following properties:

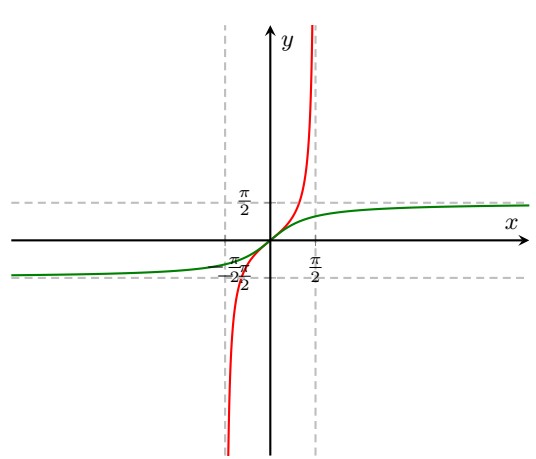
- $\text{dom}(\tan) = \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\} = \bigcup_{k \in \mathbb{Z}}]-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi[$.
- $\tan = \frac{\sin}{\cos}$ is differentiable on D because its the quotient of differentiable functions with denominator never vanishes.

$$\tan'(x) = \frac{1}{\cos^2 x} = 1 + \tan^2 x.$$

- $\tan \in C^\infty(D)$ because its the quotient of C^∞ functions with denominator never vanishes.
- $\tan$ is odd. $\begin{cases} \text{dom}(\tan) \text{ is centered at } 0. \\ \tan(-x) = -\tan(x) \quad \forall x \in D \end{cases}$
- $\tan$ is π -periodic. $\begin{cases} \tan(x + \pi) = \tan x \quad \forall x \in D. \\ \tan(x + k\pi) = \tan x \quad \forall x \in D, \quad \forall k \in \mathbb{Z}. \end{cases}$



$x \mapsto \tan x$ is π -periodic and odd on D .



$\tan :]-\frac{\pi}{2}, \frac{\pi}{2}[\mapsto \mathbb{R}, \quad \arctan : \mathbb{R} \mapsto]-\frac{\pi}{2}, \frac{\pi}{2}[$
 $x \rightarrow \tan x \quad x \rightarrow \arctan x$

The restriction of the tangent function, $\tan :]-\frac{\pi}{2}, \frac{\pi}{2}[\mapsto \mathbb{R}$ has the following properties:

- Continuous on $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$.

- Strictly increasing on $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$. ($\tan'(x) = 1 + \tan^2 x > 0 \quad \forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$)

Then by the inverse function theorem;

★ The function tangent defines a bijection from the interval $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$ onto the interval $J =]\lim_{x \rightarrow -\frac{\pi}{2}} \tan x, \lim_{x \rightarrow \frac{\pi}{2}} \tan x[=]-\infty, +\infty[= \mathbb{R}$.

$$\tan :]-\frac{\pi}{2}, \frac{\pi}{2}[\mapsto \mathbb{R}, \quad \arctan : \mathbb{R} \mapsto]-\frac{\pi}{2}, \frac{\pi}{2}[$$

$$x \mapsto y = \tan x \quad y \mapsto x = \arctan y$$

$$y = \tan x, \quad x \in]-\frac{\pi}{2}, \frac{\pi}{2}[\iff x = \arctan y, \quad y \in \mathbb{R}.$$

★ Since $\tan$ is differentiable on $I =]-\frac{\pi}{2}, \frac{\pi}{2}[$, then $\arctan$ is differentiable on

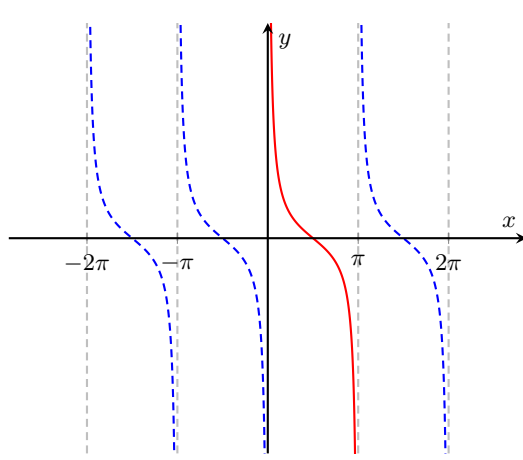
$$J \setminus \{y = \tan x : \tan'(x) = 0\} = J = \mathbb{R}.$$

★ Since $\tan$ is of class C^∞ on I , then $\arctan$ is of class C^∞ on the domain of differentiation $\mathbb{R}$.

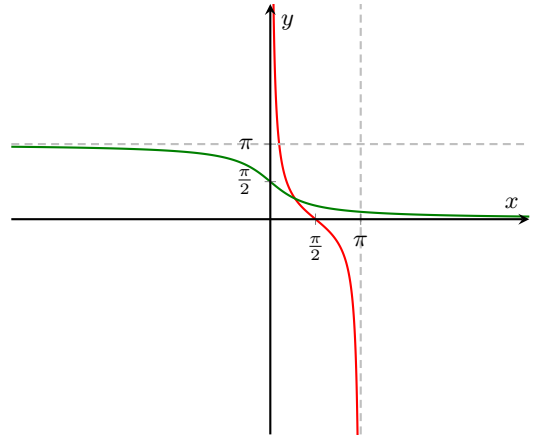
$$\arctan'(y) = (\tan^{-1})'|_{y=\tan x} = \frac{1}{\tan'(x)} = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + y^2} \quad \forall y \in \mathbb{R}.$$

5.1.4 Cotangent And Arccotangent Functions

The cotangent function is defined by $\cot x = \frac{\cos x}{\sin x}$ on $D = \mathbb{R} \setminus \{\pi + k\pi : k \in \mathbb{Z}\}$. The function $\cot : D \mapsto \mathbb{R}$ satisfies $\cot x = -\tan(x - \frac{\pi}{2})$. Hence the graph of the function $\cot$ is obtained from the graph of $\tan$ by translating horizontally towards the left and reflecting the graph of $\tan$ with respect to x axis.



$x \mapsto \cot x$ is π -periodic and odd on D .



$\cot :]0, \pi[\mapsto \mathbb{R}, \quad \arccot : \mathbb{R} \mapsto]0, \pi[$
 $x \mapsto \cot x \quad x \mapsto \arccot x$

5.2 The Power Function

The power function of exponent $\alpha \in \mathbb{R}$ is defined on $]0, \infty[$ by $p_\alpha :]0, \infty[\mapsto]0, \infty[$.
 $x \mapsto x^\alpha := e^{\alpha \ln x}$

The power function has the following properties:

- p_α is continuous, differentiable and of class C^∞ on $]0, \infty[$ and

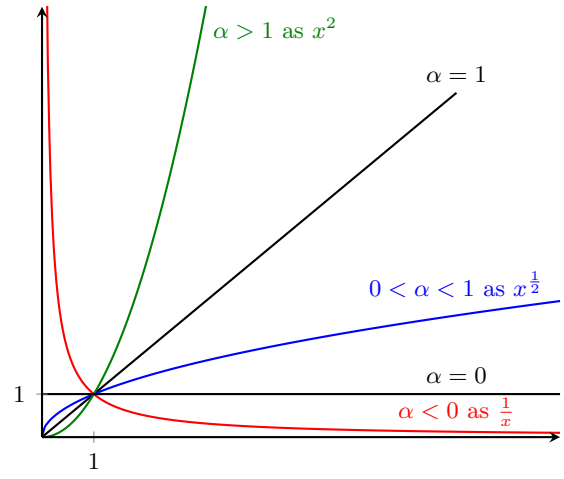
$$(x^\alpha)' = (e^{\alpha \ln x})' = \frac{\alpha}{x} e^{\alpha \ln x} = \frac{\alpha}{x} x^\alpha = \alpha x^{\alpha-1}.$$

- $x^\alpha x^\beta = x^{\alpha+\beta}$ and $(x^\alpha)^\beta = x^{\alpha\beta} \quad \forall \alpha, \beta \in \mathbb{R} \quad \forall x > 0$.

Some useful limits;

$$\lim_{x \rightarrow 0^+} x^\alpha = \lim_{x \rightarrow 0^+} e^{\alpha \ln x} = \begin{cases} 0 & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ \infty & \text{if } \alpha < 0. \end{cases}$$

$$\lim_{x \rightarrow \infty} x^\alpha = \lim_{x \rightarrow \infty} e^{\alpha \ln x} = \begin{cases} \infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0. \end{cases}$$



The graph of $x \mapsto x^\alpha$ on $]0, \infty[$.

Note that;

- If $\alpha \in \mathbb{N}$, then the power function $x \mapsto x^\alpha$ is defined and of class C^∞ on $\mathbb{R}$.
- If $\alpha \in \mathbb{Z} \setminus \mathbb{N} = \{-1, -2, \dots\}$, then the power function $x \mapsto x^\alpha$ is defined and of class C^∞ on $\mathbb{R}^*$.

5.3 Logarithmic and Exponential Functions

In this section we recall the logarithmic function and its inverse the exponential function. The comparison between Logarithm, power and exponential functions is stated as exercises.

5.3.1 Logarithm Of Base e .

- $\ln x := \int_1^x \frac{1}{t} dt$ is well defined on $]0, \infty[$.
- $\ln 1 = 0$.
- $\ln$ is differentiable on $]0, \infty[$ and $\ln' x = \frac{1}{x}$.
- $\ln' \in C^\infty(]0, \infty[) \implies \ln \in C^\infty(]0, \infty[)$.
- $\ln(xy) = \ln(x) + \ln(y) \quad \forall x, y > 0$.

Proof. Fix $y > 0$. Let $f(x) = \ln(xy) - \ln(x) - \ln(y)$. Since $f'(x) = 0$ on the interval $]0, \infty[$, then f is constant on this interval and $f(x) = f(1) = 0$.

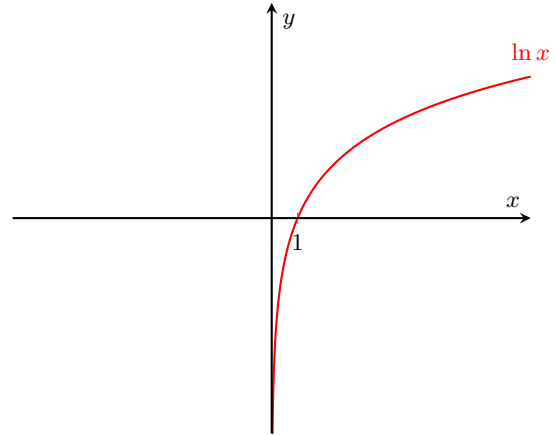
- $\ln \frac{1}{x} = -\ln x \quad \forall x > 0$.

Proof. $0 = \ln \frac{x}{x} = \ln x + \ln \frac{1}{x}$.

- $\ln \frac{x}{y} = \ln x - \ln y \quad \forall x, y > 0$.

Proof. $\ln \frac{x}{y} = \ln x + \ln \frac{1}{y} = \ln x - \ln y$.

- $\ln x^\alpha = \alpha \ln x, \quad \forall x > 0 \quad \forall \alpha \in \mathbb{R}$.



The graph of $x \mapsto \ln x$ on $]0, \infty[$.

Note that;

- If $xy > 0$, then $\ln(xy) = \ln|x| + \ln|y|$ and $\ln \frac{x}{y} = \ln|x| - \ln|y|$.
- $(\ln u(x))' = \frac{u'(x)}{u(x)}$ provided that $u(x) > 0$ and u is differentiable at x (Chain Rule).
- $\lim_{x \rightarrow 0^+} \ln x = -\infty, \quad \lim_{x \rightarrow \infty} \ln x = \infty, \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$ and $\lim_{x \rightarrow 0^+} x \ln x = 0$. (The proof left as exercise.)

5.3.2 Logarithm Of Base $a > 0, a \neq 1$.

- $\log_a(x) := \frac{\ln x}{\ln a} \quad x \in]0, \infty[$.
- $(\log_a(x))' := \frac{1}{x \ln a} \quad x \in]0, \infty[$. Hence $\log_a$ is strictly increasing if $a > 1$ and strictly decreasing if $0 < a < 1$.

5.3.3 Exponential Of Base e .

Since $\ln :]0, \infty[\mapsto \mathbb{R}$ is continuous and strictly increasing, then it is a bijection from $I =]0, \infty[$ onto $J =]\lim_{x \rightarrow 0^+} \ln x, \lim_{x \rightarrow \infty} \ln x[=]-\infty, +\infty[$. We define $\exp = \ln^{-1}$, the inverse of the function $\ln$. Usually we denote $\exp(x)$ by e^x .

$$\ln :]0, \infty[\mapsto \mathbb{R}, \quad \exp : \mathbb{R} \mapsto]0, \infty[.$$

$$x \mapsto y = \ln x \quad y \mapsto x = e^y$$

$$y = \ln x, \quad x \in]0, \infty[\iff x = e^y, \quad y \in \mathbb{R}.$$

$$e^{\ln x} = x \quad \forall x \in]0, \infty[\quad \ln(e^x) = x \quad \forall x \in \mathbb{R}.$$

★ Since $\ln$ is differentiable on $I =]0, \infty[$, then $\exp$ is differentiable on

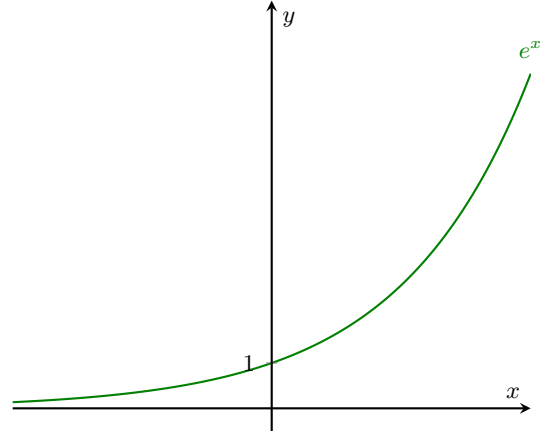
$$J \setminus \{y = \ln x : \ln'(x) = 0\} = J = \mathbb{R}.$$

★ Since $\ln$ is of class C^∞ on I , then $\exp$ is of class C^∞ on the domain of differentiation $\mathbb{R}$.

$$(\exp)'(y) = (\ln^{-1})'|_{y=\ln x} = \frac{1}{\ln'(x)} = x = e^y \quad \forall y \in \mathbb{R}.$$

Consequently, $(e^x)^{(n)} = e^x \quad \forall x \in \mathbb{R}, \forall n \in \mathbb{N}$.

- $e^x > 0 \quad \forall x \in \mathbb{R}$.
- $\exp$ is strictly increasing on $\mathbb{R}$:
 $x < 0 \implies e^x < e^0 = 1$.
 $x > 0 \implies e^x > e^0 = 1$.
- $e^{x+y} = e^x e^y \quad \forall x, y \in \mathbb{R}$.
- $e^{-x} = \frac{1}{e^x} \quad \forall x \in \mathbb{R}$.
- $e^{x-y} = \frac{e^x}{e^y} \quad \forall x, y \in \mathbb{R}$.
- $(e^x)^\alpha = e^{\alpha x} \quad \forall x, \alpha \in \mathbb{R}$.



The graph of $x \mapsto e^x$ on $\mathbb{R}$.

5.3.4 Exponential Of Base $a > 0, a \neq 1$.

Since $\log_a :]0, \infty[\mapsto \mathbb{R}$ is continuous and strictly increasing, then it is a bijection. Its inverse $\exp_a(y)$ is usually denoted by a^y and defined by $\exp_a : \mathbb{R} \mapsto]0, \infty[$.

$$\log_a :]0, \infty[\mapsto \mathbb{R}, \quad \exp_a : \mathbb{R} \mapsto]0, \infty[.$$

$$x \mapsto y = \log_a x \quad y \mapsto x = a^y$$

$$y = \log_a x, \quad x \in]0, \infty[\iff x = a^y, \quad y \in \mathbb{R}.$$

$$a^{\log_a x} = x \quad \forall x \in]0, \infty[\quad \log_a(a^x) = x \quad \forall x \in \mathbb{R}.$$

- $\exp_a$ is differentiable on $\mathbb{R}$ and $(a^x)' = (e^{x \ln a})' = \ln(a) e^{x \ln a} = (\ln a) a^x \quad \forall x \in \mathbb{R}$.
- $\lim_{x \rightarrow \infty} x^{\pm 1} a^x = \lim_{x \rightarrow \infty} x^{\pm 1} e^{x \ln a} = \begin{cases} \infty & \text{if } a > 1 \\ 0 & \text{if } a < 1. \end{cases}$

Exercise. Show, using three different ways, definition of derivative, equivalent definition of derivative and MVT the following:

1. $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$. Deduce $\lim_{x \rightarrow \pm\infty} (1 + \frac{1}{x})^x = e$ and $\lim_{x \rightarrow \pm\infty} (1 + \frac{a}{x})^x = e^a \quad (a \in \mathbb{R})$.
2. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

5.4 Hyperbolic Functions And Inverses

The functions cosine hyperbolic, sine hyperbolic and tangent hyperbolic are defined on $\mathbb{R}$ by:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

- $\cosh$ is positive ($\cosh x > 0, \forall x \in \mathbb{R}$) and even function.
- $\sinh$ and $\tanh$ are odd functions.

The hyperbolic functions satisfy the following identities:

$$\textcircled{1} \cosh x + \sinh x = e^x. \quad \textcircled{2} \cosh x - \sinh x = e^{-x}. \quad \textcircled{3} \cosh^2 x - \sinh^2 x = 1.$$

The point of coordinates $(X, Y) = (\cosh x, \sinh x)$ runs along the right branch of the hyperbola $X^2 - Y^2 = 1$ as x varies.

If we divide $\textcircled{3}$ by $\cosh^2 x > 0$, then we obtain $\textcircled{4} 1 - \tanh^2 x = \frac{1}{\cosh^2 x}$.

Also, one may verify easily that

$$\cosh(a+b) = \cosh a \cosh b + \sinh a \sinh b,$$

$$\sinh(a+b) = \cosh a \sinh b + \sinh a \cosh b$$

$$\tanh(a+b) = \frac{\tanh a + \tanh b}{1 + \tanh a \tanh b}.$$

5.4.1 Sine Hyperbolic And Inverse Sine Hyperbolic

The function **sinh** : $\mathbb{R} \mapsto \mathbb{R}$ has the following properties:

$$x \mapsto \frac{e^x - e^{-x}}{2}$$

- $\text{dom}(\sinh) = \mathbb{R}$.
- $\sinh, \cosh$ are differentiable on $\mathbb{R}$ and $\sinh' x = \cosh x, \cosh' x = \sinh x$.
- $\sinh \in C^\infty(\mathbb{R})$ and $\sinh^{(n)}(x) = \begin{cases} \cosh x & \text{if } n \text{ odd} \\ \sinh x & \text{if } n \text{ even} \end{cases}$.
- $\sinh$ is odd. $\begin{cases} \text{dom}(\sinh) = \mathbb{R} \text{ is centered at } 0. \\ \sinh(-x) = -\sinh(x) \quad \forall x \in \mathbb{R}. \end{cases}$

The sine hyperbolic function, **sinh** : $\mathbb{R} \mapsto \mathbb{R}$ has the following properties:

$$x \mapsto \sinh x$$

- Continuous on $I = \mathbb{R}$.
- Strictly increasing on $I = \mathbb{R}$:
 $\sinh'(x) = \cosh x > 0 \quad \forall x \in \mathbb{R}.$

Then by the inverse function theorem;

The function $\sinh$ is a bijection from the interval $I = \mathbb{R}$ onto the interval

$$J =]\lim_{x \rightarrow -\infty} \sinh x, \lim_{x \rightarrow \infty} \sinh x[=]-\infty, \infty[.$$

$$\sinh : \mathbb{R} \mapsto \mathbb{R}, \quad \sinh^{-1} : \mathbb{R} \mapsto \mathbb{R}.$$

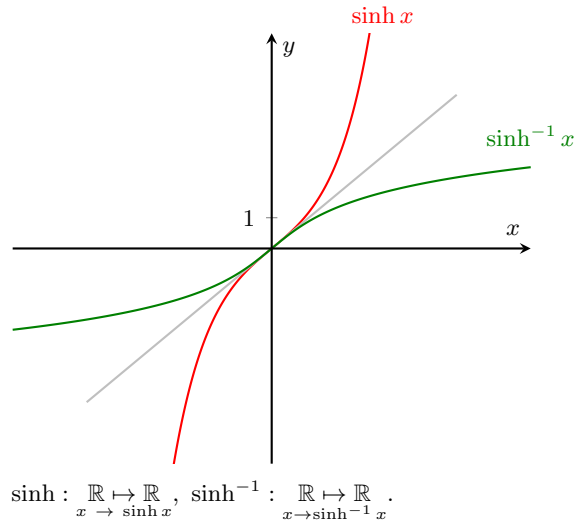
$$x \mapsto y = \sinh x, \quad y \mapsto x = \sinh^{-1} y.$$

$$y = \sinh x, \quad x \in \mathbb{R} \text{ iff } x = \sinh^{-1} y, \quad y \in \mathbb{R}.$$

★ Since $\sinh$ is differentiable on $I = \mathbb{R}$, then

$\sinh^{-1}$ is differentiable on

$$J \setminus \{y = \sinh x : \sinh'(x) = 0\} = J = \mathbb{R}.$$



★ Since $\sinh$ is of class C^∞ on $I = \mathbb{R}$, then $\sinh^{-1}$ is of class C^∞ on the domain of differentiation $\mathbb{R}$.

$$(\sinh^{-1})'(y) = (\sinh^{-1})'|_{y=\sinh x} = \frac{1}{\sinh'(x)} = \frac{1}{\cosh x} = \frac{1}{\sqrt{1 + \sinh^2(x)}} = \frac{1}{\sqrt{1 + y^2}} \quad \forall y \in \mathbb{R}.$$

Note that some references use $\operatorname{argsinh}$, arsinh or $\operatorname{arcsinh}$ to denote $\sinh^{-1}$.

5.4.2 Cosine Hyperbolic And Inverse Cosine Hyperbolic

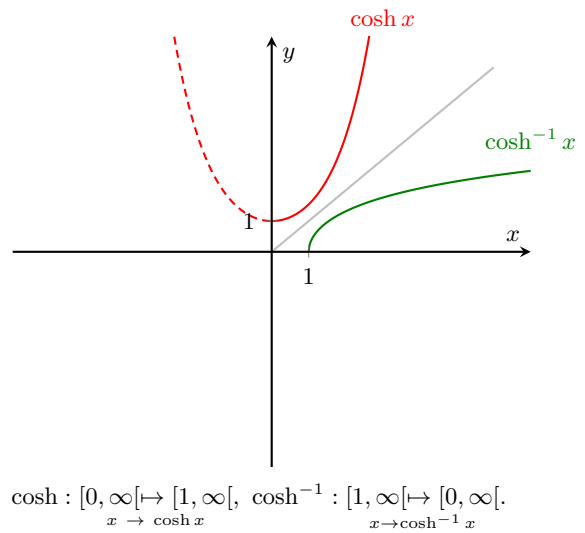
The function $\cosh : \mathbb{R} \mapsto [1, \infty[$ has the following properties:

$$x \mapsto \frac{e^x + e^{-x}}{2}$$

- $\text{dom}(\cosh) = \mathbb{R}$.
- $\cosh x = \frac{e^x + e^{-x}}{2} \geq 1 \quad \forall x \in \mathbb{R}$ and $\cosh 0 = 1$.
- $\cosh$ is differentiable on $\mathbb{R}$ and $\cosh' x = \sinh x$.
- $\cosh \in C^\infty(\mathbb{R})$ and $\cosh^{(n)}(x) = \begin{cases} \cosh x & \text{if } n \text{ even} \\ \sinh x & \text{if } n \text{ odd} \end{cases}$.
- $\cosh$ is even. $\begin{cases} \text{dom}(\cosh) = \mathbb{R} \text{ is centered at } 0. \\ \cosh(-x) = \cosh(x) \quad \forall x \in \mathbb{R}. \end{cases}$

The restriction of the cosine hyperbolic function, $\cosh : [0, \infty[\mapsto [1, \infty[$ has the following properties:

- Continuous on $I = [0, \infty[$.
 - Strictly increasing on $I = [0, \infty[$:
 $\cosh'(x) = \sinh x > 0 \quad \forall x \in]0, \infty[$.
Then by the inverse function theorem;
The function $\cosh$ is a bijection from the interval $I = [0, \infty[$ onto the interval $J = [\cosh 0, \lim_{x \rightarrow \infty} \cosh x] = [1, \infty[$.
- $\cosh : [0, \infty[\mapsto [1, \infty[$, $\cosh^{-1} : [1, \infty[\mapsto [0, \infty[$.
 $\begin{matrix} x \mapsto y = \cosh x & & y \mapsto x = \cosh^{-1} y \end{matrix}$
 $y = \cosh x$, $x \in [0, \infty[$ iff $x = \cosh^{-1} y$, $y \in [1, \infty[$.
★ Since $\cosh$ is differentiable on $I = [0, \infty[$,
then $\cosh^{-1}$ is differentiable on
 $J \setminus \{y = \cosh x : \cosh'(x) = 0\} =]1, \infty[$.



★ Since $\cosh$ is of class C^∞ on I , then $\cosh^{-1}$ is of class C^∞ on the domain of differentiation $]1, \infty[$.

$$(\cosh^{-1})'(y) = (\cosh^{-1})'|_{y=\cosh x} = \frac{1}{\cosh'(x)} = \frac{1}{\sinh x} = \frac{1}{\sqrt{\cosh^2(x) - 1}} = \frac{1}{\sqrt{y^2 - 1}} \quad \forall y \in]1, \infty[.$$

Note that some references use argcosh , arcosh or arccosh to denote $\cosh^{-1}$.

5.4.3 Tangent Hyperbolic And Inverse Tangent Hyperbolic

The function $\tanh : \mathbb{R} \mapsto]-1, 1[$ has the following properties:

$$x \mapsto \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- $\text{dom}(\tanh) = \mathbb{R}$.
- $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{1 - e^{-2x}}{1 + e^{-2x}}$.
- $\lim_{x \rightarrow -\infty} \tanh x = \lim_{x \rightarrow -\infty} \frac{e^{2x} - 1}{e^{2x} + 1} = -1$. ($y = -1$ H.A.).
- $\lim_{x \rightarrow \infty} \tanh x = \lim_{x \rightarrow \infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = 1$. ($y = 1$ H.A.).

- $\tanh$ is differentiable on $\mathbb{R}$ and

$$\tanh' x = \left(\frac{\sinh x}{\cosh x} \right)' = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x = \frac{1}{\cosh^2 x}.$$

- $\tanh \in C^\infty(\mathbb{R})$. (Quotient of C^∞ functions with denominator never vanishes)
- $\tanh$ is odd.

The tangent hyperbolic function, $\tanh : \mathbb{R} \mapsto]-1, 1[$ has the following properties:

- Continuous on $I = \mathbb{R}$.

- Strictly increasing on $I = \mathbb{R}$:

$$\tanh'(x) = \frac{1}{\cosh^2 x} > 0 \quad \forall x \in \mathbb{R}.$$

Then by the inverse function theorem;

The function $\tanh$ is a bijection from the interval $I = \mathbb{R}$ onto the interval

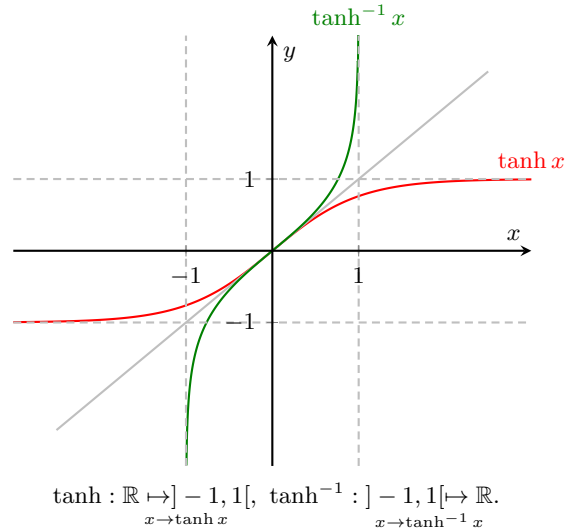
$$J =]\lim_{x \rightarrow -\infty} \tanh x, \lim_{x \rightarrow \infty} \tanh x[=]-1, 1[.$$

$$\tanh : \mathbb{R} \mapsto]-1, 1[, \quad \tanh^{-1} :]-1, 1[\mapsto \mathbb{R}.$$

$$y = \tanh x, \quad x \in \mathbb{R} \text{ iff } x = \tanh^{-1} y, \quad y \in]-1, 1[.$$

★ Since $\tanh$ is differentiable on $I = \mathbb{R}$, then $\tanh^{-1}$ is differentiable on

$$J \setminus \{y = \tanh x : \tanh'(x) = 0\} = J =]-1, 1[.$$



★ Since $\tanh$ is of class C^∞ on $I = \mathbb{R}$, then $\tanh^{-1}$ is of class C^∞ on the domain of differentiation $] -1, 1[$.

$$(\tanh^{-1})'(y) = (\tanh^{-1})'|_{y=\tanh x} = \frac{1}{\tanh'(x)} = \frac{1}{1 - \tanh^2 x} = \frac{1}{1 - y^2} \quad \forall y \in]-1, 1[.$$

Note that some references use $\operatorname{argtanh}$, artanh or $\operatorname{arctanh}$ to denote $\tanh^{-1}$.

One can show directly by a simple calculations the following identities:

1. $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1}) \quad \forall x \in \mathbb{R}.$
2. $\cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1}) \quad \forall x \in [1, \infty[.$
3. $\tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad \forall x \in]-1, 1[.$

5.5 Exercises

Exercise 1. Let $f(x) = \arctan\left(\frac{1+x}{1-x}\right)$ and $g(x) = \arctan x$.

1. Give the domains of definition of f and g .
2. Calculate $f'(x)$.
3. Give a relation between f and g .

Exercise 2. Show that

$$\arctan x + \arctan \frac{1}{x} = \begin{cases} \frac{\pi}{2} & \text{if } x > 0 \\ -\frac{\pi}{2} & \text{if } x < 0. \end{cases}$$

Exercise 3.(property) Show that $\lim_{x \rightarrow \infty} \ln x = \infty$ and $\lim_{x \rightarrow 0^+} \ln x = -\infty$.

Exercise 4.(property)

1. Show that $\ln x < \sqrt{x} \quad \forall x \in]0, \infty[$.

2. Deduce that $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$.

3. Deduce that $\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty$.

Exercise 5.(property) Let $a > 0$ (or $a \in \mathbb{R}$), $b > 0$. Show that

1. $\lim_{x \rightarrow \infty} \frac{(\ln x)^a}{x^b} = 0$.

2. $\lim_{x \rightarrow \infty} \frac{e^{bx}}{x^a} = \infty$.

3. $\lim_{x \rightarrow \infty} \frac{e^{bx}}{(\ln x)^a} = \infty$.

Exercise 6. Let $a \in \mathbb{R}, b > 0$. Show that $\lim_{x \rightarrow 0^+} x^b |\ln x|^a = 0$.

Exercise 7. Let $a \in \mathbb{R}, b > 0$. Show that $\lim_{x \rightarrow -\infty} |x|^a e^{bx} = 0$.

Exercise 8. Let $f(x) = \arcsin \frac{2x}{1+x^2} + \arccos \frac{1-x^2}{1+x^2}$.

1. Find $\text{dom}(f)$.

2. Study the continuity of f and determine $\lim_{x \rightarrow \infty} f(x)$.

3. Show that $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2} \quad \forall x > 0$.

4. Deduce that $\lim_{x \rightarrow \infty} (\pi x - 2x \arctan x) = 2$.

6.1 Taylor And Finite Expansions

The finite expansion of a function near a point $x_0 \in \mathbb{R}$ is the representation of the function as sum of a polynomial of a certain degree and an infinitesimal function ($\xrightarrow{x \rightarrow x_0} 0$). In a neighborhood of x_0 one can approximate complicated functions by polynomials which are easy to handle.

6.1 Definition. (Finite Expansion)

Let I be an open interval, $x_0 \in I$ or boundary point of I , $n \in \mathbb{N}$ and f be a function defined on I except possibly at x_0 . We say that f has finite expansion of order n near x_0 " $f.e_n(x_0)$ " if there exists a polynomial p of degree $\leq n$ and a function $\varepsilon : I \setminus \{x_0\} \mapsto \mathbb{R}$ verifies $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$ such that

$$f(x) = p(x) + (x - x_0)^n \varepsilon(x) \quad \forall x \in I \setminus \{x_0\}.$$

- $p(x)$ is called the principle part of the finite expansion.
- $(x - x_0)^n \varepsilon(x)$ is called the remainder of the finite expansion (or the error of the approximation).

6.2 Remark.

- We will use the notation "little oh" $o(x - x_0)^n = (x - x_0)^n \varepsilon(x)$ with $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$.
- The change of variable $t = x - x_0$ ($t \xrightarrow{x \rightarrow x_0} 0$) allows us to perform f.e near zero instead of $x_0 \in \mathbb{R}$.
- The change of variable $t = \frac{1}{x}$ ($t \xrightarrow{x \rightarrow \pm\infty} 0$) allows us to perform f.e near zero "instead of $\pm\infty$ ".

We will limit ourselves to study the finite expansion near zero.

6.3 Example. Show that $f(x) = \frac{1}{1-x}$ has $f.e_n(0)$, $n \in \mathbb{N}$.

Solution.

$$\begin{aligned} \frac{1}{1-x} &= \frac{1 - x^{n+1} + x^{n+1}}{1-x} \\ &= \frac{1 - x^{n+1}}{1-x} + \frac{x^{n+1}}{1-x} \\ &= \frac{(1-x)(1+x+x^2+\dots+x^n)}{1-x} + \frac{x^{n+1}}{1-x} && (\text{Factorization}) \\ &= \underbrace{1+x+x^2+\dots+x^n}_{p(x)} + x^n \underbrace{\frac{x}{1-x}}_{\varepsilon(x)} && (\varepsilon(x) = \frac{x}{1-x} \xrightarrow{x \rightarrow 0} 0) \\ &= p(x) + o(x^n). \end{aligned}$$

f.e._n(0) for the functions $x \mapsto \frac{1}{1-x}$ and $x \mapsto \frac{1}{1+x}$:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + o(x^n).$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + o(x^n). \quad (\text{replace } x \text{ by } -x)$$

6.4 Example. Show that $f(x) = \begin{cases} x^{n+1} \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ has f.e._n(0).

Solution. $f(x) = 0 + 0x + \cdots + 0x^n + x^n(x \sin \frac{1}{x}) = 0 + x^n \varepsilon(x)$ with $\varepsilon(x) = x \sin \frac{1}{x} \xrightarrow{x \rightarrow 0} 0$.

6.5 Property. (Uniqueness Of Finite Expansion)

If $f : I \setminus \{x_0\} \mapsto \mathbb{R}$ has f.e._n(0), then the principle part of the f.e is unique.

Proof. Assume that for all $x \in I \setminus \{x_0\}$ we have

$$f(x) = p_1(x) + x^n \varepsilon_1(x) \text{ with } p_1(x) = a_0 + a_1x + \cdots + a_nx^n \text{ and } \varepsilon_1(x) \xrightarrow{x \rightarrow 0} 0$$

$$f(x) = p_2(x) + x^n \varepsilon_2(x) \text{ with } p_2(x) = b_0 + b_1x + \cdots + b_nx^n \text{ and } \varepsilon_2(x) \xrightarrow{x \rightarrow 0} 0$$

Claim $p_1 = p_2$:

$$\begin{aligned} f(x) = f(x) &\implies p_1(x) + x^n \varepsilon_1(x) = p_2(x) + x^n \varepsilon_2(x) \\ &\implies (a_0 - b_0) + (a_1 - b_1)x + \cdots + (a_n - b_n)x^n = x^n(\varepsilon_1(x) - \varepsilon_2(x)) \quad (\text{let } x \rightarrow 0) \\ &\implies a_0 = b_0 \text{ and } (a_1 - b_1) + \cdots + (a_n - b_n)x^{n-1} = x^{n-1}(\varepsilon_1(x) - \varepsilon_2(x)) \end{aligned}$$

By repeating the same procedure we obtain $a_0 = b_0, a_1 = b_1, \dots, a_n = b_n$. Hence $p_1 = p_2$. $\square$

6.6 Property. Suppose $f : I \setminus \{x_0\} \mapsto \mathbb{R}$ has f.e._n(0). Then

1. The principle part of the f.e of an even function involves only even powers.
2. The principle part of the f.e of an odd function involves only odd powers.

Proof.

1. Since f has f.e._n(0), then $f(x) = a_0 + a_1x + \cdots + a_nx^n + x^n \varepsilon(x) \quad \forall x \in I \setminus \{x_0\}$ with $\varepsilon(x) \xrightarrow{x \rightarrow 0} 0$.

Now,

$$\begin{aligned} f(x) = f(-x) &\implies f(x) = a_0 - a_1x + a_2x^2 - a_3x^3 + \cdots + (-1)^n a_nx^n + o(x^n) \\ &\implies a_1 = -a_1, \quad a_3 = -a_3, \dots \quad (\text{uniqueness of f.e}) \\ &\implies a_k = 0 \text{ for any odd integer } k \in [1, n] \\ &\implies f(x) = a_0 + a_2x^2 + \cdots + (-1)^n a_nx^n + o(x^n) \end{aligned}$$

2. Similarly as 1.

$\square$

6.7 Property. If f has $f.e_n(0)$ and $m < n$, then f has $f.e_m(0)$.

Proof. Assume f has $f.e_n(0)$. Then

$f(x) = a_0 + a_1x + \cdots + a_mx^m + a_{m+1}x^{m+1} + \cdots + a_nx^n + x^n\varepsilon(x)$ with $\varepsilon(x) \xrightarrow{x \rightarrow 0} 0$. Hence,

$$\begin{aligned} f(x) &= a_0 + a_1x + \cdots + a_mx^m + a_{m+1}x^{m+1} + \cdots + a_nx^n + x^n\varepsilon(x) \\ &= a_0 + a_1x + \cdots + a_mx^m + x^m[a_{m+1}x^1 + \cdots + a_nx^{n-m} + x^{n-m}\varepsilon(x)] \\ &= a_0 + a_1x + \cdots + a_mx^m + o(x^m). \end{aligned}$$

□

If $f(x) = 1 + 2x + x^4 + 3x^7 + o(x^7)$, then for example f has $f.e_5(0)$ and $f(x) = 1 + 2x + x^4 + o(x^5)$.

Recall.

Let I be open interval, $x_0 \in I$ and $f : I \mapsto \mathbb{R}$. Then f is differentiable at x_0 iff f has $f.e_1(x_0)$.

Exercise 1. Let I be open interval, $x_0 \in I$ and $f : I \setminus \{x_0\} \mapsto \mathbb{R}$. Assume that

$$f(x) = a_0 + a_1(x - x_0) + (x - x_0)\varepsilon(x) \quad \forall x \in I \setminus \{x_0\}.$$

Show that f has an extension by continuity $\tilde{f}$ at x_0 and $\tilde{f}$ is differentiable at x_0 with $\tilde{f}'(x_0) = a_1$.

6.8 Property. (Taylor Formula With Peano's Remainder)

If f is n -times differentiable at x_0 , then f has $f.e_n(x_0)$ given by

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!}f^{(n)}(x_0) + o(x - x_0)^n.$$

The case $x_0 = 0$:

If f is n -times differentiable at 0, then f has $f.e_n(0)$ given by

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + o(x^n).$$

Proof. Let us proof the case $x_0 = 0$. Let

$$p(x) := f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0).$$

$$r(x) := f(x) - p(x) = f(x) - f(0) - xf'(0) - \frac{x^2}{2!}f''(0) - \cdots - \frac{x^n}{n!}f^{(n)}(0).$$

$$\varepsilon(x) := \frac{r(x)}{x^n}.$$

Note that r is n -times differentiable at 0 and $r(0) = r'(0) = \cdots = r^{(n)}(0) = 0$. Assume without loss of generality $x > 0$, then by repeated application of the Mean Value Theorem we obtain

$$\begin{aligned} r(x) &= r(x) - r(0) = xr'(c_1) & (0 < c_1 < x) \\ &= x(r'(c_1) - r'(0)) = xc_1r''(c_2) & (0 < c_2 < c_1 < x) \\ &\vdots & \vdots \\ &= xc_1c_2 \cdots c_{n-2}r^{(n-1)}(c_{n-1}) & (0 < c_{n-1} < \cdots < c_2 < c_1 < x) \end{aligned}$$

Hence, by the help of the substitution theorem we obtain

$$|\varepsilon(x)| = \left| \frac{r(x)}{x^n} \right| \leq \left| \frac{r^{(n-1)}(c_{n-1})}{x} \right| \leq \left| \frac{r^{(n-1)}(c_{n-1})}{c_{n-1}} \right| = \left| \frac{r^{(n-1)}(c_{n-1}) - r^{(n-1)}(0)}{c_{n-1} - 0} \right| \xrightarrow{x \rightarrow 0} r^{(n)}(0) = 0.$$

□

Note that;

- The converse of Taylor Formula With Peano's Remainder is not necessarily true for $n \geq 2$. The function $f(x) = \begin{cases} x^3 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ has $f.e_2(0)$ but it is not twice differentiable at 0.
- If f is n -times differentiable at 0 and $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n + o(x^n)$, then by uniqueness of f.e we obtain $a_0 = f(0)$, $a_1 = f'(0)$, $a_2 = \frac{f''(0)}{2!}, \dots, a_n = \frac{f^{(n)}(0)}{n!}$.

Finite expansion of elementary functions near $x_0 = 0$:

$$\begin{aligned}
 \textcircled{0} f(x) &= f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + o(x^n) \quad (\text{provided } f^{(n)}(0) \exists) \\
 \textcircled{1} e^x &= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + o(x^n) \quad (f^{(n)}(0) = 1) \\
 \textcircled{2} \cosh x &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2n}}{(2n)!} + o(x^{2n}) \quad (f^{(2n)}(0) = 1 \wedge f^{(2n+1)}(0) = 0) \\
 \textcircled{3} \sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+1}) \quad (f^{(2n)}(0) = 0 \wedge f^{(2n+1)}(0) = 1) \\
 \textcircled{4} \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n}) \quad (f^{(n)}(0) = \cos(n\frac{\pi}{2})) \\
 \textcircled{5} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+1}) \quad (f^{(n)}(0) = \sin(n\frac{\pi}{2})) \\
 \textcircled{6} \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n!} + o(x^n) \quad \left(f^{(n)}(0) = \frac{(-1)^{n-1}(n-1)!}{(1+0)^n}\right) \\
 \textcircled{7} (1+x)^\alpha &= 1 + \alpha x + \alpha(\alpha-1) \frac{x^2}{2!} + \alpha(\alpha-1)(\alpha-2) \frac{x^3}{3!} + \dots + \alpha(\alpha-1) \dots (\alpha-n+1) \frac{x^n}{n!} + o(x^n). \\
 \textcircled{7} \frac{1}{1+x} &= (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + o(x^n). \\
 \textcircled{7}' \frac{1}{1-x} &= 1 + x + x^2 + x^3 + \dots + x^n + o(x^n).
 \end{aligned}$$

6.2 Operations On Finite Expansions

6.9 Property. (Linear Combinations And Products)

Let f, g be two functions defined on I and have $f.e_n(0)$ given by $f(x) = p(x) + o(x^n)$ and $g(x) = q(x) + o(x^n)$ where p, q are polynomials of degree $\leq n$. Then

1. $f + g$ has $f.e_n(0)$ and $f(x) + g(x) = p(x) + q(x) + o(x^n)$.
2. $\alpha f, (\alpha \in \mathbb{R})$, has $f.e_n(0)$ and $\alpha f(x) = \alpha p(x) + o(x^n)$.
3. fg has $f.e_n(0)$ and $f(x)g(x) = [p(x)q(x) \text{ discarding powers } > n] + o(x^n)$.

Proof.

1. $f(x) + g(x) = p(x) + q(x) + o(x^n) + o(x^n) = p(x) + q(x) + o(x^n)$.
2. $\alpha f(x) = \alpha p(x) + \alpha o(x^n) = \alpha p(x) + o(x^n)$.
3. $f(x)g(x) = \left[\begin{array}{l} \text{part of } p(x)q(x) \text{ with deg } \leq n \\ \text{part of } p(x)q(x) \text{ with deg } > n \end{array} \right] + o(x^n) = \left[\begin{array}{l} \text{part of } p(x)q(x) \text{ with deg } \leq n \\ \text{part of } p(x)q(x) \text{ with deg } > n \end{array} \right] + o(x^n)$.

6.10 Examples.

1. Find $f.e_3(0)$ of $f(x) = \sin x + \cos x$.

$$f(x) = \left[x - \frac{x^3}{3!}\right] + \left[1 - \frac{x^2}{2!}\right] + o(x^3) = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + o(x^3).$$

2. Find $f.e_3(0)$ of $f(x) = 3 \sin x + 2 \cos x$.

$$f(x) = 3\left[x - \frac{x^3}{3!}\right] + 2\left[1 - \frac{x^2}{2!}\right] + o(x^3) = 2 + 3x - x^2 - \frac{x^3}{2} + o(x^3).$$

3. Find $f.e_3(0)$ of $f(x) = \sin x \cos x$.

$$f(x) = \left[x - \frac{x^3}{3!}\right]\left[1 - \frac{x^2}{2!}\right] + o(x^3) = x - \frac{x^3}{2} - \frac{x^3}{6} + o(x^3) = x - \frac{2}{3}x^3 + o(x^3).$$

4. Find $f.e_7(0)$ of $f(x) = (\cos x - 1)(\sin x - x)$.

$$f(x) = \left[-\frac{x^2}{2!} + \frac{x^4}{4!}\right]\left[-\frac{x^3}{3!} + \frac{x^5}{5!}\right] + o(x^7) = \frac{x^5}{2!3!} - \frac{x^7}{2!5!} - \frac{x^7}{3!4!} + o(x^7).$$

6.11 Property. (Composition Of Finite Expansions)

If f, u have $f.e_n(0)$ and $u(x) \xrightarrow{x \rightarrow 0} 0$, then $f \circ u(x) = f(u(x))$ has $f.e_n(0)$ given by

$$f(u(x)) = [p(q(x)) \text{ discarding powers } > n] + o(x^n)$$

where p and q are the principle parts of the $f.e_n(0)$ for f and u respectively.

For $u(x) \xrightarrow{x \rightarrow 0} 0$ and u has $f.e_n(0)$ given by $u(x) = q(x) + o(x^n)$, we have

$$\frac{1}{1-u(x)} = \frac{1}{1-q(x)+o(x^n)} = [1 + q(x) + q^2(x) + \cdots + q^n(x) \text{ discarding powers } > n] + o(x^n)$$

6.12 Example.

1. Find $f.e_5(0)$ of $\frac{1}{\cos x}$.

2. Deduce the $f.e_5(0)$ of $\tan x$.

Solution.

$$\begin{aligned} 1. \frac{1}{\cos x} &= \frac{1}{1 - \underbrace{(x^2/2 - x^4/24)}_{q(x)} + o(x^5)} = [1 + q(x) \cdots + q^5(x) \text{ discarding powers } > n] + o(x^5) \\ &= 1 + (x^2/2 - x^4/24) + (x^4/4) + o(x^5) = 1 + x^2/2 + 5x^4/24 + o(x^5). \end{aligned}$$

$$\begin{aligned} 2. \tan x &= \sin x \frac{1}{\cos x} = \left[x - \frac{x^3}{6} + \frac{x^5}{120}\right] \left[1 + x^2/2 + 5x^4/24\right] + o(x^5) \\ &= x - \frac{x^3}{6} + \frac{x^5}{120} + x^3/2 - x^5/12 + 5x^5/24 + o(x^5) \\ &= x + 1/3x^3 + 2/15x^5 + o(x^5). \end{aligned}$$

6.13 Example. Find $f.e_3(0)$ of $f(x) = \ln(1 + \cos x)$.

Solution.

$$\begin{aligned} f(x) &= \ln(1 + \cos x) = \ln(1 + 1 - \frac{x^2}{2!} + o(x^3)) = \ln(2[1 - \frac{x^2}{4} + o(x^2)]) \\ &= \ln 2 + \ln(1 - \frac{x^2}{4} + o(x^2)) \quad (\ln(1 + u(x)); u(x) \xrightarrow{x \rightarrow 0} 0) \\ &= \ln 2 - \frac{x^2}{4} + o(x^3). \end{aligned}$$

6.14 Property. (Quotient Of Finite Expansion)

If ① N , D have $f.e_n(0)$ and ② $D(x) \xrightarrow{x \rightarrow 0} \ell \neq 0$, then $\frac{N}{D}$ has $f.e_n(0)$ obtained by performing algorithm like the division algorithm for polynomials but following increasing powers of x .

Proof. $\frac{N(x)}{D(x)} = \frac{1}{\ell} N(x) \frac{1}{D(x)/\ell} = \frac{1}{\ell} N(x) \frac{1}{1-u(x)}$ with $u(x) = 1 - D(x)/\ell \xrightarrow{x \rightarrow 0} 0$. □

6.15 Example. Find the $f.e_5(0)$ of $\tan x = \frac{\sin x}{\cos x}$.

- $\sin x$ has $f.e_5(0)$ and

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5).$$

- $\cos x$ is continuous at 0 and $\cos(0) \neq 0$.

- $\cos x$ has $f.e_5(0)$ and

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^5).$$

- Then $\tan x$ has $f.e_5(0)$ and

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + o(x^5).$$

- Since $\tan$ is odd function, then

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + o(x^6).$$

$$\begin{array}{r} x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5) \quad | \quad 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^5) \\ \hline x - \frac{x^3}{2} + \frac{x^5}{24} + o(x^5) \quad | \quad x + \frac{x^3}{3} + \frac{2}{15}x^5 + o(x^5) \\ \hline \frac{x^3}{3} - \frac{x^5}{30} + o(x^5) \\ \hline \frac{x^3}{3} - \frac{x^5}{6} + o(x^5) \\ \hline \frac{2}{15}x^5 + o(x^5) \\ \hline \frac{2}{15}x^5 + o(x^5) \\ \hline 0 + o(x^5) \end{array}$$

Similarly on can obtain by two different ways $\tanh x = x - \frac{x^3}{3} + \frac{2}{15}x^5 + o(x^6)$.

Special Cases By Examples: Quotients with $D(0) = 0$.

- Find the $f.e_5(0)$ of $\frac{\cos x - 1}{x^2}$.

Solution. $\frac{\cos x - 1}{x^2} = \frac{-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + 0x^7}{x^2} + o(x^5) = -\frac{1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + o(x^5).$

- Find the $f.e_3(0)$ of $f(x) = \frac{\sinh x - x}{\cosh x - 1}$.

Solution. Since $D(0) = 0$, $D(x) = \frac{x^2}{2!} + \dots = x^2[1/2 + \dots]$ and $N(x) = \frac{x^3}{3!} + \dots = x^2[x/6 + \dots]$, then we can cancel by x^2 . So to obtain $f.e_3(0)$ for $f(x) = \frac{N(x)}{D(x)}$ we need $f.e$ of order $3 + 2 = 5$ for both functions N and D .

$$\begin{aligned} f(x) &= \frac{\sinh x - x}{\cosh x - 1} = \frac{\frac{x^3}{3!} + \frac{x^5}{5!} + o(x^5)}{\frac{x^2}{2!} + \frac{x^4}{4!} + o(x^5)} \\ &= \frac{x^2 \left[\frac{x}{6} + \frac{x^3}{120} + o(x^3) \right]}{x^2 \left[\frac{1}{2} + \frac{x^2}{24} + o(x^3) \right]} \\ &= \frac{\frac{x}{3} + \frac{x^3}{60} + o(x^3)}{1 + \frac{x^2}{12} + o(x^3)} \\ &= \left[\frac{x}{3} + \frac{x^3}{60} + o(x^3) \right] \left[1 - \frac{x^2}{12} + o(x^3) \right] \\ &= \frac{x}{3} - \frac{x^3}{36} + \frac{x^3}{60} + o(x^3) \\ &= \frac{x}{3} - \frac{x^3}{90} + o(x^3). \end{aligned}$$

Or by using division algorithm:

$$\begin{array}{r} \frac{x}{3} + \frac{x^3}{60} + o(x^3) \quad | \quad 1 + \frac{x^2}{12} + o(x^3) \\ \hline \frac{x}{3} + \frac{x^3}{36} + o(x^3) \quad | \quad \frac{x}{3} - \frac{x^3}{90} + o(x^3) \\ \hline -\frac{x^3}{90} + o(x^3) \\ \hline -\frac{x^3}{90} + o(x^3) \\ \hline 0 + o(x^3) \end{array}$$

Asymptotic Finite Expansions By Example: In the case $f(x) \xrightarrow{x \rightarrow 0} \infty$, It is possible to find an asymptotic expansion of $f(x)$ in increasing negative and positive powers of x :

$$f(x) = \frac{a_{-m}}{x^m} + \frac{a_{-m+1}}{x^{m-1}} + \cdots + \frac{a_{-1}}{x} + a_0 + a_1x + \cdots + a_nx^n + o(x^n).$$

This form helps to understand better how f tends to infinity.

Find the asymptotic expansion to order 2 near 0 of $f(x) = \frac{\sinh x}{\cosh x - 1}$.

Solution.

$$\begin{aligned} f(x) &= \frac{\sinh x}{\cosh x - 1} = \frac{x[1 + \cdots]}{x^2[\frac{1}{2} + \cdots]} = \frac{x \text{ f.e}_3(0)}{x^2 \text{ f.e}_3(0)} = \frac{\text{f.e}_4(0)}{\text{f.e}_5(0)}. \\ f(x) &= \frac{x + \frac{x^3}{3!} + o(x^4)}{\frac{x^2}{2!} + \frac{x^4}{4!} + o(x^5)} = \frac{x[1 + \frac{x^2}{3!} + o(x^3)]}{x^2[\frac{1}{2} + \frac{x^2}{4!} + o(x^3)]} = \frac{2}{x} \frac{1 + \frac{x^2}{6} + o(x^3)}{1 + \frac{x^2}{12} + o(x^3)} \\ &= \frac{2}{x} \left[1 + \frac{x^2}{6} + o(x^3)\right] \left[1 - \frac{x^2}{12} + o(x^3)\right] = \frac{2}{x} \left[1 - \frac{x^2}{12} + \frac{x^2}{6} + o(x^3)\right] = \frac{2}{x} + \frac{1}{6}x + o(x^2). \end{aligned}$$

$f(x) = \frac{2}{x} + o(1)$ shows that f is asymptotic to the function $h(x) = \frac{2}{x}$.

6.16 Property. (Finite Expansion Of Primitives)

If f has $\text{f.e}_n(0)$ given by $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + o(x^n)$, then $F(x) = \int f(x) dx$ the primitive of f near zero has $\text{f.e}_{n+1}(0)$ and

$$F(x) = F(0) + a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} + \cdots + a_n\frac{x^{n+1}}{n+1} + o(x^{n+1}).$$

Proof. Let $g(x) = f(x) - [a_0 + a_1x + a_2x^2 + \cdots + a_nx^n]$, then $\frac{g(x)}{x^n} \xrightarrow{x \rightarrow 0} 0$. Let $G(x) = F(x) - \left[F(0) + a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} + \cdots + a_n\frac{x^{n+1}}{n+1}\right]$. Then $\lim_{x \rightarrow 0} \frac{|G(x)|}{|x^{n+1}|} = 0$ by mvt and substitution theorem. $\square$

6.17 Corollary.

- ① $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n-1} \frac{x^n}{n!} + o(x^n)$
- ② $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$
- ③ $\operatorname{argtanh} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots + \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$
- ④ $\arcsin x = x + \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \cdots + o(x^{2n+2})$
- ⑤ $\operatorname{argsinh} x = x - \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \cdots + o(x^{2n+2})$
- ⑥ $\arccos x = \frac{\pi}{2} - \arcsin x.$

Proof. Exercise. $\square$

6.3 Equivalent And Finite Expansion

In this section the symbol x_0 will denote any of $x_0, x_0^+, x_0^-, \infty, -\infty$.

6.18 Definition. Let f and g be two functions defined on $I(x_0) \setminus \{x_0\}$. We say that f is equivalent to g in a neighborhood of x_0 , and we write $f \underset{x \rightarrow x_0}{\sim} g$, if there exists a function ε defined on $I(x_0) \setminus \{x_0\}$ such that

- $f(x) = (\varepsilon(x) + 1)g(x)$ on $I(x_0) \setminus \{x_0\}$
- $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$.

Practical Characterization

$$f \underset{x \rightarrow x_0}{\sim} g \Leftrightarrow \frac{f(x)}{g(x)} \xrightarrow{x \rightarrow x_0} 1 \quad (\text{provided } g(x) \neq 0 \text{ in a neighborhood of } x_0 \text{ except possibly at } x_0).$$

Since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, then $\sin x \underset{x \rightarrow 0}{\sim} x$.

6.19 Property. Let $x_0 \in \overline{\mathbb{R}}$.

1. $f \underset{x \rightarrow x_0}{\sim} g \implies f$ and g have same sign in some $I(x_0) \setminus \{x_0\}$ and on $I(x_0)$ if both defined at x_0 .
2. $\lim_{x \rightarrow x_0} f(x) = \ell$ ($\ell \neq 0$) $\implies f \underset{x \rightarrow x_0}{\sim} \ell$.
3. $f \underset{x \rightarrow x_0}{\sim} g$ and $\lim_{x \rightarrow x_0} g(x) = \ell \implies \lim_{x \rightarrow x_0} f(x) = \ell$.
4. $f \underset{x \rightarrow x_0}{\sim} g \iff g \underset{x \rightarrow x_0}{\sim} f$.

6.20 Property. Let $f, g, \tilde{f}$ and $\tilde{g}$ be functions defined on $I(x_0) \setminus \{x_0\}$. If $f \underset{x \rightarrow x_0}{\sim} g$ and $\tilde{f} \underset{x \rightarrow x_0}{\sim} \tilde{g}$,

1. $f \cdot \tilde{f} \underset{x \rightarrow x_0}{\sim} g \cdot \tilde{g}$.
2. $\frac{f}{\tilde{f}} \underset{x \rightarrow x_0}{\sim} \frac{g}{\tilde{g}}$. (provided $\tilde{f}, \tilde{g} \neq 0$ in a neighborhood of x_0 except possibly at x_0)
3. $f^\alpha \underset{x \rightarrow x_0}{\sim} g^\alpha$. (provided that: $f, g > 0$ in a neighborhood of x_0 . In particular, no conditions if $\alpha \in \mathbb{N}$ and $f, g \neq 0$ in a neighborhood of x_0 except possibly at x_0 if $\alpha \in \mathbb{Z} \setminus \mathbb{N}$.)

Attentions:

- Writing $f(x) \underset{x \rightarrow x_0}{\sim} 0$ means that $f(x) = (1 + \varepsilon(x))0 = 0$ in a neighborhood of x_0 which is possible, but in general when you write $f(x) \underset{x \rightarrow x_0}{\sim} 0$ you are making a mistake.

- You should never add equivalents without checking. $x \underset{x \rightarrow \infty}{\sim} x+1$ and $-x \underset{x \rightarrow \infty}{\sim} -x+5$ but $0 \not\underset{x \rightarrow \infty}{\sim} 6$.

- You should never compose equivalents without checking. $x \underset{x \rightarrow \infty}{\sim} x+1$ but $e^x \not\underset{x \rightarrow \infty}{\sim} e^{x+1}$.

Logarithm: $f(x) \underset{x \rightarrow x_0}{\sim} g(x)$ and $\lim_{x \rightarrow x_0} g(x) = \ell \neq 1 \implies \ln(f(x)) \underset{x \rightarrow x_0}{\sim} \ln(g(x))$.

$1+x \underset{x \rightarrow 0}{\sim} 1+x^2$ but $\ln(1+x) \not\underset{x \rightarrow 0}{\sim} \ln(1+x^2)$.

6.21 Property. (Finite Expansion And Equivalent)

If in a neighborhood of zero $f(x) = a_k x^k + a_{k+1} x^{k+1} + \cdots + a_n x^n + o(x^n)$ with $a_k \neq 0$, then $f(x) \underset{x \rightarrow 0}{\sim} a_k x^k$. (i.e. f is equivalent to the first non zero term in the finite expansion)

For example $e^x \underset{x \rightarrow 0}{\sim} 1$, $e^x - 1 \underset{x \rightarrow 0}{\sim} x$, $e^x - 1 - x \underset{x \rightarrow 0}{\sim} \frac{x^2}{2!}$, $\sin x \underset{x \rightarrow 0}{\sim} x$ and $\cos x - 1 \underset{x \rightarrow 0}{\sim} -\frac{x^2}{2!}$.

6.22 Property. (Finite Expansion And Tangent)

If in a neighborhood of zero $f(x) = a_0 + a_1 x + \cdots + a_n x^n + o(x^n)$, then the equation of the tangent (T) to the representative curve (C) of f at $x = 0$ is given by

$$Y_T = a_0 + a_1 x.$$

The relative position of the tangent (T) with respect to (C) near 0 is determined by the first nonzero term after $a_1 x$ in the finite expansion.

Proof. Evident. ($Y_T = f(0) + (x - 0)f'(0) = a_0 + a_1 x$) □

6.23 Examples.

1. Let $f(x) = 1 - x + 3x^2 + o(x^2)$. Then the equation of the tangent (T) to the representative curve (C) of f at $x = 0$ is given by $Y_T = 1 - x$. Since $f(x) - Y_T \underset{x \rightarrow 0}{\sim} 3x^2 \geq 0$, then the curve (C) is above (T) near 0.
2. Let $f(x) = 1 - x - 3x^3 + o(x^3)$. Then the equation of the tangent (T) to the representative curve (C) of f at $x = 0$ is given by $Y_T = 1 - x$. Since $f(x) - Y_T \underset{x \rightarrow 0}{\sim} -3x^3 \begin{cases} > 0 & \text{if } x < 0 \\ < 0 & \text{if } x > 0. \end{cases}$, then the curve (C) is above (T) near 0^- and then the curve (C) is below (T) near 0^+ .

How to find $\lim_{x \rightarrow 0} \frac{N(x)}{D(x)}$ using finite expansion:

We start by finding an equivalent for $D(x)$. Suppose $D(x) \underset{x \rightarrow 0}{\sim} b_k x^k$. Then it is sufficient to find $f.e_k(0)$ for $N(x)$. Three cases may occur:

First case: $N(x) \underset{x \rightarrow 0}{\sim} a_p x^p$ with $p < k$, then

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{a_p x^p}{b_k x^k} = \pm \infty \text{ if } (k - p) \text{ is even.}$$

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{a_p x^p}{b_k x^k} \nexists \text{ if } (k - p) \text{ is odd.}$$

Second case: $N(x) \underset{x \rightarrow 0}{\sim} a_p x^p$ with $p = k$, then

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{a_p x^p}{b_k x^k} = \frac{a_k}{b_k}.$$

Third case: $N(x) = o(x^k)$, then

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{x^k \varepsilon(x)}{b_k x^k} = 0.$$

Don't forget that, if $\frac{N(x)}{D(x)} \underset{x \rightarrow 0}{\sim} \frac{N_1(x)}{D_1(x)}$, then $\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{N_1(x)}{D_1(x)}$.

6.24 Examples. Compute the following limits

1. $\lim_{x \rightarrow 0} \frac{\sin x - x}{(\ln(1+x))^4}.$

• $D(x) = (\ln(1+x))^4 \underset{x \rightarrow 0}{\sim} x^4$. Hence, it is sufficient to find $f.e_4(0)$ for $N(x)$.

• $N(x) = \sin x - x = -\frac{x^3}{3!} + o(x^4) \implies \sin x - x \underset{x \rightarrow 0}{\sim} -\frac{x^3}{3!}.$

Thus $\lim_{x \rightarrow 0} \frac{\sin x - x}{(\ln(1+x))^4} = \lim_{x \rightarrow 0} \frac{-\frac{x^3}{3!}}{x^4} = \lim_{x \rightarrow 0} \frac{-1}{6x} \nexists.$

2. $\lim_{x \rightarrow 0} \frac{(e^x - 1)(\cos x - 1 + \frac{x^2}{2})}{x(\cos x - 1)^2}.$

• $D(x) = x(\cos x - 1)^2 \underset{x \rightarrow 0}{\sim} x(-\frac{x^2}{2})^2 = \frac{x^5}{4}$. Hence, it is sufficient to find $f.e_5(0)$ for $N(x)$.

• $N(x) = (e^x - 1)(\cos x - 1 + \frac{x^2}{2}) \underset{x \rightarrow 0}{\sim} x(\frac{x^4}{4!}) = \frac{x^5}{24}.$

Thus $\lim_{x \rightarrow 0} \frac{(e^x - 1)(\cos x - 1 + \frac{x^2}{2})}{x(\cos x - 1)^2} = \lim_{x \rightarrow 0} \frac{\frac{x^5}{24}}{\frac{x^5}{4}} = \frac{1}{6}.$

3. $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{(\ln(1+x))^2}.$

• $D(x) = (\ln(1+x))^2 \underset{x \rightarrow 0}{\sim} x^2$. Hence, it is sufficient to find $f.e_2(0)$ for $N(x)$.

• $N(x) = \sin x - x + \frac{x^3}{6} = o(x^2) + \frac{x^3}{6} = o(x^2) = x^2\varepsilon(x)$ with $\varepsilon(x) \xrightarrow{x \rightarrow 0} 0$.

Thus $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{(\ln(1+x))^2} = \lim_{x \rightarrow 0} \frac{x^2\varepsilon(x)}{x^2} = \lim_{x \rightarrow 0} \varepsilon(x) = 0.$

4. $\lim_{x \rightarrow 0} (\ln|x| \ln(\cos x)).$

Since $\ln(\cos x) = \ln(1 - \frac{x^2}{2} + o(x^2)) \underset{x \rightarrow 0}{\sim} -\frac{x^2}{2}$, then $\ln|x| \ln(\cos x) \underset{x \rightarrow 0}{\sim} -\frac{x^2}{2} \ln|x|.$

Thus $\lim_{x \rightarrow 0} (\ln|x| \ln(\cos x)) = \lim_{x \rightarrow 0} (-\frac{x^2}{2} \ln|x| = 0).$

5. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 4x + 1} - x) = \lim_{x \rightarrow \infty} (\sqrt{x^2(1 + \frac{4}{x} + \frac{1}{x^2})} - x)$
 $= \lim_{t \rightarrow 0} \frac{1}{t} (\sqrt{1 + 4t + t^2} - 1) = \lim_{t \rightarrow 0} \frac{1}{t} ((1 + 4t + t^2)^{\frac{1}{2}} - 1)$
 $= \lim_{t \rightarrow 0} \frac{1}{t} (1 + \frac{1}{2}4t + o(t) - 1) = \lim_{t \rightarrow 0} \frac{2t + o(t)}{t}$
 $= \lim_{t \rightarrow 0} \frac{2t}{t} = 2.$

$(t = \frac{1}{x} \xrightarrow{x \rightarrow \infty} 0)$

6. $\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - 3}{2 - \sqrt{2x-2}} = \lim_{t \rightarrow 0} \frac{\sqrt{t+9} - 3}{2 - \sqrt{2t+4}} = \lim_{t \rightarrow 0} \frac{\sqrt{9(\frac{t}{9} + 1)} - 3}{2 - \sqrt{4(\frac{t}{2} + 1)}}$
 $= \lim_{t \rightarrow 0} \frac{3(\frac{t}{9} + 1)^{\frac{1}{2}} - 3}{2 - 2(\frac{t}{2} + 1)^{\frac{1}{2}}} = \frac{3}{2} \lim_{t \rightarrow 0} \frac{(1 + \frac{t}{9})^{\frac{1}{2}} - 1}{1 - (1 + \frac{t}{2})^{\frac{1}{2}}} = \frac{3}{2} \lim_{t \rightarrow 0} \frac{\frac{t}{18} + o(t)}{-\frac{t}{4}}$
 $= \frac{3}{2} \lim_{t \rightarrow 0} \frac{\frac{t}{18}}{-\frac{t}{4}} = -\frac{1}{3}.$

$(t = x - 3 \xrightarrow{x \rightarrow 3} 0)$

Let $f(x) = 1 + a_n x^n + o(x^n)$ and $g(x) = \frac{1}{b_m x^m + o(x^m)}$. How to find $\lim_{x \rightarrow 0} [f(x)]^{g(x)}$?

Note that

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} = \lim_{x \rightarrow 0} e^{\frac{\ln(1 + a_n x^n + o(x^n))}{b_m x^m + o(x^m)}} = \lim_{x \rightarrow 0} e^{\frac{a_n x^n}{b_m x^m}} = \lim_{x \rightarrow 0} e^{\frac{a_n}{b_m} x^{n-m}}.$$

Three cases may occur:

First case: If $n < m$, then

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} = +\infty \text{ or } 0 \text{ if } (m - n) \text{ is even.}$$

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} \nexists \text{ if } (m - n) \text{ is odd.}$$

Second case: If $n = m$, then

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} = e^{\frac{a_n}{b_m}}.$$

Third case: If $n > m$, then

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} = e^0 = 1.$$

6.25 Property. (Substitution And Equivalent)

Let $x_0, t_0 \in \mathbb{R}$. If $f(x) \underset{x \rightarrow x_0}{\sim} g(x)$ and $u(t) \xrightarrow[t \rightarrow t_0]{} x_0$ with one of the following conditions in case $x_0 \in \mathbb{R}$

1. $u(t) \neq x_0$ in some deleted neighborhood of t_0 ,
 2. g, f are continuous at x_0 with g or f never vanishes in some neighborhood of x_0
- then $f(u(t)) \underset{t \rightarrow t_0}{\sim} g(u(t))$.

Proof.

1. We have $f(x) \underset{x \rightarrow x_0}{\sim} g(x)$, then $f(x) = (\varepsilon(x) + 1)g(x)$ on $I_{x_0} \setminus \{x_0\}$ with $\varepsilon(x) \xrightarrow{x \rightarrow x_0} 0$. Since $u(t) \xrightarrow[t \rightarrow t_0]{} x_0$, then there exists $I_{t_0} \setminus \{t_0\}$ deleted neighborhood of t_0 such that for all $t \in I_{t_0} \setminus \{t_0\}$ we have $u(t) \in I_{x_0} \setminus \{x_0\}$ and $f(u(t)) = (\varepsilon(u(t)) + 1)g(u(t))$ with $\varepsilon(u(t)) \xrightarrow[t \rightarrow t_0]{} 0$. (substitution theorem)
2. We have ε is continuous at x_0 and thus by the help of the substitution theorem we obtain $f(u(t)) = (\varepsilon(u(t)) + 1)g(u(t))$ with $\varepsilon(u(t)) \xrightarrow[t \rightarrow t_0]{} 0$. $\square$

For examples:

- $\sin x \underset{x \rightarrow 0}{\sim} x$ and $(t - 1) \ln(t - 1) \xrightarrow[t \rightarrow 1]{} 0$ and $(t - 1) \ln(t - 1) \neq 0$ in some deleted neighborhood of 1 implies $\sin((t - 1) \ln(t - 1)) \underset{t \rightarrow 1}{\sim} (t - 1) \ln(t - 1)$.
- If $u(t) \xrightarrow[t \rightarrow t_0]{} 0$ and $u(t) \neq 0$ in some deleted neighborhood of t_0 , then

$$\ln(1 + u(t)) \underset{t \rightarrow t_0}{\sim} u(t), \quad \sin(u(t)) \underset{t \rightarrow t_0}{\sim} u(t), \quad \cos(u(t)) - 1 \underset{t \rightarrow t_0}{\sim} \frac{(u(t))^2}{2}.$$

6.4 Exercises

Exercise 1. Let I be open interval, $x_0 \in I$ and $f : I \setminus \{x_0\} \mapsto \mathbb{R}$. Assume that

$$f(x) = a_0 + a_1(x - x_0) + (x - x_0)\varepsilon(x) \quad \forall x \in I \setminus \{x_0\} \quad \text{with } \varepsilon(x) \xrightarrow{x \rightarrow x_0} 0.$$

Show that f has an extension by continuity $\tilde{f}$ at x_0 and $\tilde{f}$ is differentiable at x_0 with $\tilde{f}'(x_0) = a_1$.

Exercise 2. Let $f(x) = (\sqrt{1 + x^3} + \sin(ax))^{\frac{1}{\sinh(ax)}}$, $a \in \mathbb{R}^*$.

1. Give the $f.e_2(0)$ of $f(x)$.

2. Show that f can be extended by continuity at 0 and give its extension g .
3. Justify the differentiability of g at 0 and give $g'(0)$.
4. Give the equation of the tangent (T) to (C) the representative curve of g at $x = 0$. Discuss the relative positions according to the values of a .

Exercise 3. Let $f(x) = 2 + ax^2 + bx^3 + x^4 + o(x^5)$, $a, b \in \mathbb{R}$. Give the equation of the tangent (T) to (C) the representative curve of f at $x = 0$. Discuss the relative positions according to the values of a and b .

Exercise 4. Prove Corollary 6.17 up to order 5.

Exercise 5. Let $y \in \mathbb{R}$. Show that

$$\lim_{x \rightarrow \infty} \left(1 + \frac{y}{x}\right)^x = e^y, \quad \lim_{x \rightarrow 0^+} x^x = 1, \quad \lim_{x \rightarrow 0^+} x^{-1/x} = 0, \quad \lim_{x \rightarrow 0^+} x^{\ln x} = \infty, \quad \lim_{x \rightarrow 0^+} x^{-1/\ln x} = 1/e.$$

Exercise 6. Let $f(x) = x \left[\sqrt{x^2 - ax + 1} - x \left(\cos \frac{1}{x} \right)^x \right]$.

Determine the value of a so that $\lim_{x \rightarrow \infty} f(x)$ exists and finite. Give the value of this limit.

Exercise 7. Compute $\lim_{x \rightarrow 0^+} \frac{e^{x/2} - \cosh(\sqrt{x})}{(x + \sqrt{x})^\alpha}$ for all values $\alpha \in \mathbb{R}$.

Exercise 8. Let $f(x) = \sqrt{x} + \ln x$ and $g(x) = \sqrt[4]{x^4 + x^3} - \sqrt[4]{x^4 - x^3}$.

Give $f.e_3(1)$ of $f(x)$ and $f.e_2(-\infty)$ of $g(x)$.

Exercise 9. Let $f(x) = \arctan(\sin x)$, $g(x) = \arctan(e^x)$ and $h(x) = \arctan\left(\frac{1+x}{1-x}\right)$.

Give $f.e_3(0)$ of $f(x)$, $g(x)$ and $h(x)$.

Exercise 10. Let $f(x) = xe^{x^2}$, $x \in \mathbb{R}$.

1. Show that f has an inverse function $g = f^{-1}$ and determine D_g .
2. Show that g has $f.e_3(0)$ and give this finite expansion.
Hint: use $g \circ f(x) = x$ and the uniqueness of finite expansion.